DIAMOND

MV B- BUILD FINAL

2003/08/21

Changed by: EE2	Date Changed: Thursday, August 21, 2003	Time Changed: 11:19:38 am	QA CHK	VER: A02	Model Number: PC8803	Sheet 1 of 67

Engineer: David Du Drawn by: David Du R&D CHK DOC CTRL CHK MFG ENGR CHK	INVENTEC	Size: A3
	TITLE: DIAMOND	

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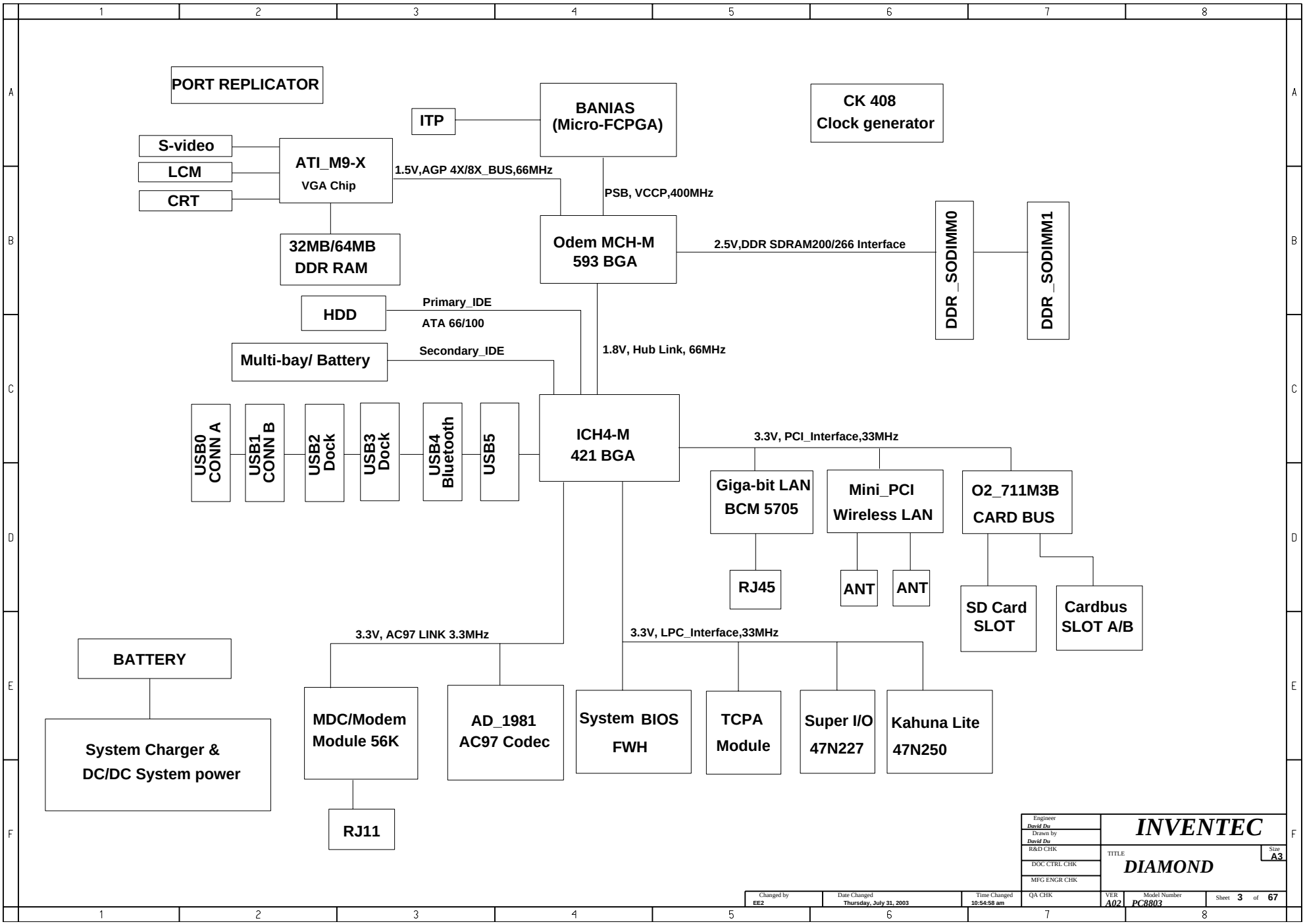
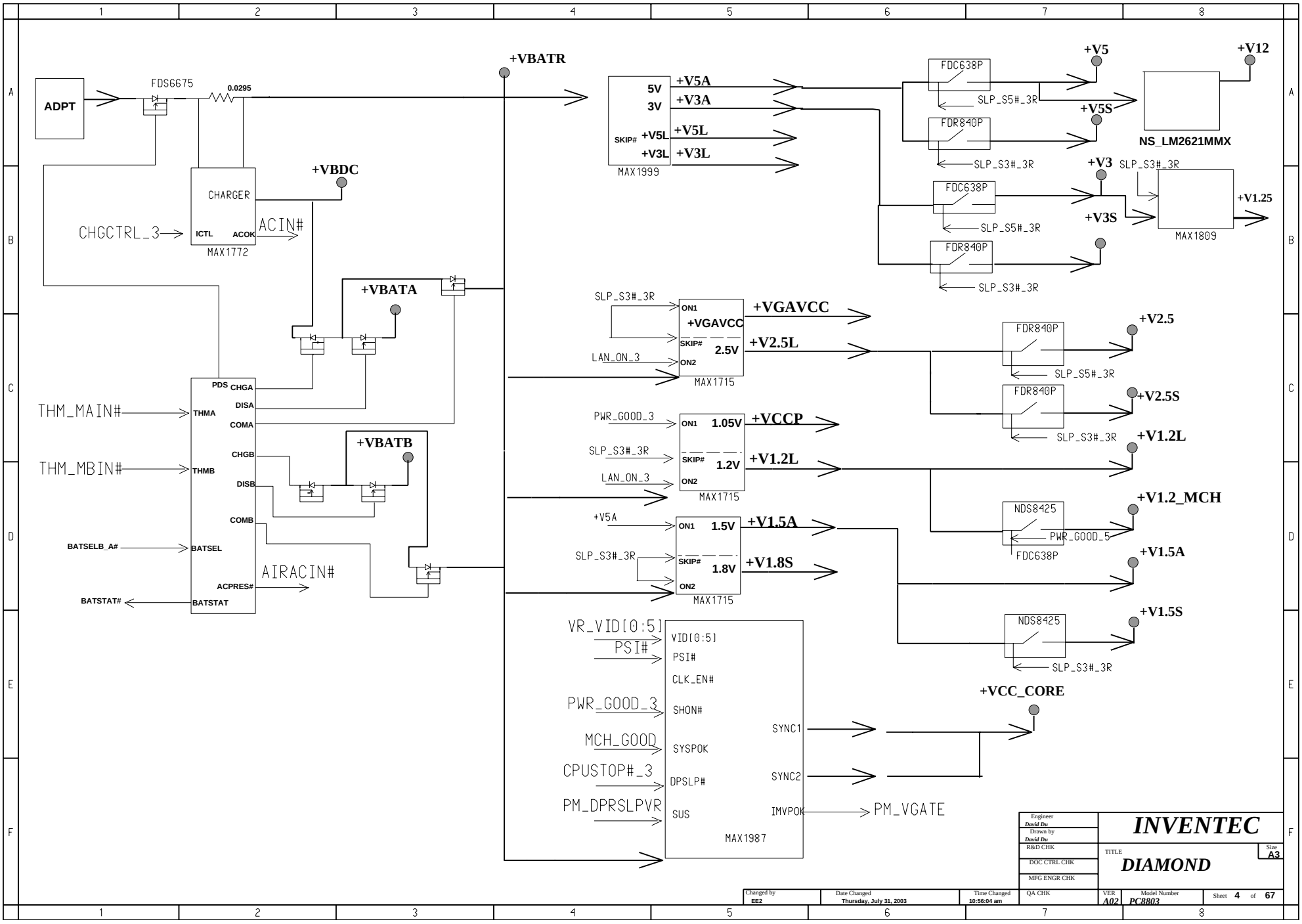
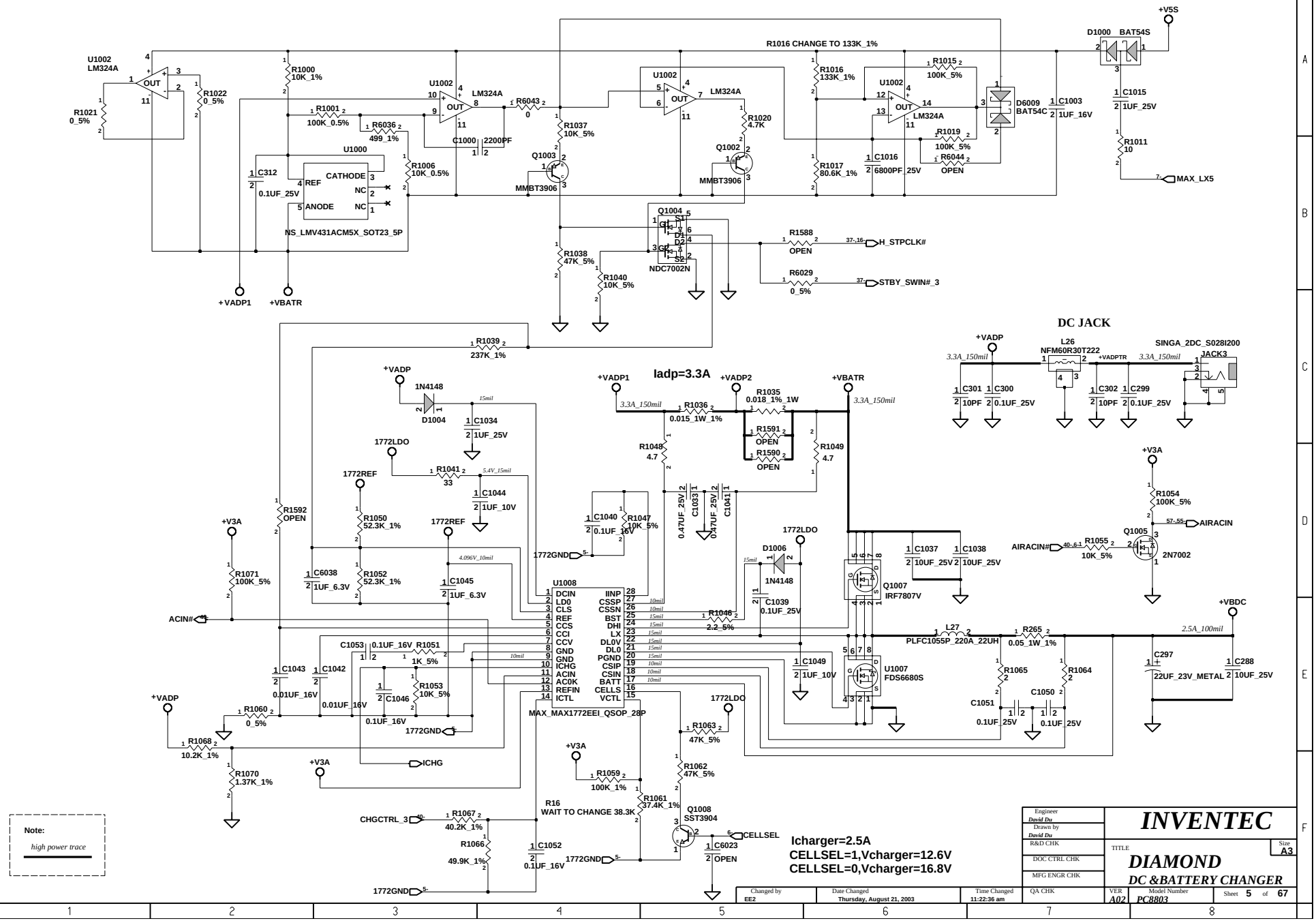
Engineer: David Du	INVENTEC	
Drawn by: David Du	TITLE: DIAMOND	Size: A3
R&D CHK		
DOC CTRL CHK		
MFG ENGR CHK		

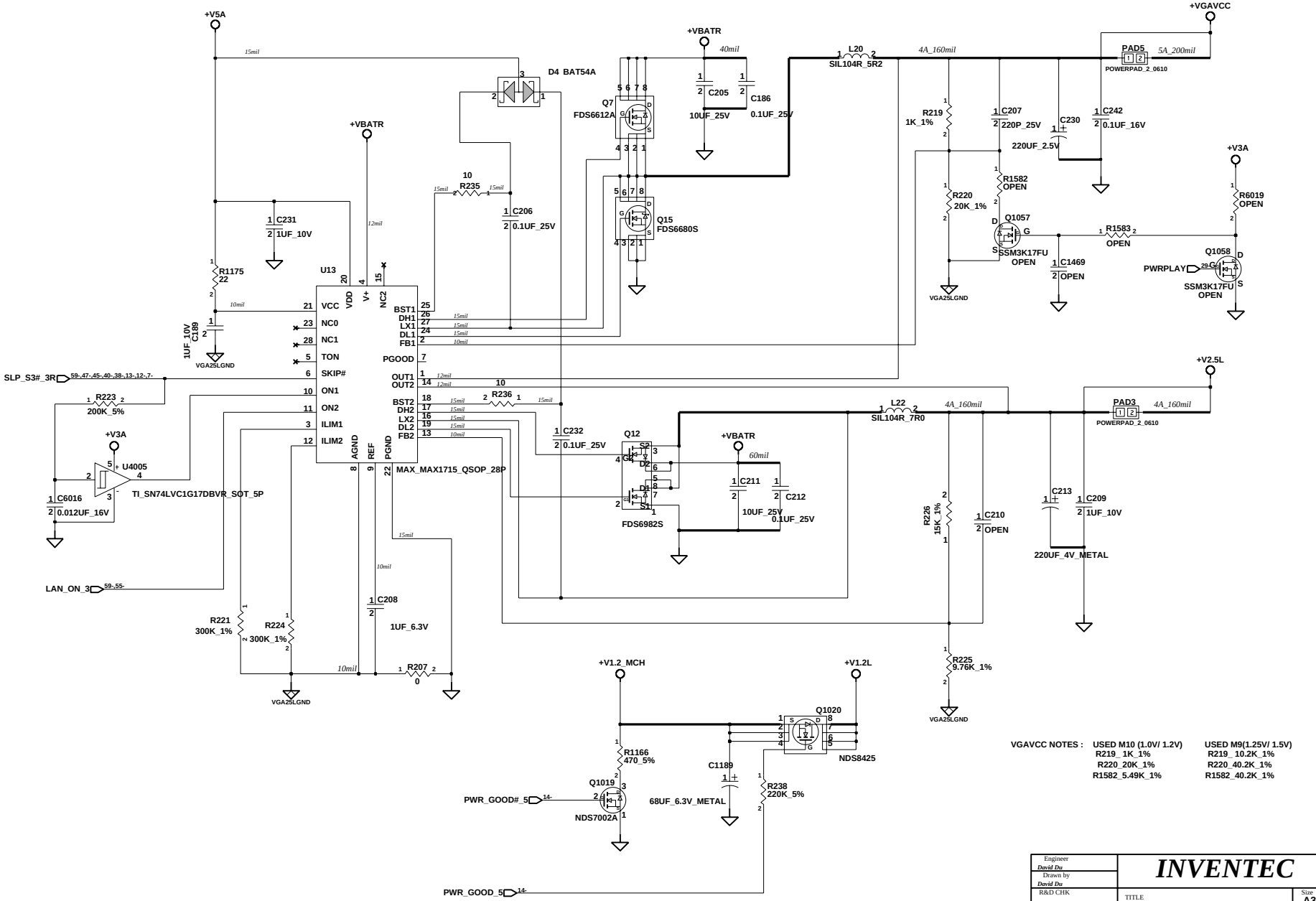

PORT REPLICATOR
ITP
BANIAS
(Micro-FCPGA)
CK 408
Clock generator
S-video
LCM
CRT
ATI_M9-X
VGA Chip
1.5V,AGP 4X/8X_BUS,66MHz
PSB, VCCP,400MHz
32MB/64MB
DDR RAM
Odem MCH-M
593 BGA
2.5V,DDR SDRAM200/266 Interface
DDR_SODIMM0
DDR_SODIMM1
HDD
Primary_IDE
ATA 66/100
Multi-bay/ Battery
Secondary_IDE
1.8V, Hub Link, 66MHz
USB0
CONN A
USB1
CONN B
USB2
Dock
USB3
Dock
USB4
Bluetooth
USB5
ICH4-M
421 BGA
3.3V, PCI_Interface,33MHz
Giga-bit LAN
BCM 5705
Mini_PCI
Wireless LAN
O2_711M3B
CARD BUS
RJ45
ANT
ANT
SD Card
SLOT
Cardbus
SLOT A/B
3.3V, AC97 LINK 3.3MHz
3.3V, LPC_Interface,33MHz
BATTERY
System Charger &
DC/DC System power
MDC/Modem
Module 56K
AD_1981
AC97 Codec
System BIOS
FWH
TCPA
Module
Super I/O
47N227
Kahuna Lite
47N250
RJ11
Engineer
David Du
Drawn by
David Du
R&D CHK
DOC CTRL CHK
MFG ENGR CHK
INVENTEC
TITLE
DIAMOND
Size
A3
Changed by
EE2
Date Changed
Thursday, July 31, 2003
Time Changed
10:54:58 am
QA CHK
VER
A02
Model Number
PC8803
Sheet 3 of 67



Changed by EE2	Date Changed Thursday, July 31, 2003	Time Changed 10:56:04 am	QA CHK	VER A02	Model Number PC8803	Sheet 4 of 67

Engineer David Du	**INVENTEC**	
Drawn by David Du	TITLE **DIAMOND**	Size A3
R&D CHK		
DOC CTRL CHK		
MFG ENGR CHK		


R1016 CHANGE TO 133K_1%
DC JACK
ladp=3.3A
U1008
MAX_MAX1772EEI_QSOP_28P
Q1007 IRF7807V
U1007 FDS6680S
Q1008 SST3904
Q1005 2N7002
D1000 BAT54S
D6009 BAT54C
D1004 1N4148
D1006 1N4148
L26 NFM60R30T222
L27 PLFC1055P_220A_22UH
JACK3 SINGA_2DC_S028I200
NS_LMV431ACM5X_SOT23_5P
NDC7002N
MMBT3906
Icharger=2.5A
CELLSEL=1,Vcharger=12.6V
CELLSEL=0,Vcharger=16.8V
R16 WAIT TO CHANGE 38.3K
Note: high power trace
INVENTEC
DIAMOND
DC &BATTERY CHANGER
PC8803
A02
Sheet 5 of 67



VGAVCC NOTES :

USED M10 (1.0V/ 1.2V)	USED M9(1.25V/ 1.5V)
R219_1K_1%	R219_10.2K_1%
R220_20K_1%	R220_40.2K_1%
R1582_5.49K_1%	R1582_40.2K_1%

Changed by: EE2	Date Changed: Thursday, August 21, 2003	Time Changed: 11:26:35 am	QA CHK	VER: A02	Model Number: PC8803

Engineer: David Du
Drawn by: David Du
R&D CHK
DOC CTRL CHK
MFG ENGR CHK

INVENTEC

TITLE: **DIAMOND**
SYSTEM POWER (2.5V/1.25V)

Size: A3

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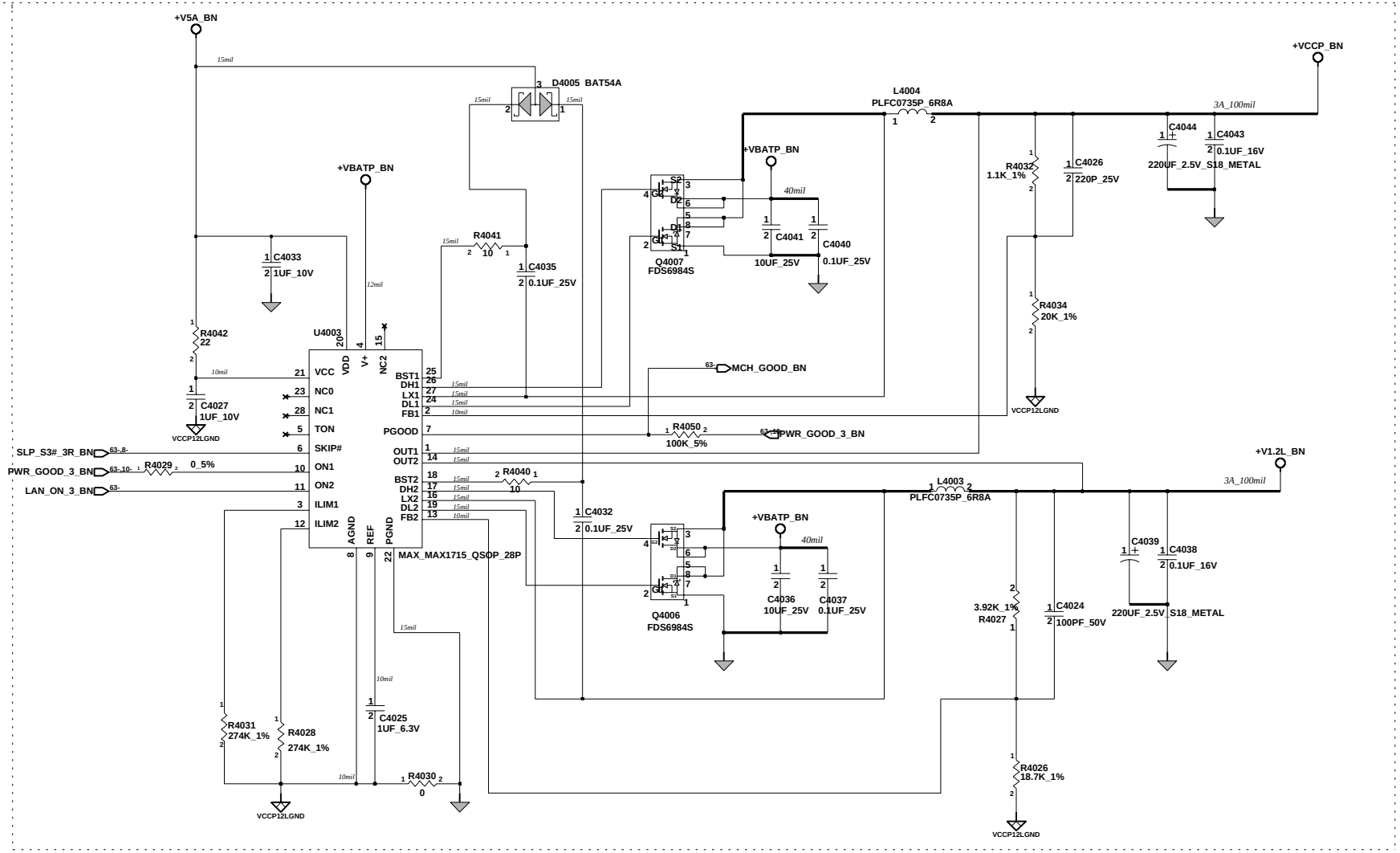
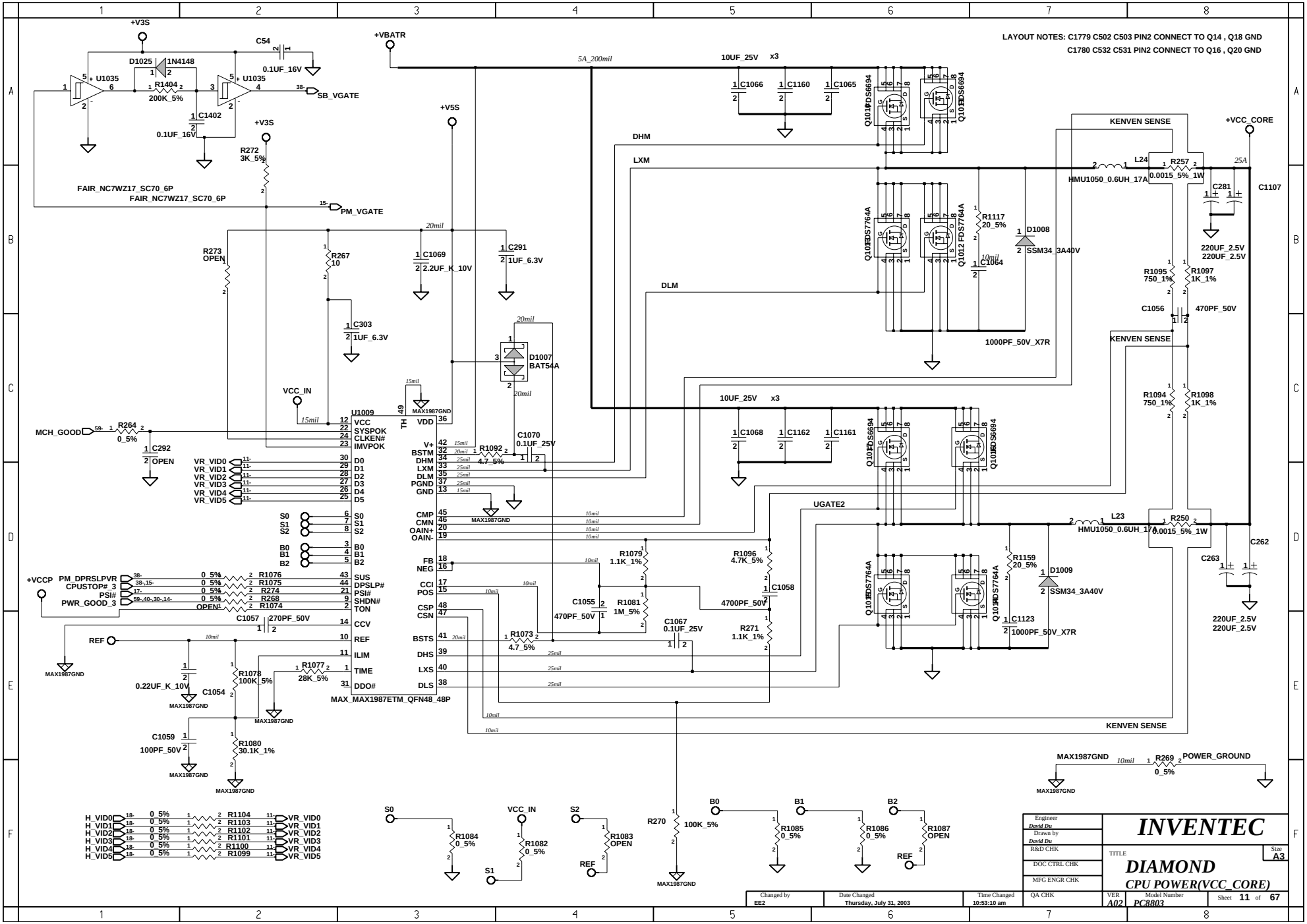
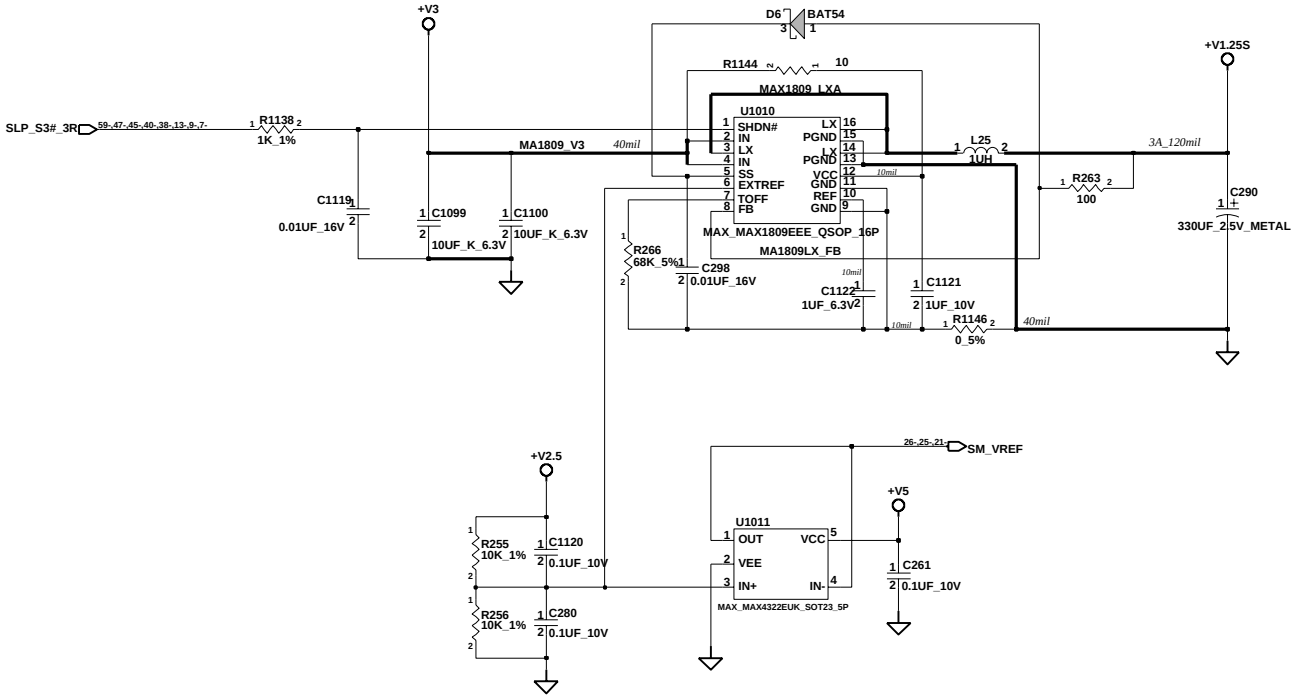


SYSTEM POWER (VCCP&1.2) ON BUTTON DAUGHTER BOARD

Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet
EE2	Thursday, July 31, 2003	6:06:02 pm		A02	PC8803	10 of 67

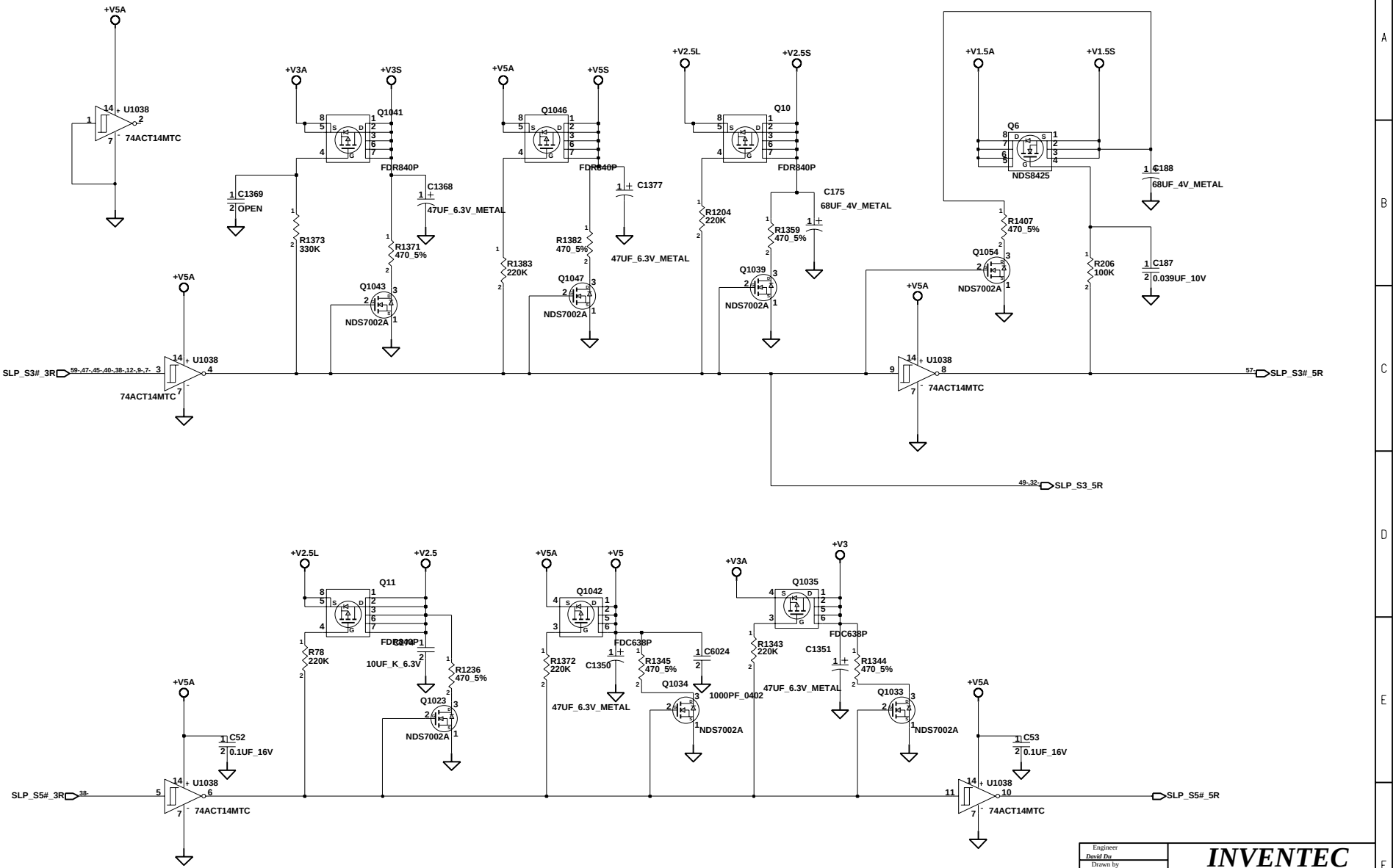
Engineer: David Du	INVENTEC
Drawn by: David Du	
R&D CHK	TITLE: DIAMOND
DOC CTRL CHK	SYSTEM POWER(VCCP/1.2V)
MFG ENGR CHK	Size: A3


LAYOUT NOTES: C1779 C502 C503 PIN2 CONNECT TO Q14 , Q18 GND
C1780 C532 C531 PIN2 CONNECT TO Q16 , Q20 GND
+V3S
+VBATR
+V5S
+VCC_CORE
SB_VGATE
PM_VGATE
VCC_IN
MCH_GOOD
KENVEN SENSE
DHM
LXM
DLM
UGATE2
10UF_25V x3
1000PF_50V_X7R
220UF_2.5V
220UF_2.5V
470PF_50V
SSM34_3A40V
HMU1050_0.6UH_17A
0.0015_5%_1W
MAX_MAX1987ETM_QFN48_48P
MAX1987GND
POWER_GROUND
INVENTEC
DIAMOND
CPU POWER(VCC_CORE)
A02
PC8803
Thursday, July 31, 2003
10:53:10 am
EE2
Sheet 11 of 67



Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet
EE2	Thursday, July 31, 2003	10:53:16 am		A02	PC8803	12 of 67

Engineer: David Du	**INVENTEC**	
Drawn by: David Du	TITLE: **DIAMOND**	Size: A3
R&D CHK	*VGA POWER(VDD_CORE)*	
DOC CTRL CHK		
MFG ENGR CHK		



Engineer	David Du
Drawn by	David Du
R&D CHK	
DOC CTRL CHK	
MFG ENGR CHK	

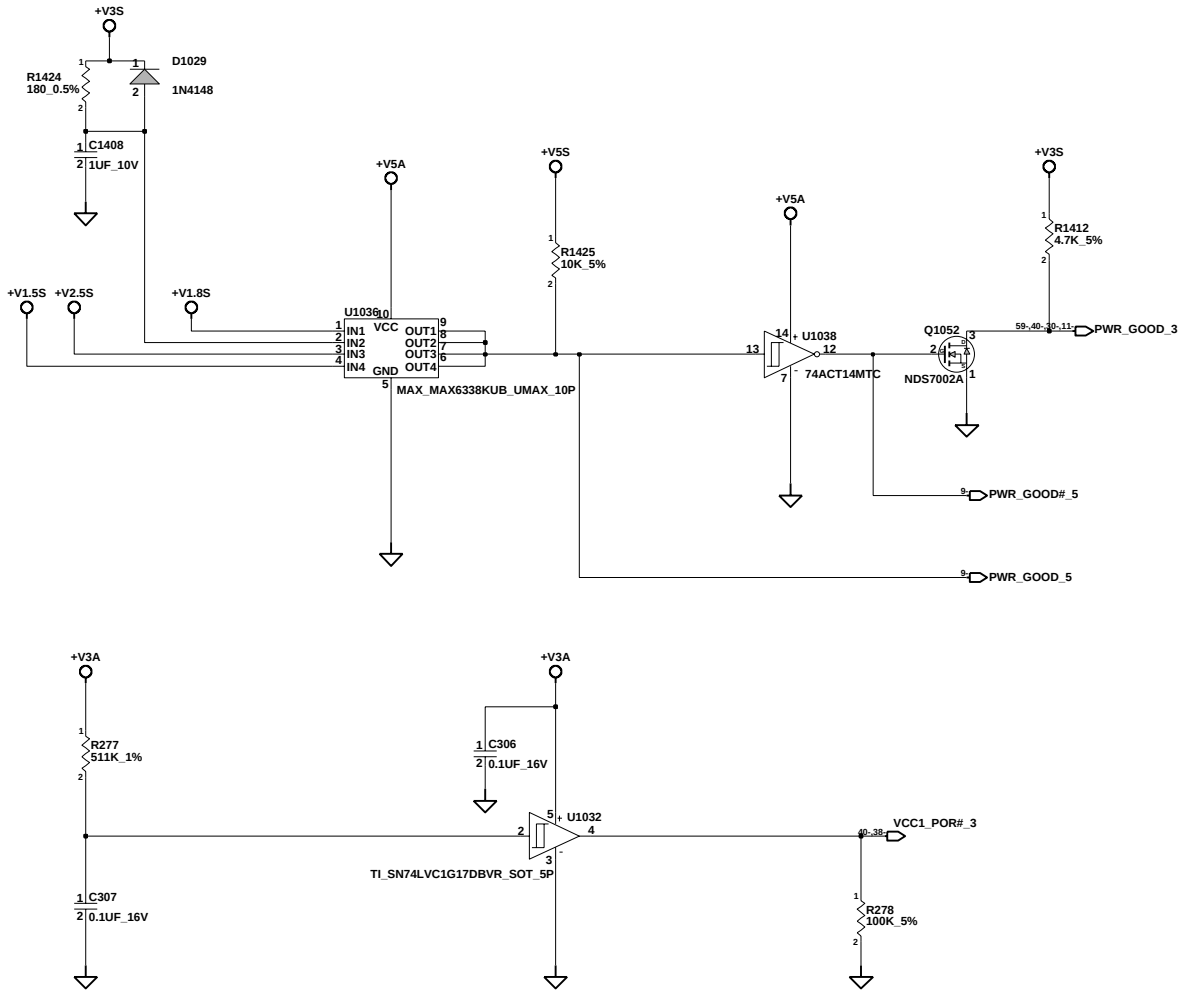
INVENTEC

TITLE: **DIAMOND**

POWER(SLEEP)

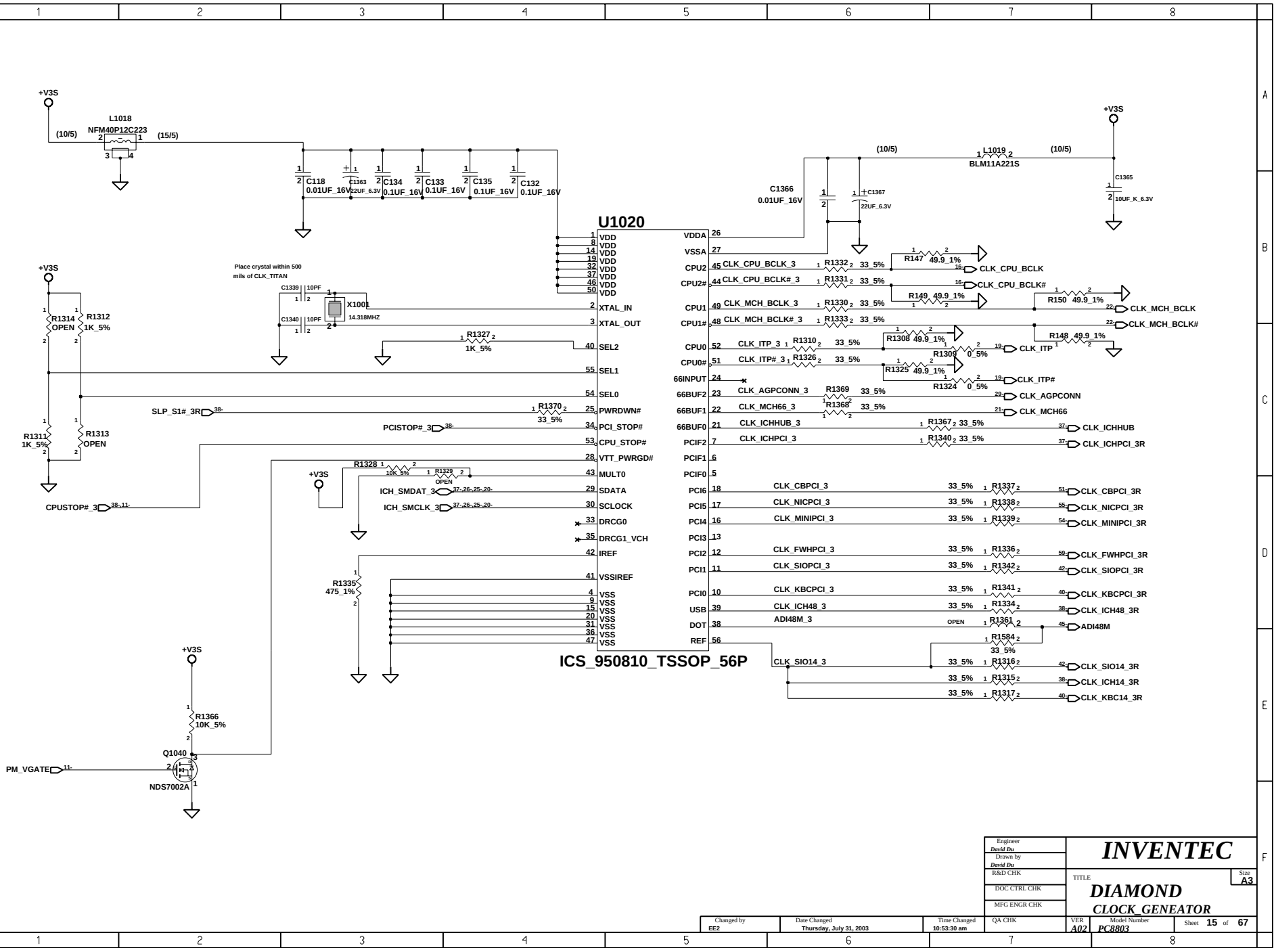
Size: A3

Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet
EE2	Thursday, August 21, 2003	11:28:38 am		A02	PC8803	13 of 67



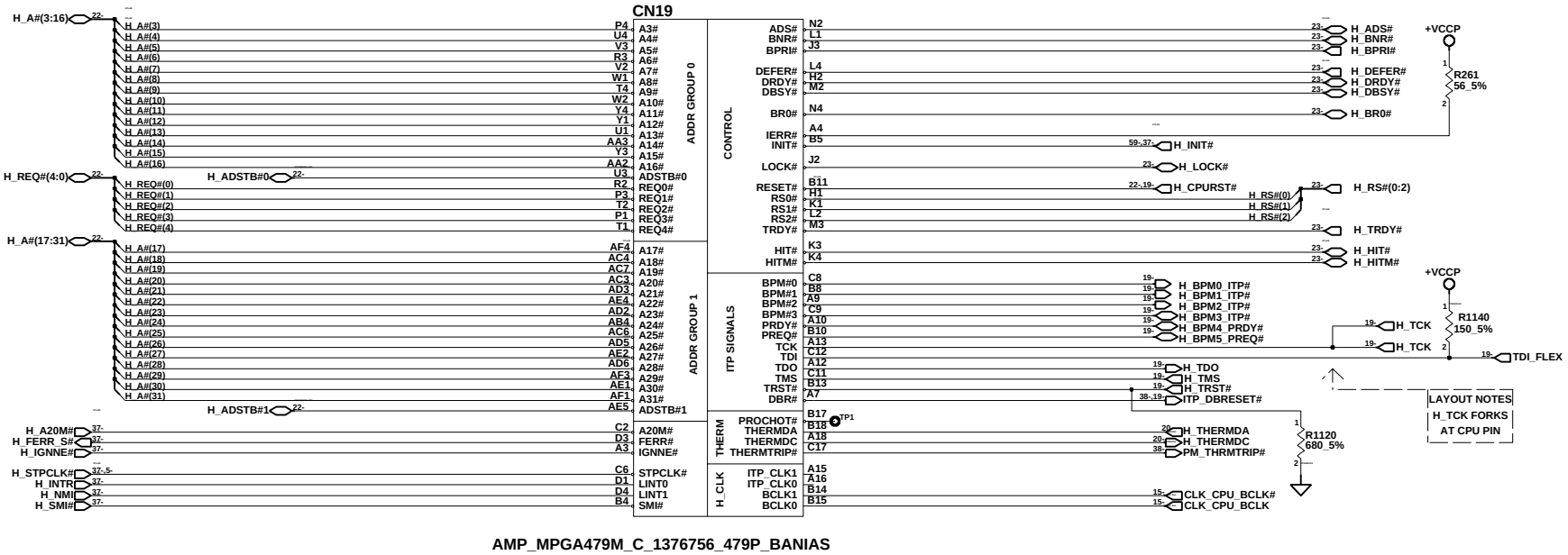
Changed by EE2	Date Changed Thursday, July 31, 2003	Time Changed 10:53:25 am	QA CHK	VER A02	Model Number PC8803	Sheet 14 of 67

Engineer *David Du*	**INVENTEC**	
Drawn by *David Du*		
R&D CHK	TITLE *DIAMOND*	Size A3
DOC CTRL CHK	*POWER(SEQUENCE)*	
MFG ENGR CHK		



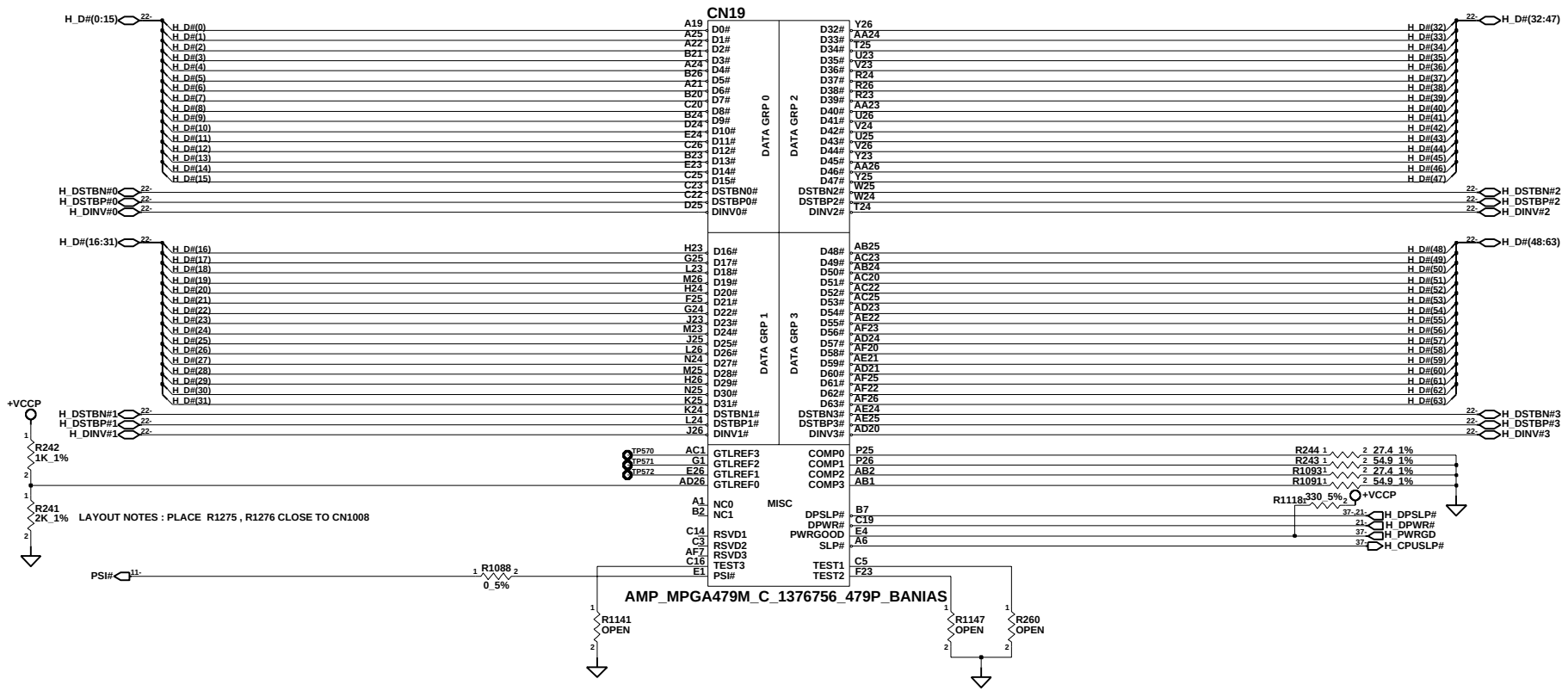
Changed by EE2	Date Changed Thursday, July 31, 2003	Time Changed 10:53:30 am	QA CHK	VER A02	Model Number PC8803	Sheet 15 of 67

Engineer David Du	INVENTEC	
Drawn by David Du	TITLE DIAMOND CLOCK_GENEATOR	Size A3
R&D CHK		
DOC CTRL CHK		
MFG ENGR CHK		



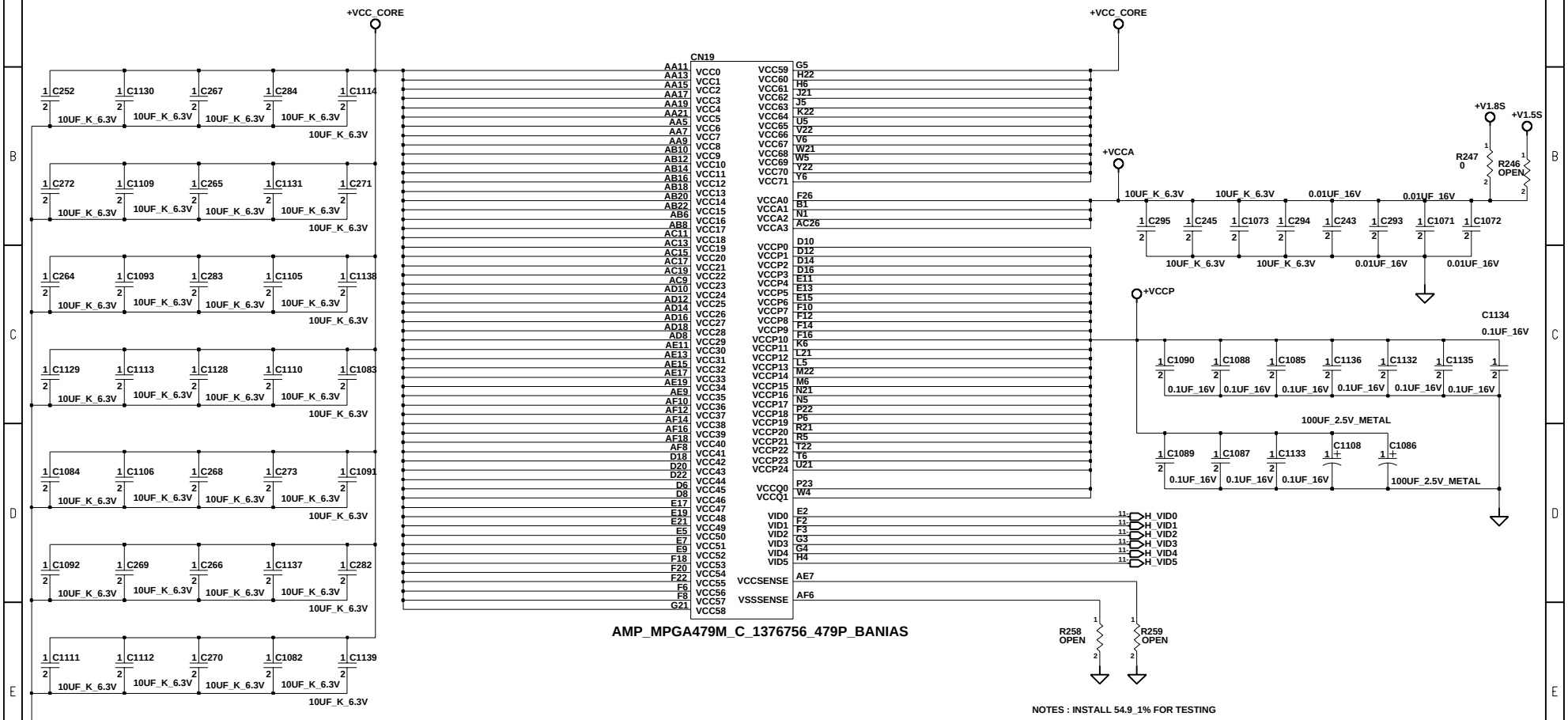
Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet
EE2	Thursday, July 31, 2003	10:53:36 am		A02	PC8803	16 of 67

Engineer: David Du	INVENTEC	
Drawn by: David Du	TITLE: DIAMOND / BANIAS-1	Size: A3
R&D CHK		
DOC CTRL CHK		
MFG ENGR CHK		



Changed by EE2	Date Changed Thursday, July 31, 2003	Time Changed 10:53:41 am	QA CHK	VER A02	Model Number PC8803	Sheet 17 of 67

Engineer David Du	INVENTEC	
Drawn by David Du	TITLE DIAMOND BANIAS-2	Size A3
R&D CHK		
DOC CTRL CHK		
MFG ENGR CHK		



NOTES : INSTALL 54.9_1% FOR TESTING

Changed by: EE2	Date Changed: Thursday, July 31, 2003	Time Changed: 10:53:46 am	QA CHK	VER: A02	Model Number: PC8803

Engineer: David Du
Drawn by: David Du
R&D CHK
DOC CTRL CHK
MFG ENGR CHK

INVENTEC

TITLE: **DIAMOND**
BANIAS-3

Size: A3

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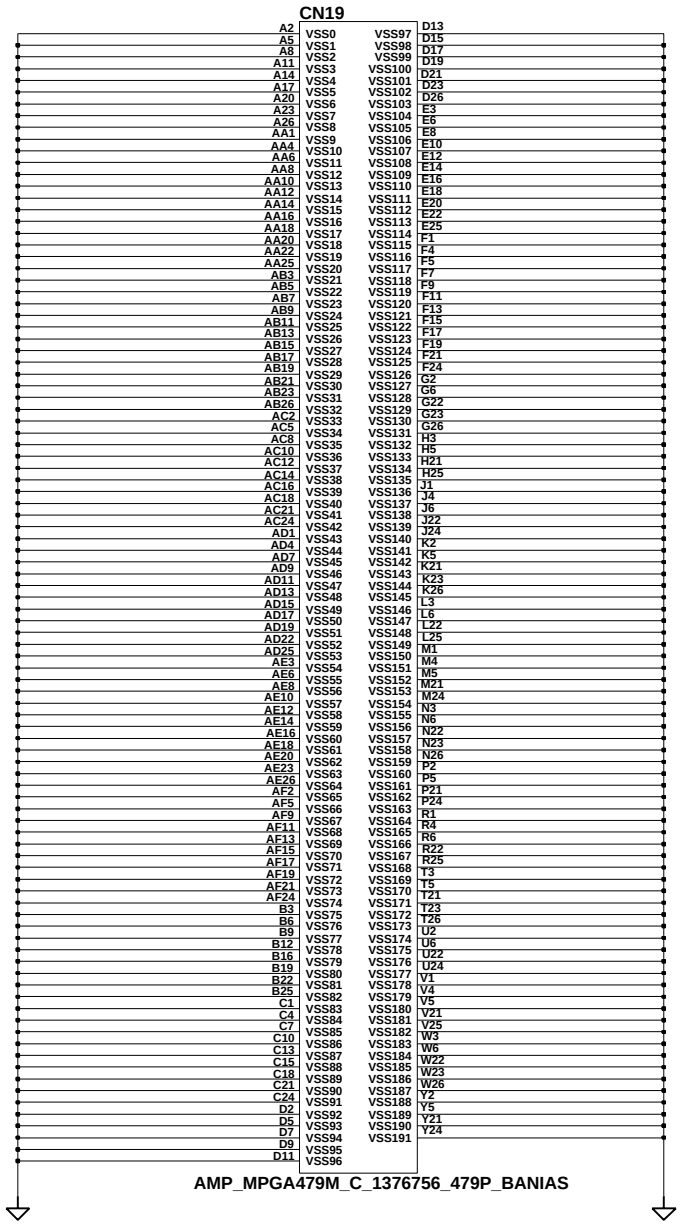
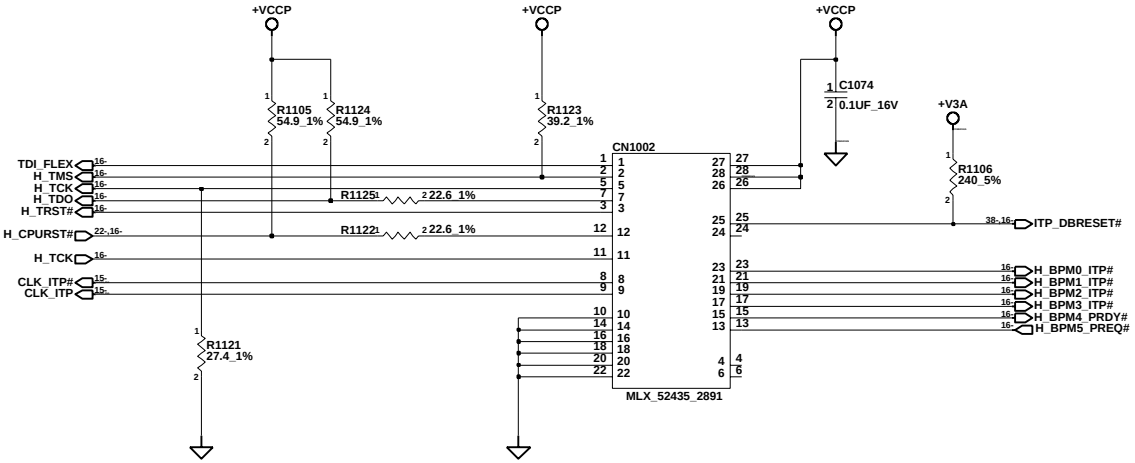


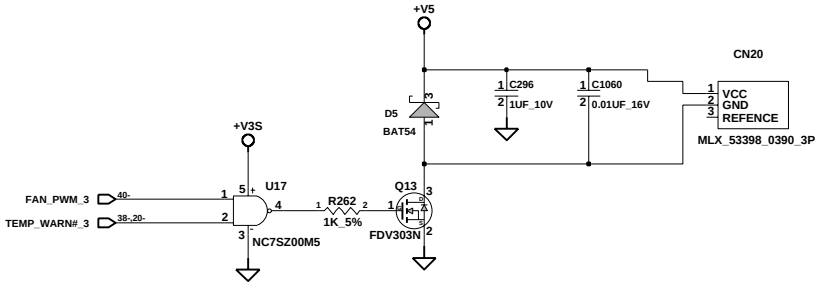
In-Target Probe

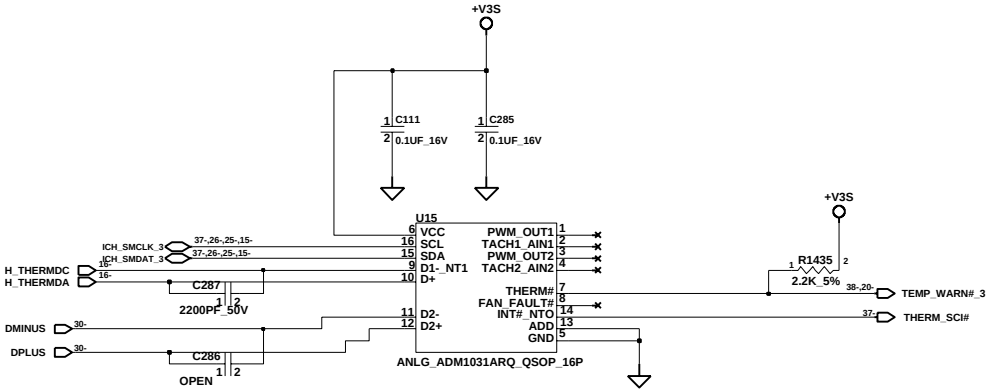


LAYOUT NOTES : H_CPURST# PIN OF MCH_M FORK OUT INTO TWO BRANCHES ON CPU AND ITP CONNECTOR

Changed by EE2	Date Changed Thursday, July 31, 2003	Time Changed 10:53:51 am	QA CHK	VER A02	Model Number PC8803	Sheet 19 of 67

Engineer *David Du*	**INVENTEC**	
Drawn by *David Du*	TITLE *DIAMOND* *BANIAS-4*	Size A3
R&D CHK		
DOC CTRL CHK		
MFG ENGR CHK		





LAYOUT NOTES: PUT THE THERMAL SENSOR CLOSE TO CPU

Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number
EE2	Wednesday, August 13, 2003	10:20:09 pm		A02	PC8803

Engineer: *David Du*
Drawn by: *David Du*
R&D CHK
DOC CTRL CHK
MFG ENGR CHK

INVENTEC

TITLE: ***DIAMOND***
THERMAL&FAN CONTROLLER

Size: A3

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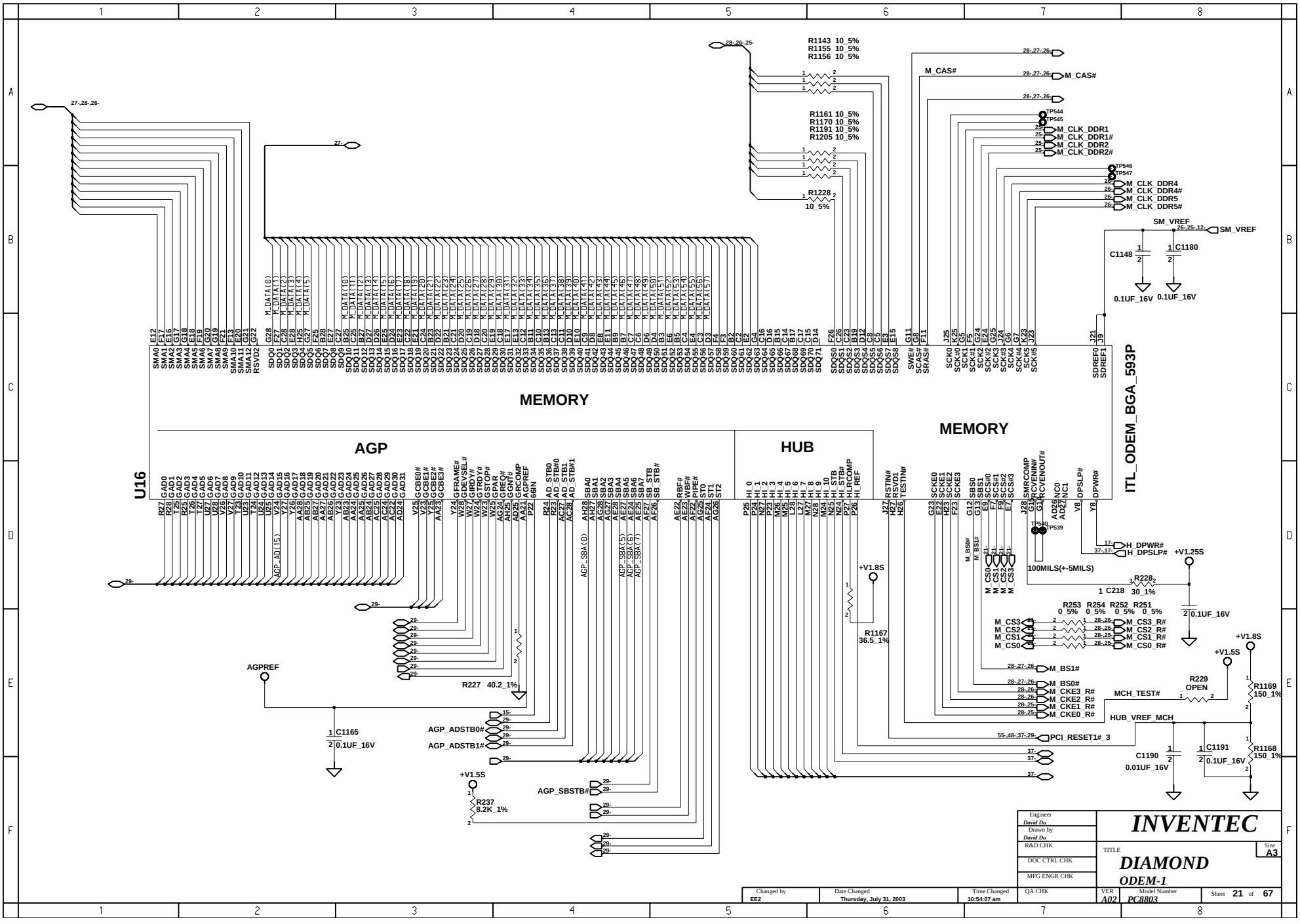
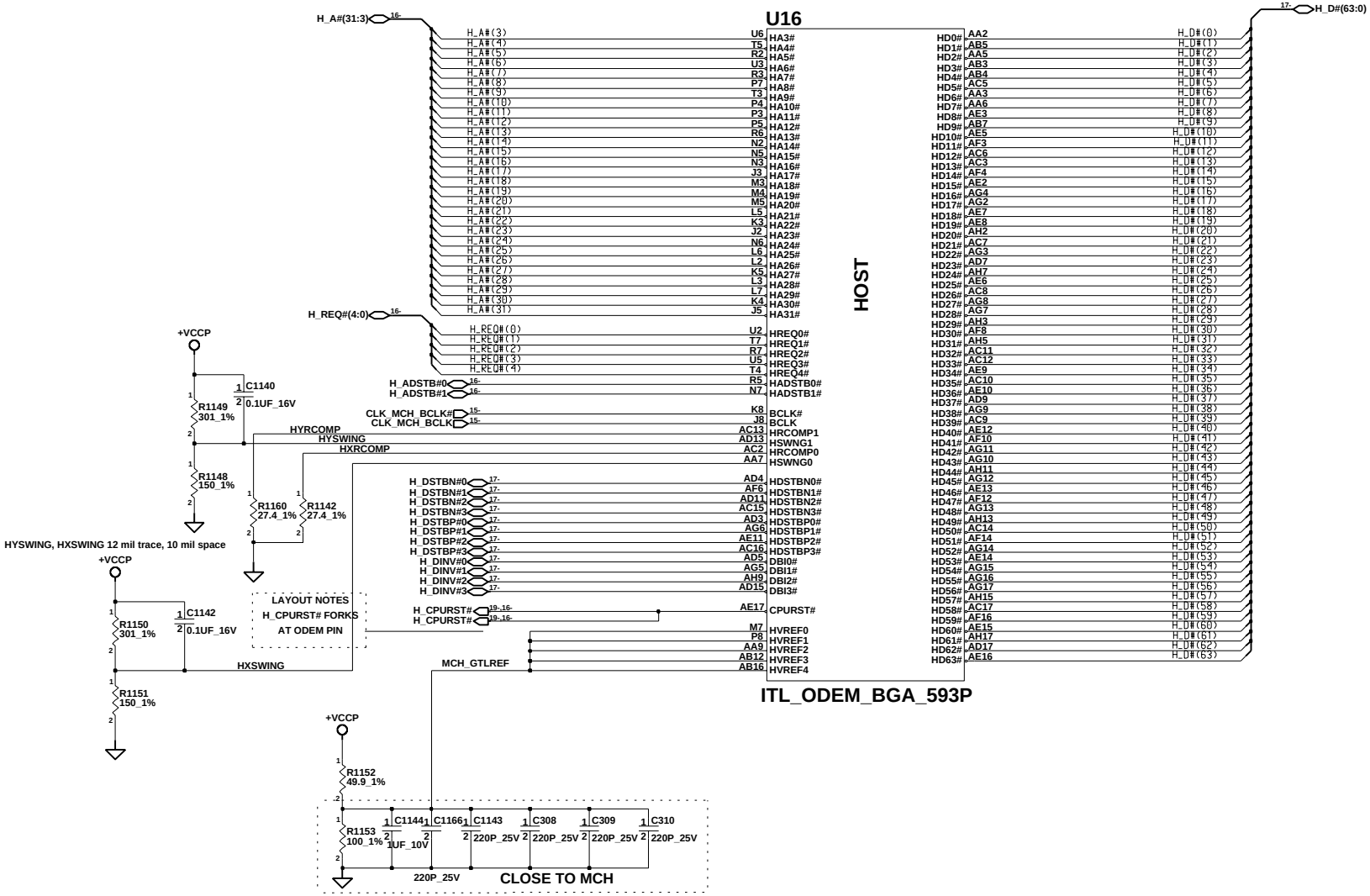


Changed by: EE2	Date Changed: Thursday, July 31, 2003	Time Changed: 10:54:07 am	QA CHK	VER: A02	Model Number: PC8803	Sheet 21 of 67

Engineer: David Du	INVENTEC
Drawn by: David Du	
R&D CHK	TITLE: DIAMOND
DOC CTRL CHK	ODEM-1
MFG ENGR CHK	Size: A3



Engineer	David Du
Drawn by	David Du
R&D CHK	
DOC CTRL CHK	
MFG ENGR CHK	

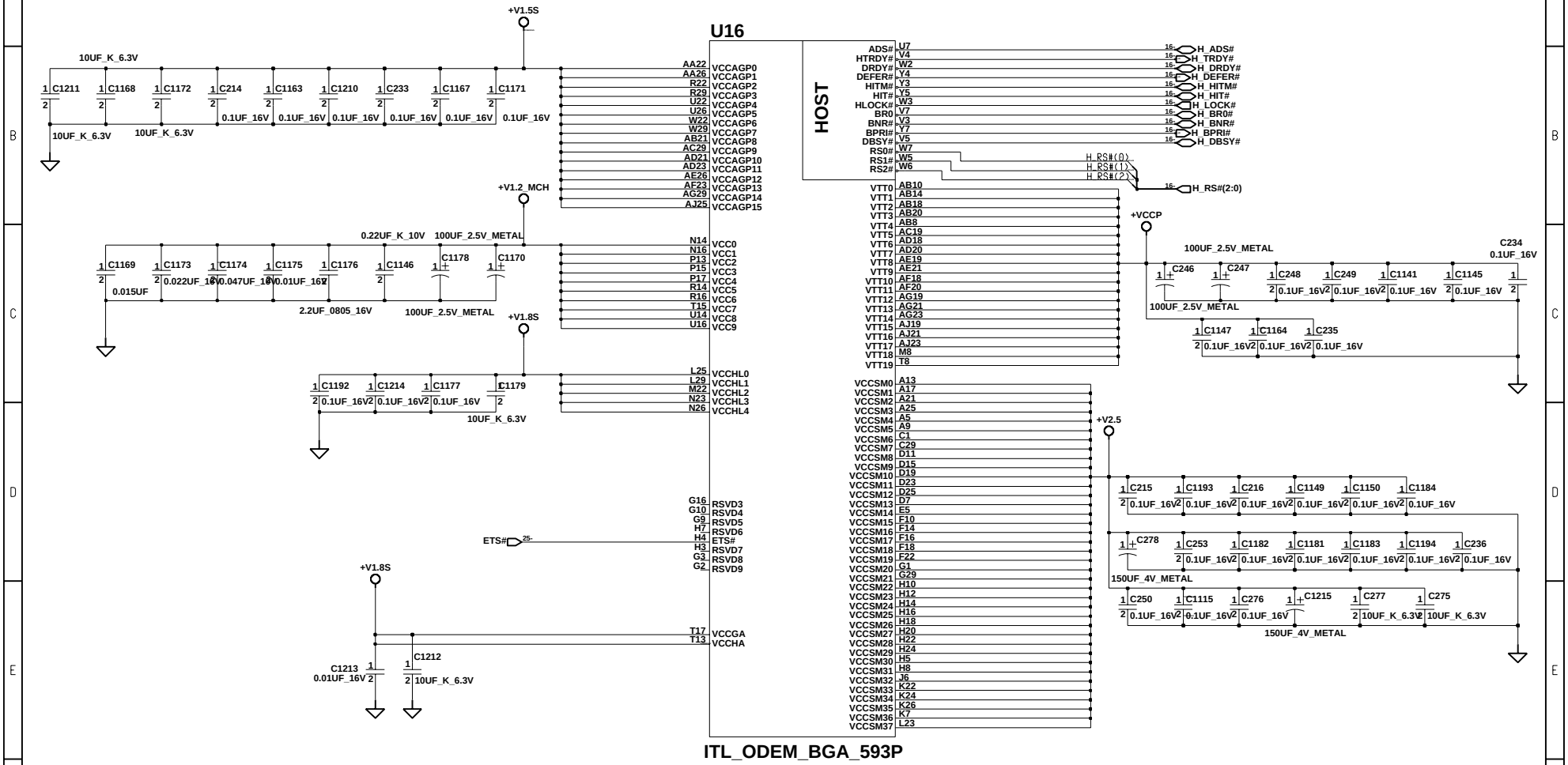
INVENTEC

TITLE: **DIAMOND**

ODEM-2

Size: A3

Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet
EE2	Thursday, July 31, 2003	10:54:13 am		A02	PC8803	22 of 67



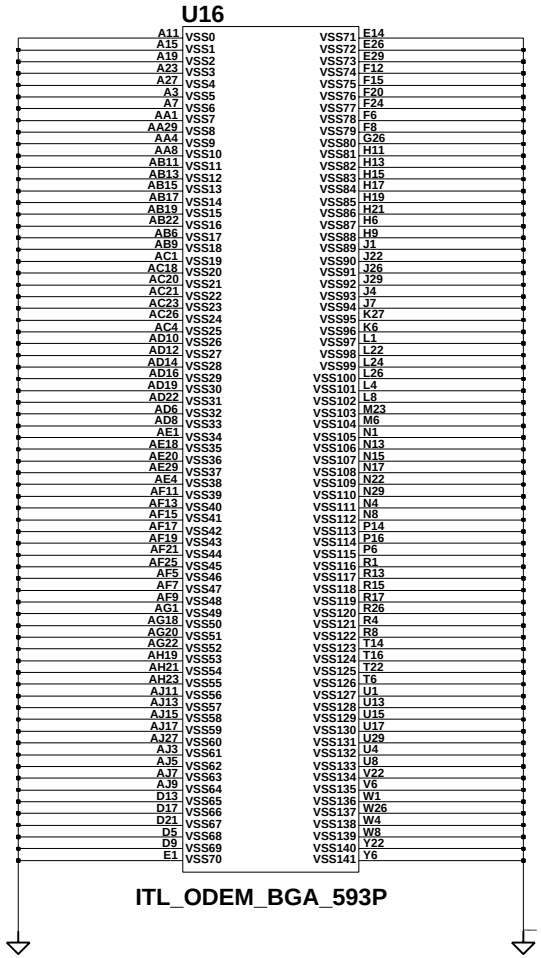
Changed by EE2	Date Changed Thursday, July 31, 2003	Time Changed 10:54:19 am	QA CHK	VER A02	Model Number PC8803	Sheet 23 of 67

Engineer: David Du
Drawn by: David Du
R&D CHK
DOC CTRL CHK
MFG ENGR CHK

INVENTEC

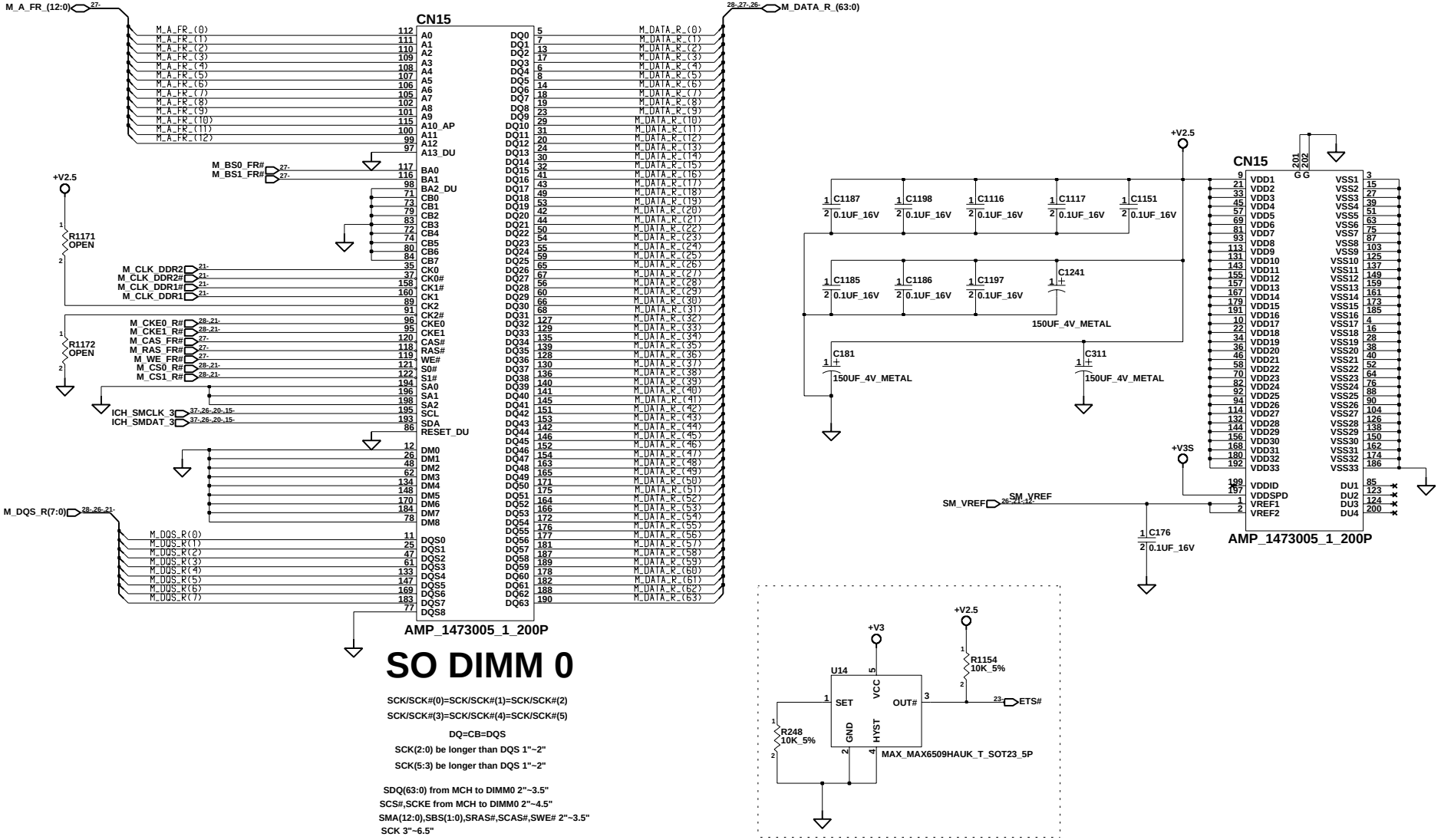
TITLE: DIAMOND
ODEM-3

Size: A3



Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet
EE2	Thursday, July 31, 2003	10:54:24 am		A02	PC8803	24 of 67

Engineer: David Du	INVENTEC	
Drawn by: David Du	TITLE: DIAMOND ODEM-4	Size: A3
R&D CHK		
DOC CTRL CHK		
MFG ENGR CHK		



SO DIMM 0

SCK/SCK#(0)=SCK/SCK#(1)=SCK/SCK#(2)

SCK/SCK#(3)=SCK/SCK#(4)=SCK/SCK#(5)

DQ=CB=DQS

SCK(2:0) be longer than DQS 1"~2"

SCK(5:3) be longer than DQS 1"~2"

SDQ(63:0) from MCH to DIMM0 2"~3.5"

SCS#,SCKE from MCH to DIMM0 2"~4.5"

SMA(12:0),SBS(1:0),SRAS#,SCAS#,SWE# 2"~3.5"

SCK 3"~6.5"

LAYOUT NOTES : LOCATION MUST BE CENTER OF BOTTON SODIMM 0

Changed by: EE2	Date Changed: Thursday, July 31, 2003	Time Changed: 10:54:29 am	QA CHK	VER: A02	Model Number: PC8803

Engineer: David Du
Drawn by: David Du
R&D CHK
DOC CTRL CHK
MFG ENGR CHK

INVENTEC

TITLE: DIAMOND
DDR-SDRAM-1

Size: A3

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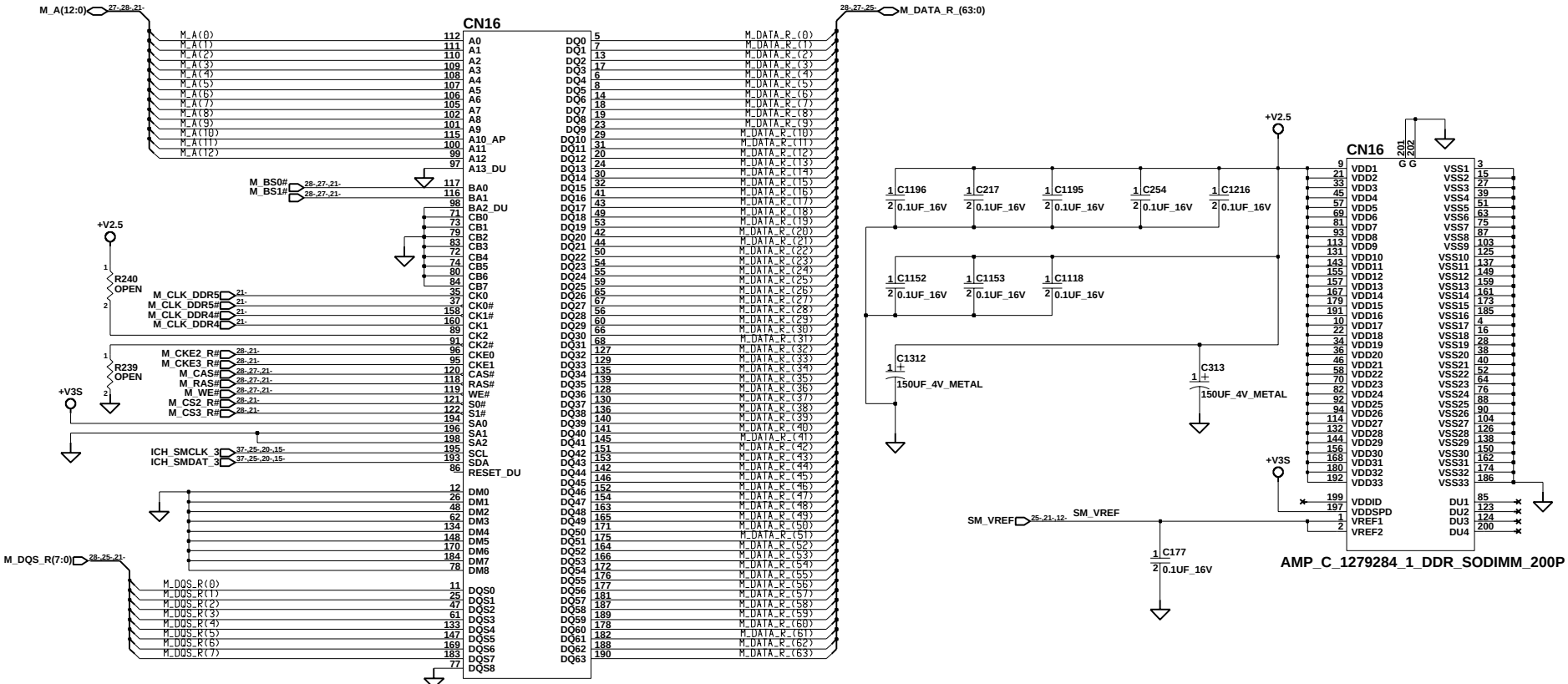
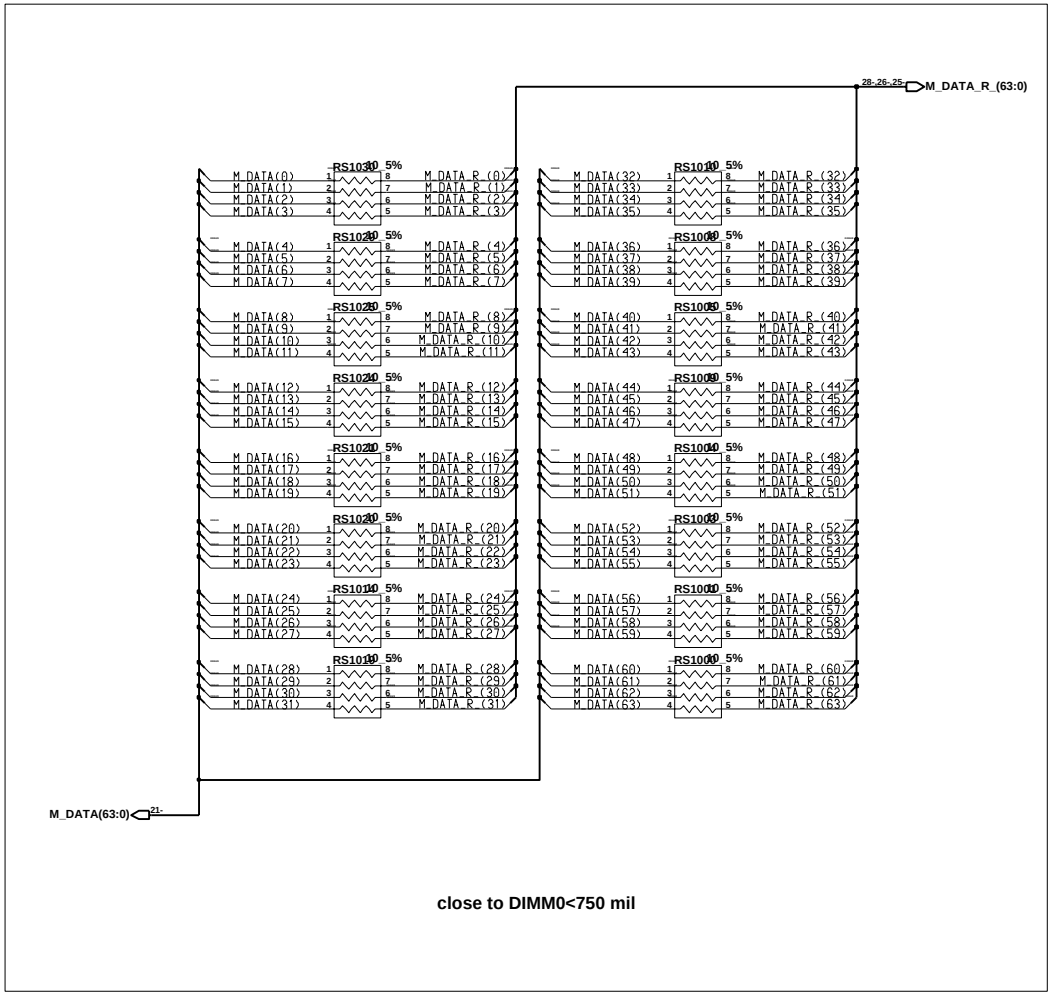


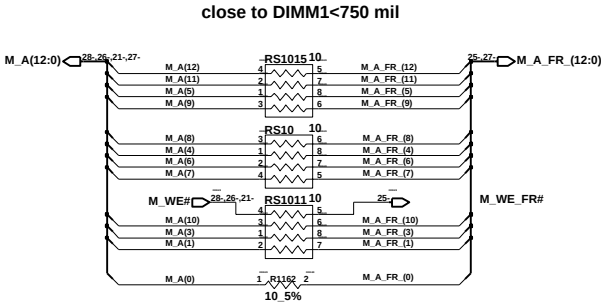
SO DIMM 1

SCK(2:0) be longer than	SCS(1:0),CKE(1:0) SMA(12:0),SBS(1:0) RAS#,CAS#,WE#	1"~3"
SCK(5:3) be longer than	SCS(3:2),CKE(3:2) SMA(12:0),SBS(1:0) RAS#,CAS#,WE#	1"~3"

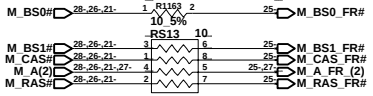
Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet 26 of 67
EE2	Thursday, July 31, 2003	10:54:35 am		A02	PC8803	

Engineer: David Du	INVENTEC
Drawn by: David Du	
R&D CHK	TITLE: DIAMOND
DOC CTRL CHK	DDR-SDRAM-2
MFG ENGR CHK	Size: A3





ALL RESISTORS ARE 10 ohm



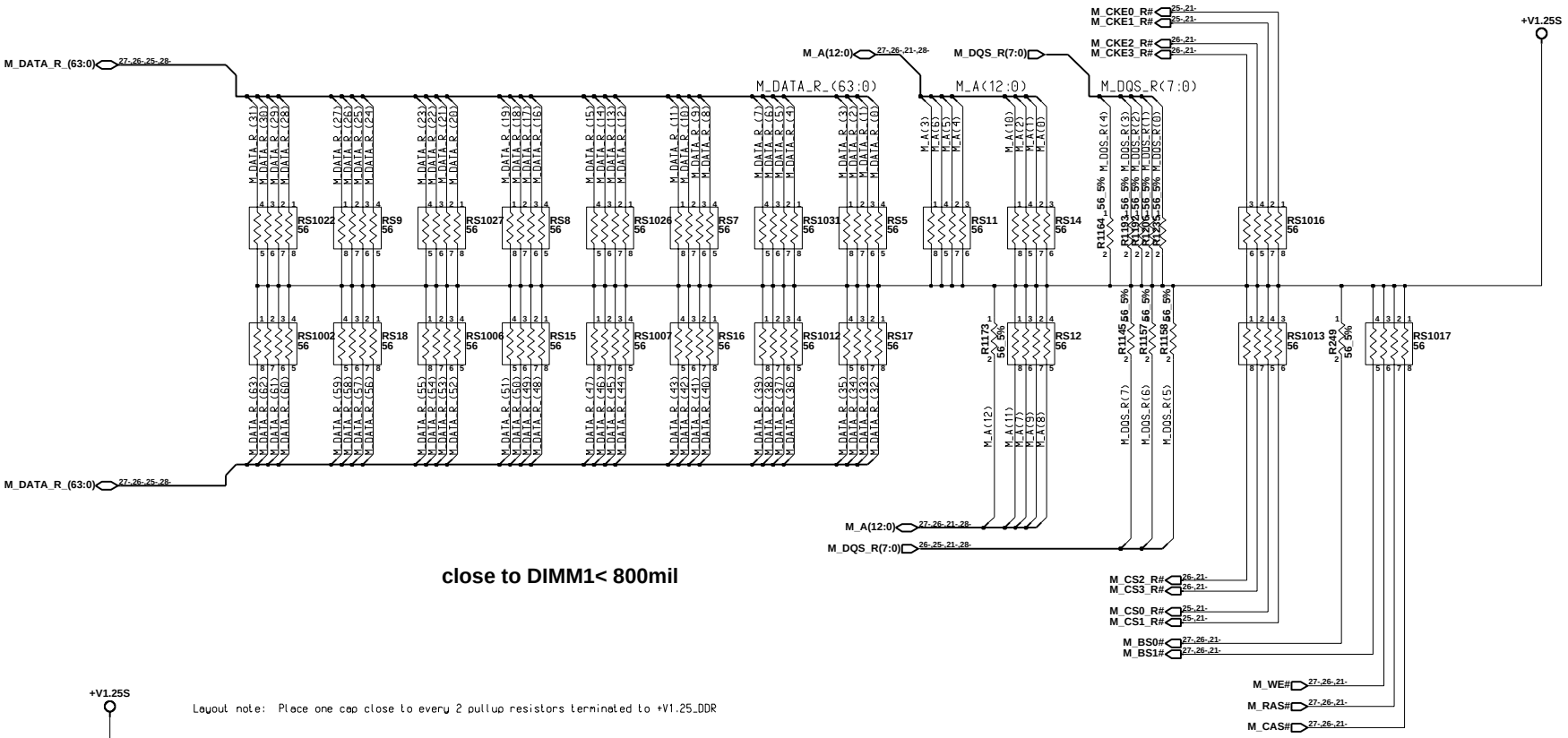
Changed by: EE2	Date Changed: Thursday, July 31, 2003	Time Changed: 10:54:40 am	QA CHK	VER: A02	Model Number: PC8803	Sheet 27 of 67

Engineer: David Du
Drawn by: David Du
R&D CHK
DOC CTRL CHK
MFG ENGR CHK

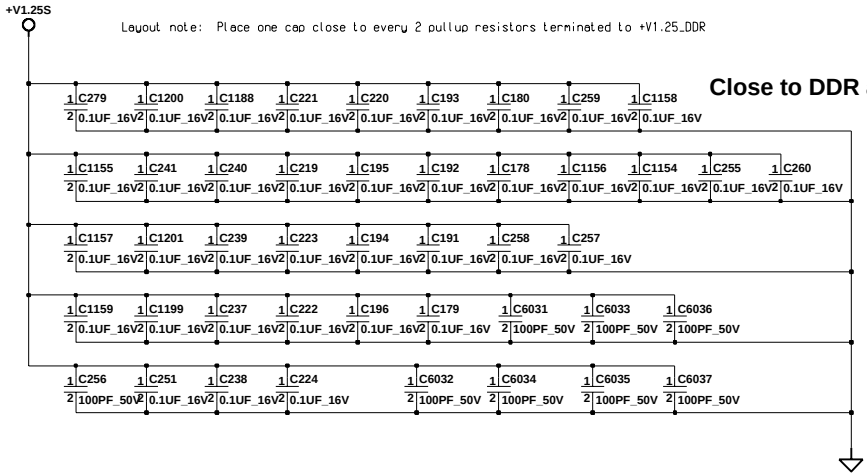
INVENTEC

TITLE: **DIAMOND**
DDR-SDRAM-3

Size: A3

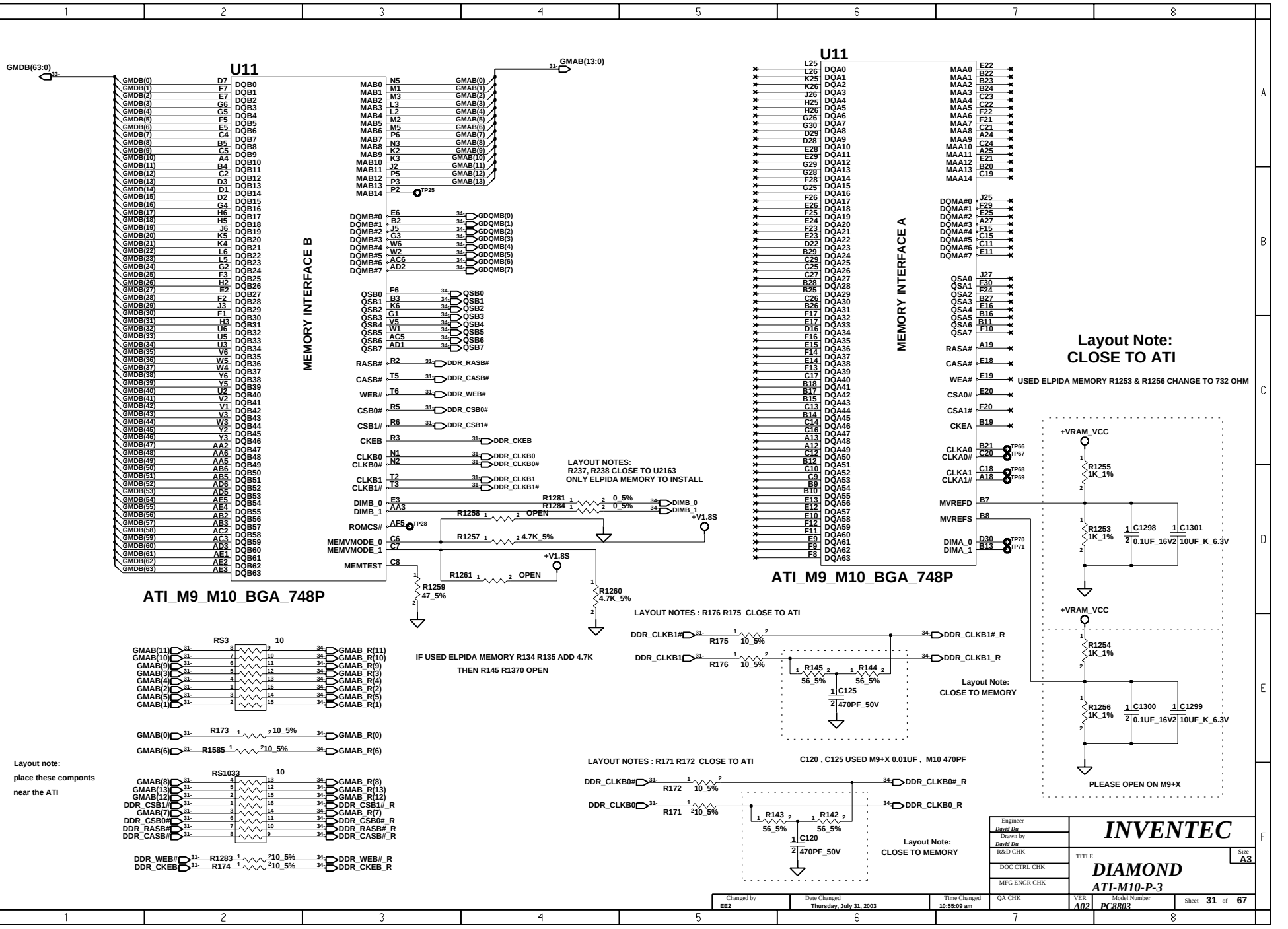


close to DIMM1< 800mil



Close to DDR as passible

Engineer: *David Du*	INVENTEC				
Drawn by: *David Du*					
R&D CHK	TITLE: *DIAMOND* *DDR-SDRAM-4*				Size: A3
DOC CTRL CHK					
MFG ENGR CHK					
Changed by: EE2	Date Changed: Thursday, July 31, 2003	Time Changed: 10:54:46 am	QA CHK	VER: *A02* Model Number: *PC8803*	Sheet 28 of 67


U11
MEMORY INTERFACE B
ATI_M9_M10_BGA_748P
GMDB(63:0)
GMAB(13:0)
LAYOUT NOTES:
R237, R238 CLOSE TO U2163
ONLY ELPIDA MEMORY TO INSTALL
+V1.8S
R1258 OPEN
R1257 4.7K_5%
R1261 OPEN
R1259 47_5%
R1260 4.7K_5%
MEMVMODE_0
MEMVMODE_1
MEMTEST
ROMCS#
DIMB_0
DIMB_1
CKEB
CLKB0
CLKB0#
CLKB1
CLKB1#
RASB#
CASB#
WEB#
CSB0#
CSB1#
QSB0
QSB1
QSB2
QSB3
QSB4
QSB5
QSB6
QSB7
DQMB#0
DQMB#1
DQMB#2
DQMB#3
DQMB#4
DQMB#5
DQMB#6
DQMB#7
MAB0
MAB1
MAB2
MAB3
MAB4
MAB5
MAB6
MAB7
MAB8
MAB9
MAB10
MAB11
MAB12
MAB13
MAB14
DQB0
DQB1
DQB2
DQB3
DQB4
DQB5
DQB6
DQB7
DQB8
DQB9
DQB10
DQB11
DQB12
DQB13
DQB14
DQB15
DQB16
DQB17
DQB18
DQB19
DQB20
DQB21
DQB22
DQB23
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DQB46
DQB47
DQB48
DQB49
DQB50
DQB51
DQB52
DQB53
DQB54
DQB55
DQB56
DQB57
DQB58
DQB59
DQB60
DQB61
DQB62
DQB63
U11
MEMORY INTERFACE A
ATI_M9_M10_BGA_748P
RASA#
CASA#
WEA#
CSA0#
CSA1#
CKEA
CLKA0
CLKA0#
CLKA1
CLKA1#
MVREFD
MVREFS
DIMA_0
DIMA_1
QSA0
QSA1
QSA2
QSA3
QSA4
QSA5
QSA6
QSA7
DQMA#0
DQMA#1
DQMA#2
DQMA#3
DQMA#4
DQMA#5
DQMA#6
DQMA#7
MAA0
MAA1
MAA2
MAA3
MAA4
MAA5
MAA6
MAA7
MAA8
MAA9
MAA10
MAA11
MAA12
MAA13
MAA14
DQA0
DQA1
DQA2
DQA3
DQA4
DQA5
DQA6
DQA7
DQA8
DQA9
DQA10
DQA11
DQA12
DQA13
DQA14
DQA15
DQA16
DQA17
DQA18
DQA19
DQA20
DQA21
DQA22
DQA23
DQA24
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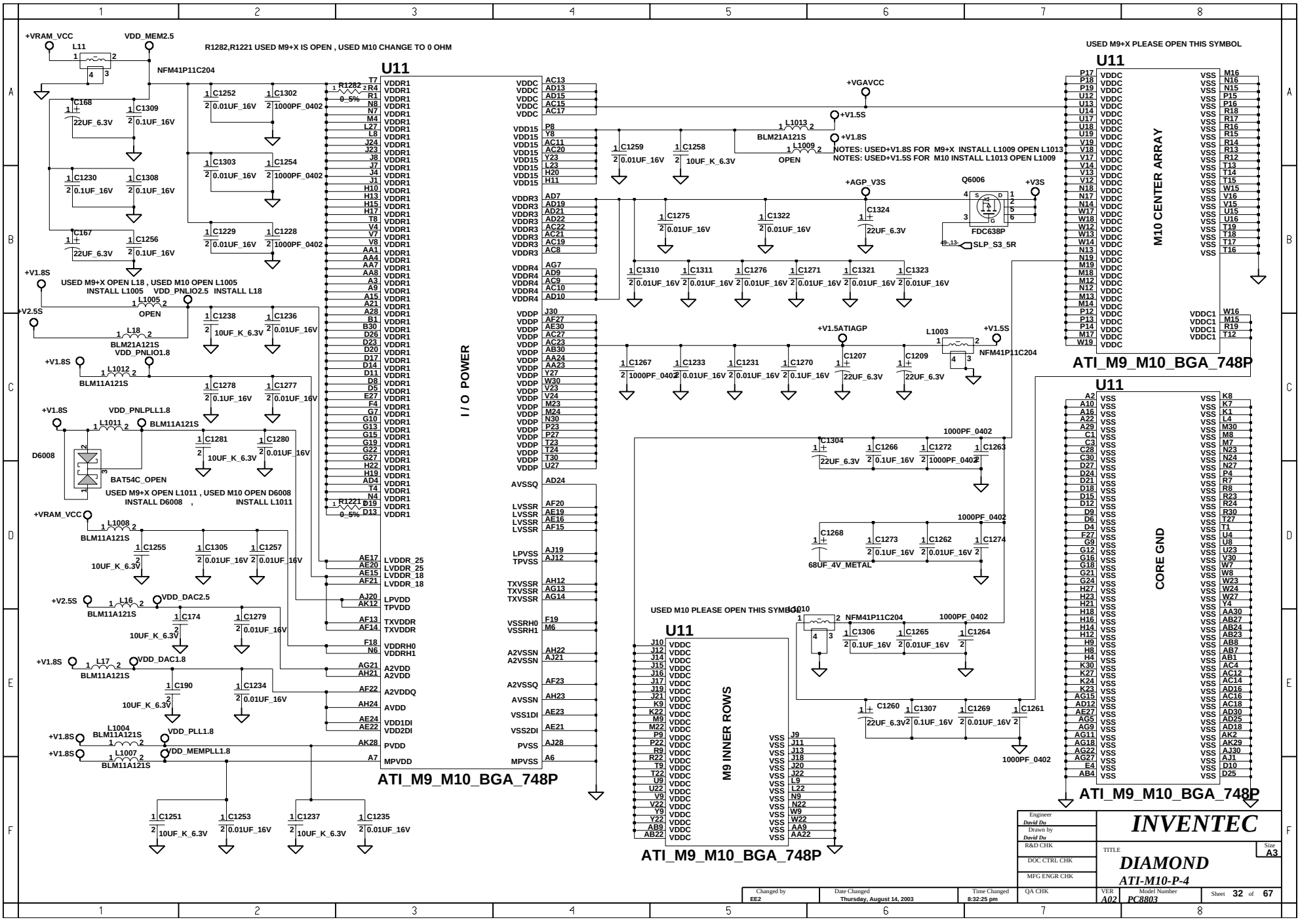
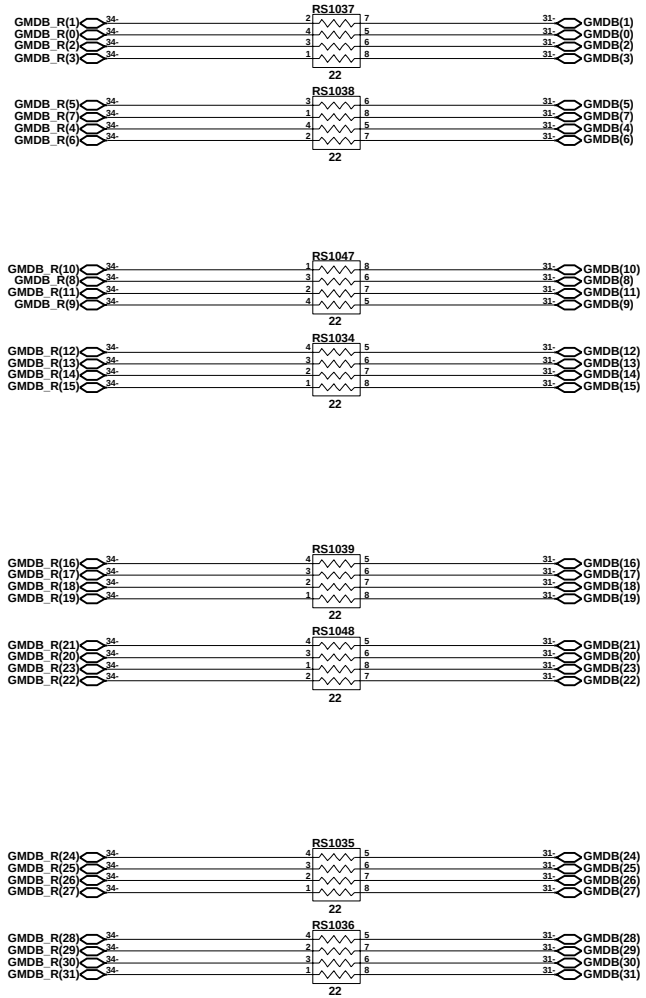
DQA34
DQA35
DQA36
DQA37
DQA38
DQA39
DQA40
DQA41
DQA42
DQA43
DQA44
DQA45
DQA46
DQA47
DQA48
DQA49
DQA50
DQA51
DQA52
DQA53
DQA54
DQA55
DQA56
DQA57
DQA58
DQA59
DQA60
DQA61
DQA62
DQA63
Layout Note:
CLOSE TO ATI
USED ELPIDA MEMORY R1253 & R1256 CHANGE TO 732 OHM
+VRAM_VCC
R1255 1K_1%
R1253 1K_1%
C1298 0.1UF_16V2
C1301 10UF_K_6.3V
R1254 1K_1%
R1256 1K_1%
C1300 0.1UF_16V2
C1299 10UF_K_6.3V
PLEASE OPEN ON M9+X
LAYOUT NOTES : R176 R175 CLOSE TO ATI
R175 10_5%
R176 10_5%
R145 56_5%
R144 56_5%
C125 470PF_50V
Layout Note:
CLOSE TO MEMORY
IF USED ELPIDA MEMORY R134 R135 ADD 4.7K
THEN R145 R1370 OPEN
RS3 10
RS1033 10
R173 10_5%
R1585 10_5%
R1283 10_5%
R174 10_5%
Layout note:
place these componts
near the ATI
LAYOUT NOTES : R171 R172 CLOSE TO ATI
C120 , C125 USED M9+X 0.01UF , M10 470PF
R172 10_5%
R171 10_5%
R143 56_5%
R142 56_5%
C120 470PF_50V
Layout Note:
CLOSE TO MEMORY
Engineer
David Du
Drawn by
David Du
R&D CHK
DOC CTRL CHK
MFG ENGR CHK
INVENTEC
TITLE
DIAMOND
ATI-M10-P-3
Size
A3
Changed by
EE2
Date Changed
Thursday, July 31, 2003
Time Changed
10:55:09 am
QA CHK
VER
A02
Model Number
PC8803
Sheet 31 of 67

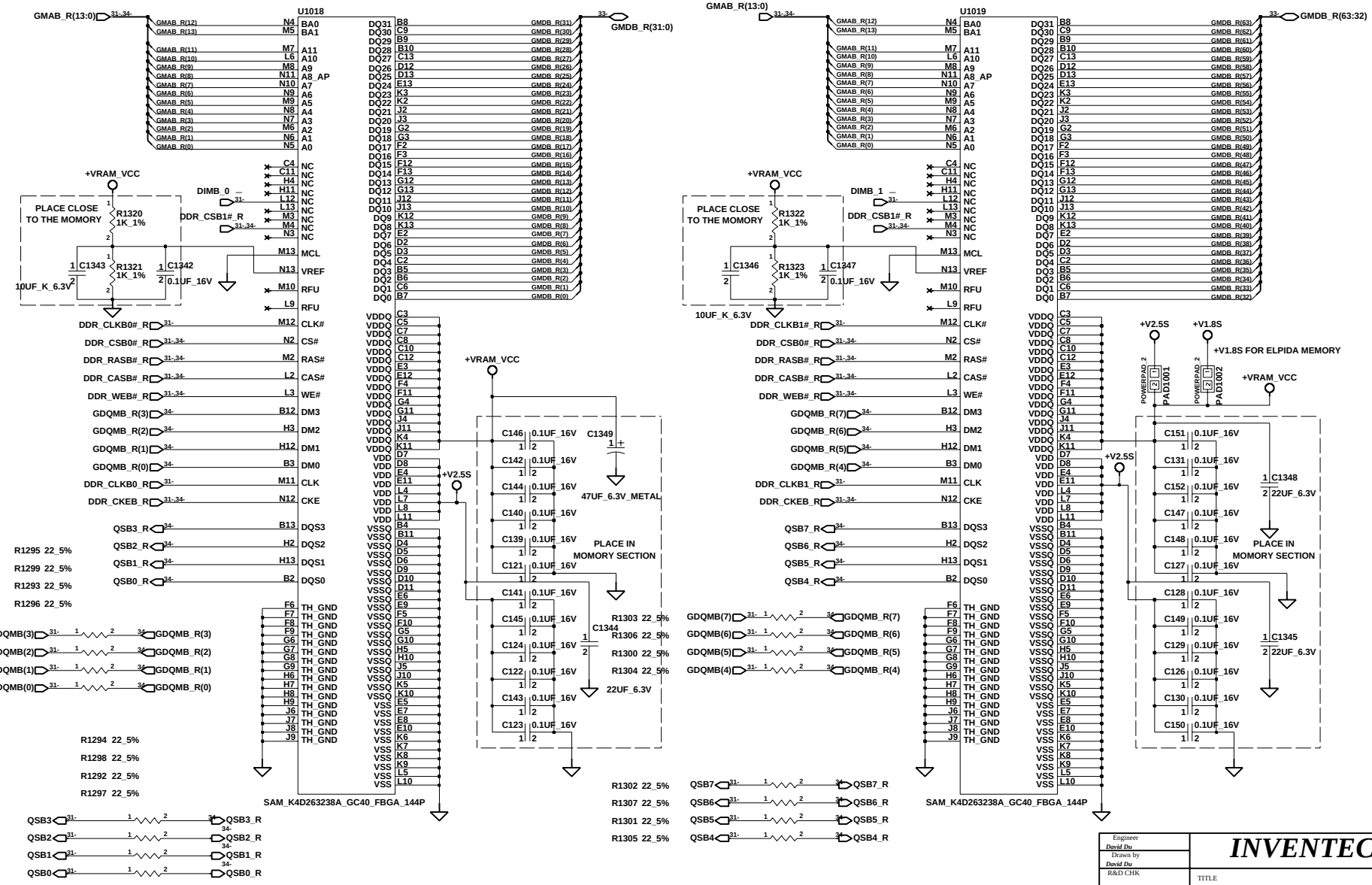

R1282,R1221 USED M9+X IS OPEN , USED M10 CHANGE TO 0 OHM
USED M9+X PLEASE OPEN THIS SYMBOL
U11
I / O POWER
M10 CENTER ARRAY
CORE GND
M9 INNER ROWS
ATI_M9_M10_BGA_748P
NFM41P11C204
BLM11A121S
BLM21A121S
BAT54C_OPEN
FDC638P
SLP_S3_5R
USED M9+X OPEN L18 , USED M10 OPEN L1005
INSTALL L1005 VDD_PNLIO2.5 INSTALL L18
USED M9+X OPEN L1011 , USED M10 OPEN D6008
INSTALL D6008 , INSTALL L1011
USED M10 PLEASE OPEN THIS SYMBOL
NOTES: USED+V1.8S FOR M9+X INSTALL L1009 OPEN L1013
NOTES: USED+V1.5S FOR M10 INSTALL L1013 OPEN L1009
+VRAM_VCC
VDD_MEM2.5
+V1.8S
+V2.5S
+V1.5S
+VGAVCC
+AGP_V3S
+V3S
+V1.5ATIAGP
VDD_PNLPLL1.8
VDD_PNLIO1.8
VDD_DAC2.5
VDD_DAC1.8
VDD_PLL1.8
VDD_MEMPLL1.8
Engineer David Du
Drawn by David Du
R&D CHK
DOC CTRL CHK
MFG ENGR CHK
INVENTEC
TITLE
DIAMOND
ATI-M10-P-4
Size A3
Changed by EE2
Date Changed Thursday, August 14, 2003
Time Changed 8:32:25 pm
QA CHK
VER A02
Model Number PC8803
Sheet 32 of 67

LAYOUT NOTES : THIS IS DEPEND RESISTOR NEED CLOSE TO VIDEO RAM



Changed by EE2	Date Changed Thursday, July 31, 2003	Time Changed 10:55:23 am	QA CHK	VER A02	Model Number PC8803	Sheet 33 of 67

Engineer David Du	INVENTEC
Drawn by David Du	
R&D CHK	TITLE DIAMOND
DOC CTRL CHK	VGA DEPEND RESISTOR
MFG ENGR CHK	Size A3



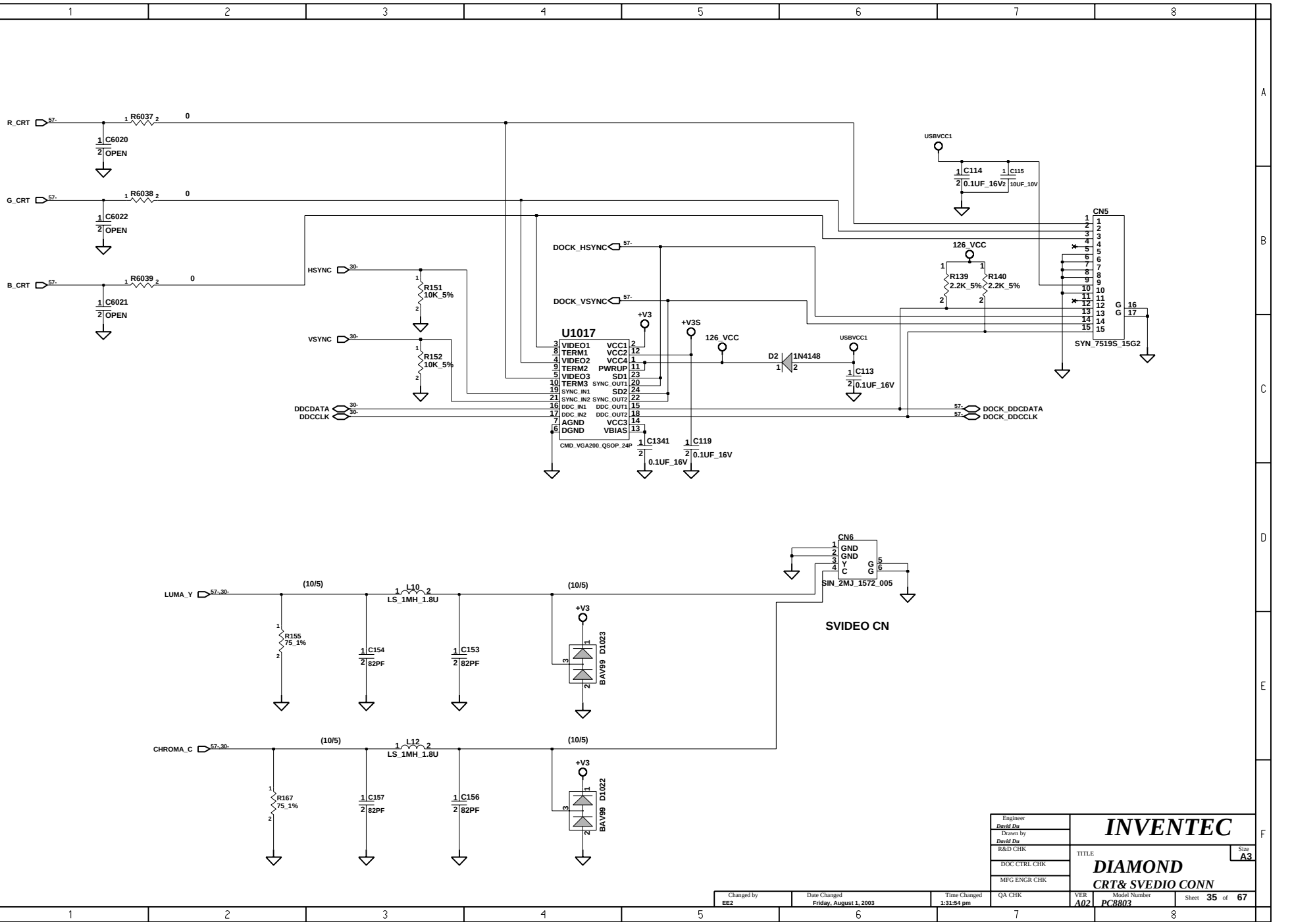
Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet 34 of 67
EE2	Thursday, July 31, 2003	10:55:28 am		A02	PC8803	

Engineer	David Du
Drawn by	David Du
R&D CHK	
DOC CTRL CHK	
MFG ENGR CHK	

INVENTEC

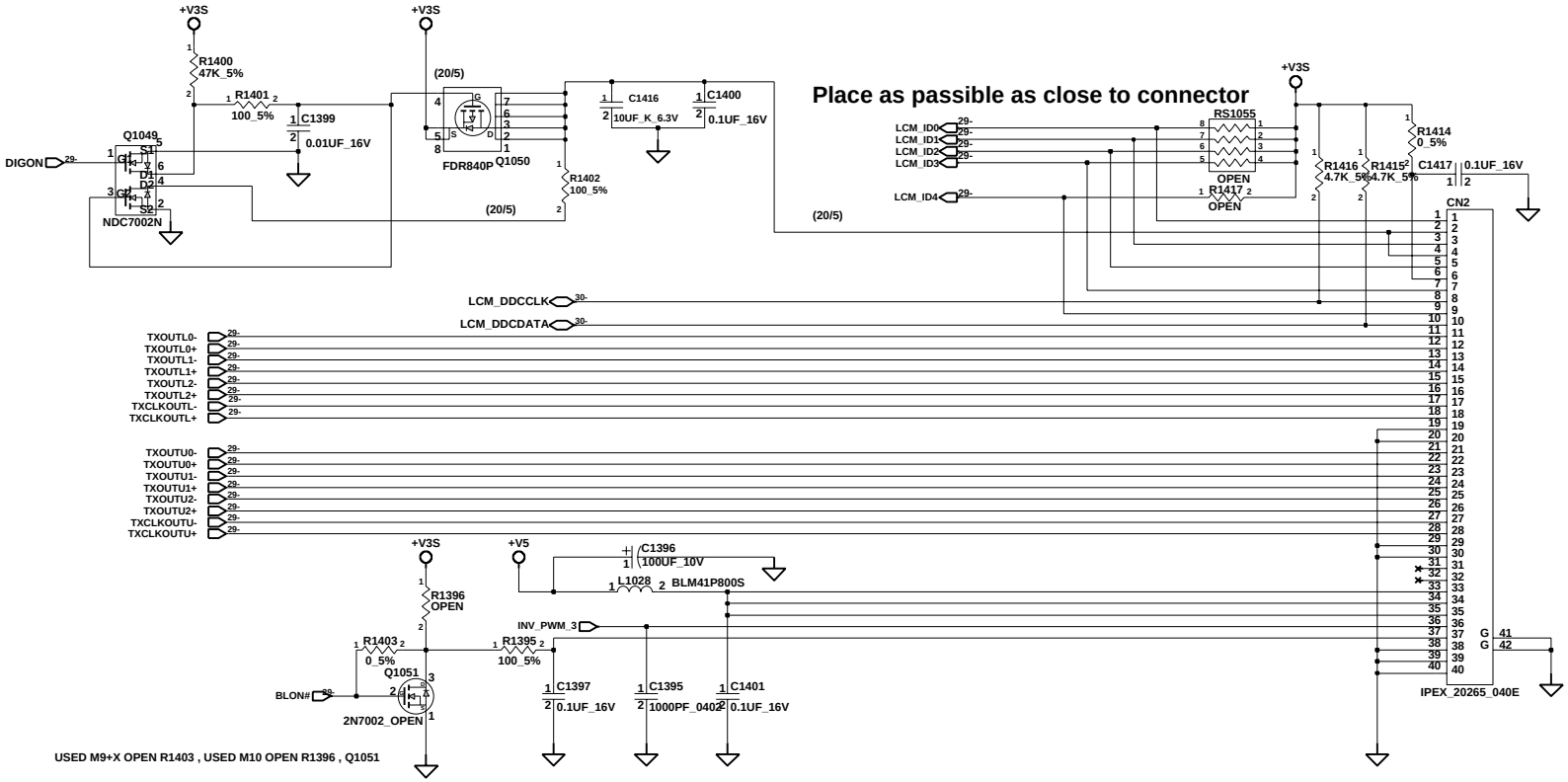
TITLE: **DIAMOND**
VIDEO RAM-2

Size: A3



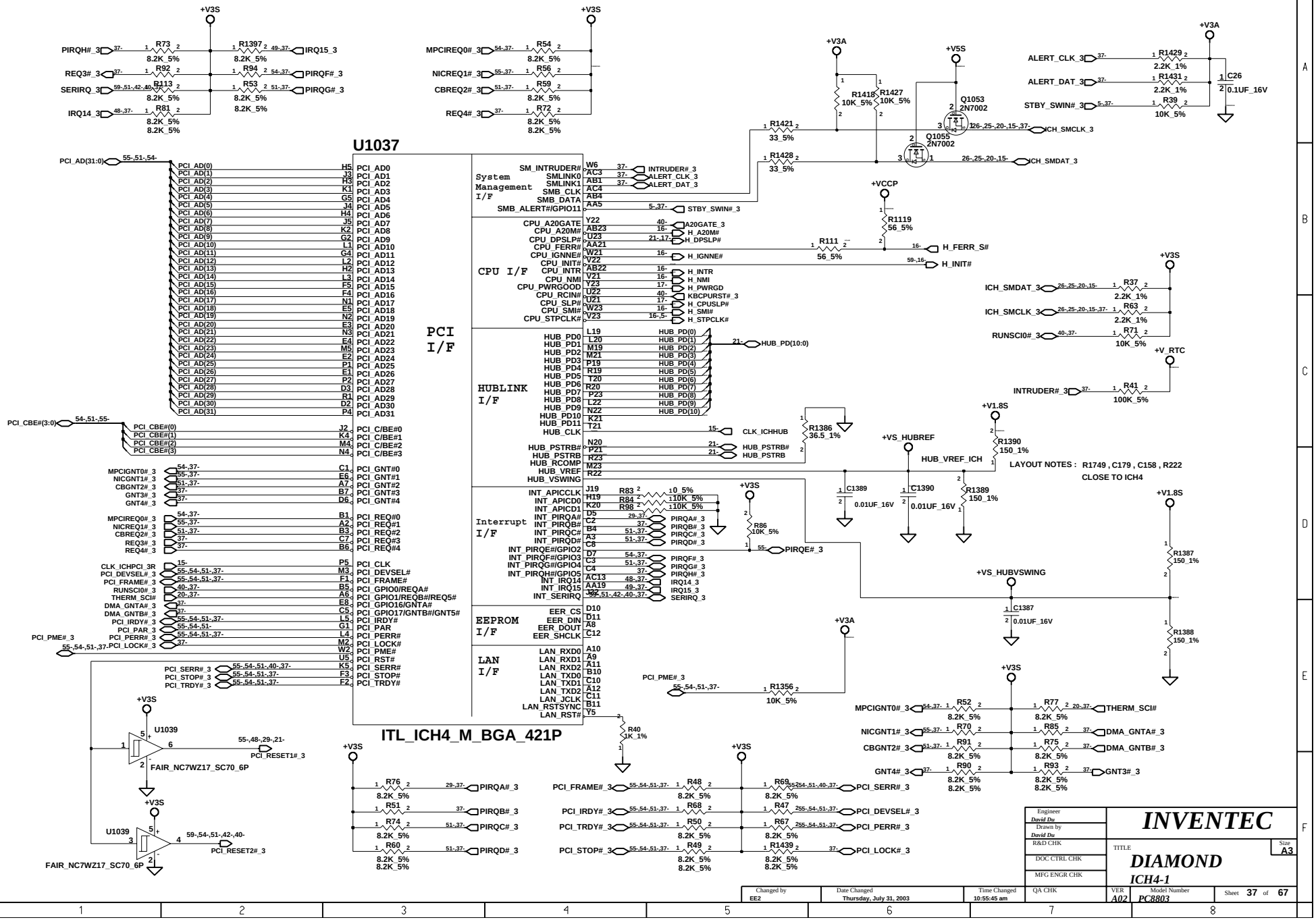
Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet	of
EE2	Friday, August 1, 2003	1:31:54 pm		A02	PC8803	35	67

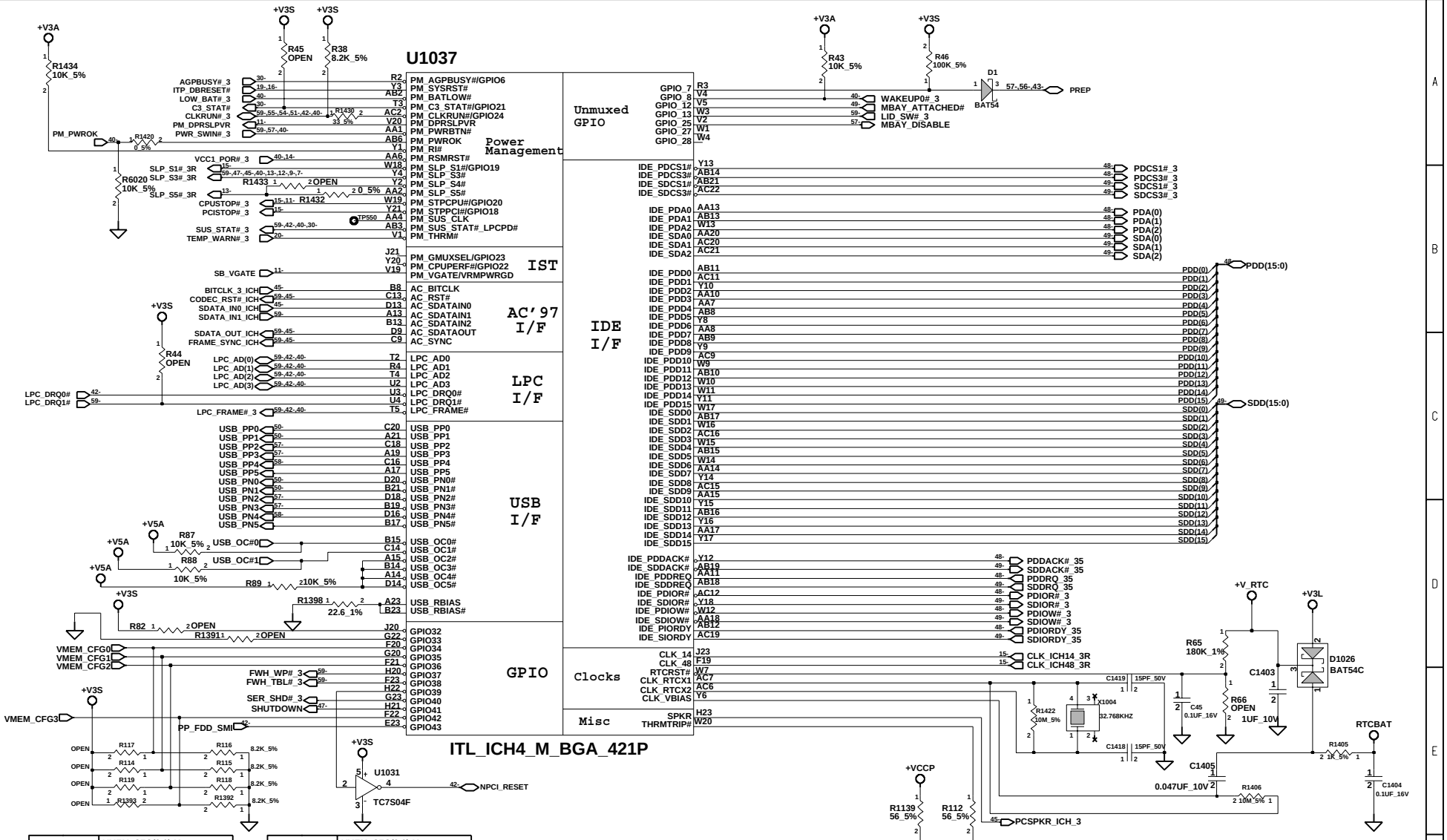
Engineer: David Du	**INVENTEC**	
Drawn by: David Du	TITLE: DIAMOND	Size: A3
R&D CHK	CRT& SVEDIO CONN	
DOC CTRL CHK		
MFG ENGR CHK		



Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet
EE2	Thursday, July 31, 2003	10:55:39 am		A02	PC8803	36 of 67

Engineer: David Du	**INVENTEC**	
Drawn by: David Du	TITLE: **DIAMOND** *LCD CONN*	Size: A3
R&D CHK		
DOC CTRL CHK		
MFG ENGR CHK		


U1037
PCI I/F
System Management I/F
CPU I/F
HUBLINK I/F
Interrupt I/F
EEPROM I/F
LAN I/F
ITL_ICH4_M_BGA_421P
+V3S
+V3A
+V5S
+VCCP
+V1.8S
+V_RTC
+VS_HUBREF
+VS_HUBVSWING
PCI_AD(31:0)
PCI_CBE#(3:0)
PIRQH#_3
REQ3#_3
SERIRQ_3
IRQ14_3
MPCIREQ0#_3
NICREQ1#_3
CBREQ2#_3
REQ4#_3
SM_INTRUDER#
SMLINK0
SMLINK1
SMB_CLK
SMB_DATA
SMB_ALERT#/GPIO11
CPU_A20GATE
CPU_A20M#
CPU_DPSLP#
CPU_FERR#
CPU_IGNNE#
CPU_INIT#
CPU_INTR
CPU_NMI
CPU_PWRGOOD
CPU_RCIN#
CPU_SLP#
CPU_SMI#
CPU_STPCLK#
HUB_PD(10:0)
HUB_CLK
HUB_PSTRB#
HUB_PSTRB
HUB_RCOMP
HUB_VREF
HUB_VSWING
INT_APICCLK
INT_APICD0
INT_APICD1
INT_PIRQA#
INT_PIRQB#
INT_PIRQC#
INT_PIRQD#
INT_PIRQE#/GPIO2
INT_PIRQF#/GPIO3
INT_PIRQG#/GPIO4
INT_PIRQH#/GPIO5
INT_IRQ14
INT_IRQ15
INT_SERIRQ
EER_CS
EER_DIN
EER_DOUT
EER_SHCLK
LAN_RXD0
LAN_RXD1
LAN_RXD2
LAN_TXD0
LAN_TXD1
LAN_TXD2
LAN_JCLK
LAN_RSTSYNC
LAN_RST#
PCI_AD0
PCI_AD1
PCI_AD2
PCI_AD3
PCI_AD4
PCI_AD5
PCI_AD6
PCI_AD7
PCI_AD8
PCI_AD9
PCI_AD10
PCI_AD11
PCI_AD12
PCI_AD13
PCI_AD14
PCI_AD15
PCI_AD16
PCI_AD17
PCI_AD18
PCI_AD19
PCI_AD20
PCI_AD21
PCI_AD22
PCI_AD23
PCI_AD24
PCI_AD25
PCI_AD26
PCI_AD27
PCI_AD28
PCI_AD29
PCI_AD30
PCI_AD31
PCI_C/BE#0
PCI_C/BE#1
PCI_C/BE#2
PCI_C/BE#3
PCI_GNT#0
PCI_GNT#1
PCI_GNT#2
PCI_GNT#3
PCI_GNT#4
PCI_REQ#0
PCI_REQ#1
PCI_REQ#2
PCI_REQ#3
PCI_REQ#4
PCI_CLK
PCI_DEVSEL#
PCI_FRAME#
PCI_GPIO0/REQA#
PCI_GPIO1/REQB#/REQ5#
PCI_GPIO16/GNTA#
PCI_GPIO17/GNTB#/GNT5#
PCI_IRDY#
PCI_PAR
PCI_PERR#
PCI_LOCK#
PCI_PME#
PCI_RST#
PCI_SERR#
PCI_STOP#
PCI_TRDY#
INTRUDER#_3
ALERT_CLK_3
ALERT_DAT_3
STBY_SWIN#_3
A20GATE_3
H_A20M#
H_DPSLP#
H_FERR_S#
H_IGNNE#
H_INIT#
H_INTR
H_NMI
H_PWRGD
KBCPURST#_3
H_CPUSLP#
H_SMI#
H_STPCLK#
HUB_PD(0)
HUB_PD(1)
HUB_PD(2)
HUB_PD(3)
HUB_PD(4)
HUB_PD(5)
HUB_PD(6)
HUB_PD(7)
HUB_PD(8)
HUB_PD(9)
HUB_PD(10)
CLK_ICHHUB
HUB_PSTRB#
HUB_PSTRB
PIRQA#_3
PIRQB#_3
PIRQC#_3
PIRQD#_3
PIRQE#_3
PIRQF#_3
PIRQG#_3
PIRQH#_3
IRQ14_3
IRQ15_3
SERIRQ_3
ICH_SMCLK_3
ICH_SMDAT_3
RUNSCI0#_3
CLK_PCI_ICH
CLK_PCI_TPM
CLK_PCI_FWH
CLK_PCI_LAN
CLK_48_ICH
CLK_14_ICH
PCI_RESET1#_3
PCI_RESET2#_3
PCI_SERR#_3
PCI_DEVSEL#_3
PCI_PERR#_3
PCI_LOCK#_3
THERM_SCI#
DMA_GNTA#_3
DMA_GNTB#_3
GNT3#_3
LAYOUT NOTES : R1749 , C179 , C158 , R222 CLOSE TO ICH4
FAIR_NC7WZ17_SC70_6P
INVENTEC
DIAMOND
ICH4-1
Engineer: David Du
Drawn by: David Du
R&D CHK
DOC CTRL CHK
MFG ENGR CHK
QA CHK
Changed by: EE2
Date Changed: Thursday, July 31, 2003
Time Changed: 10:55:45 am
Size: A3
VER: A02
Model Number: PC8803
Sheet 37 of 67

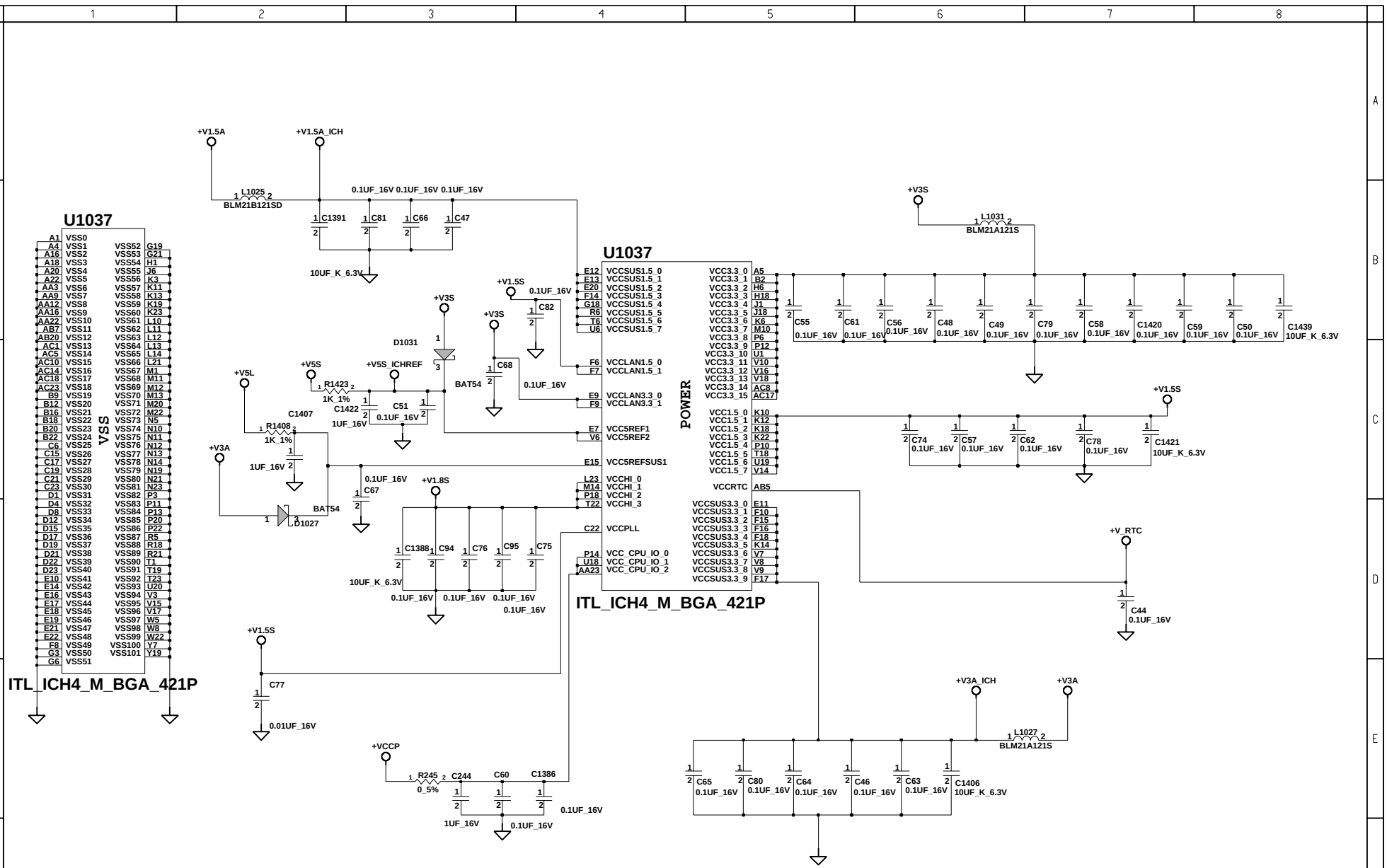


GPIO[34:36,42]		VMEM_CFG(3:0)-Memory type
00	00h	Samsung 64MB
01	01h	Samsung 128MB
02	02h	Hynix 64MB
03	03h	Hynix 128MB
04	04h	Infineon 64MB
05	05h	Infineon 128MB
06	06h	Samsung 32MB
07	07h	Hynix 32MB

GPIO[34:36,42]		VMEM_CFG(3:0)-Memory type
08	08h	Infineon 32MB
09	09h	TBD
10	Ah	TBD
11	Bh	TBD
12	Ch	TBD
13	Dh	TBD
14	Eh	TBD
15	Fh	TBD

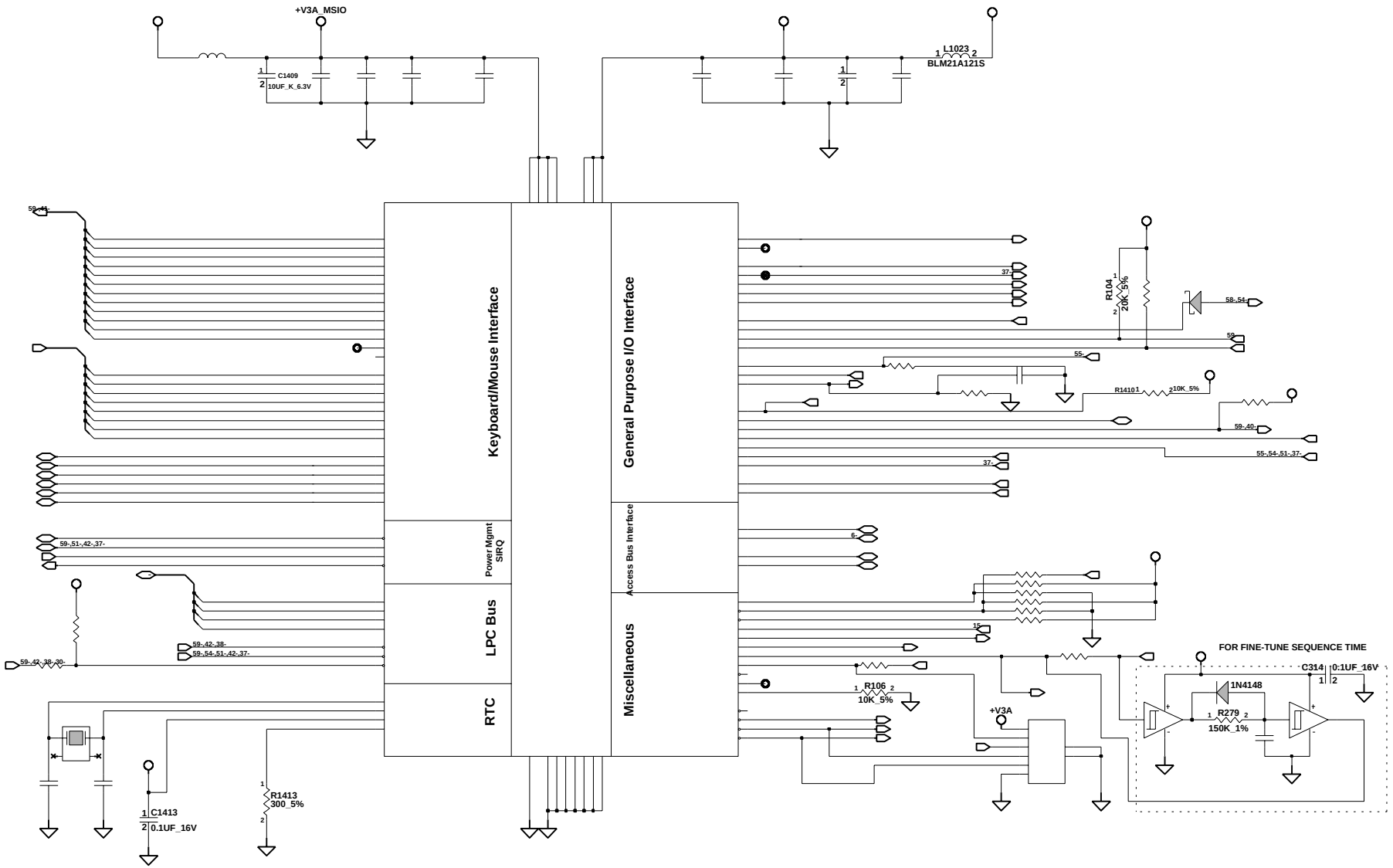
Engineer: David Du	INVENTEC				
Drawn by: David Du					
R&D CHK	TITLE: DIAMOND	Size: A3			
DOC CTRL CHK	ICH4-2				
MFG ENGR CHK					
Changed by: EE2	Date Changed: Thursday, August 21, 2003	Time Changed: 11:35:30 am	QA CHK	VER: A02	Model Number: PC8803

Sheet 38 of 67



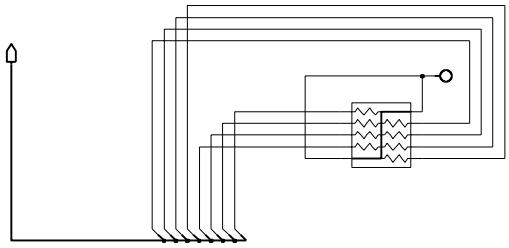
Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet
EE2	Thursday, July 31, 2003	10:55:59 am		A02	PC8803	39 of 67

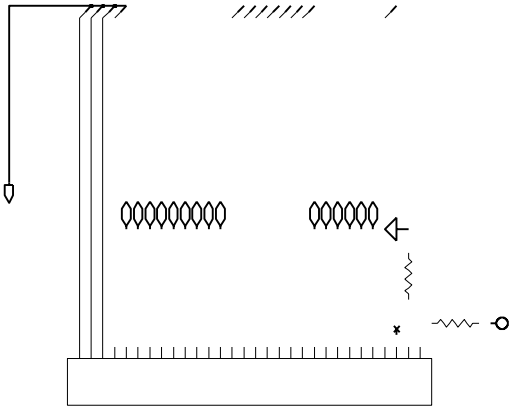
Engineer: David Du	INVENTEC	
Drawn by: David Du	TITLE: DIAMOND	Size: A3
R&D CHK	ICH4-3	
DOC CTRL CHK		
MFG ENGR CHK		

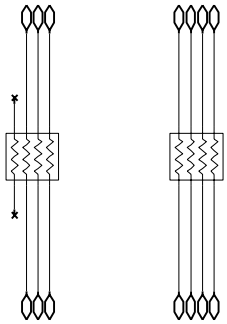


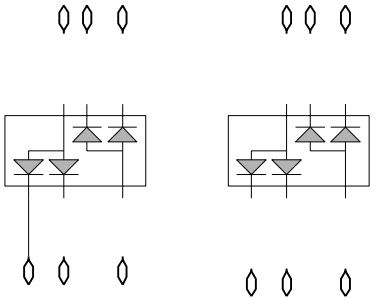
Changed by: EE2	Date Changed: Thursday, July 31, 2003	Time Changed: 10:56:09 am	QA CHK	VER: A02	Model Number: PC8803	Sheet 40 of 67

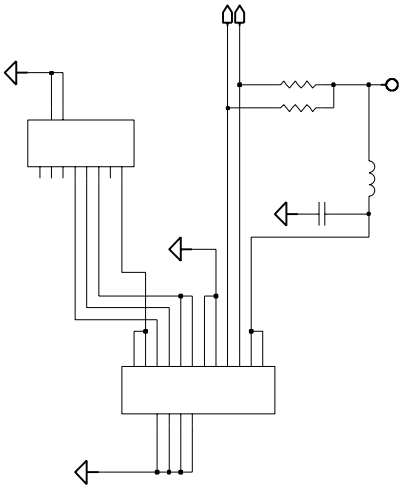
Engineer: David Du	**INVENTEC**	
Drawn by: David Du	TITLE: **DIAMOND**	Size: A3
R&D CHK	**KBC**	
DOC CTRL CHK		
MFG ENGR CHK		

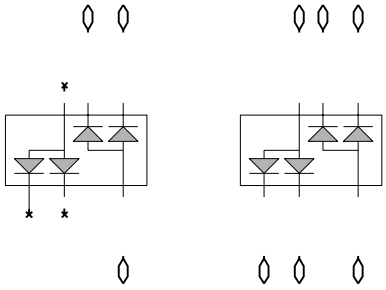


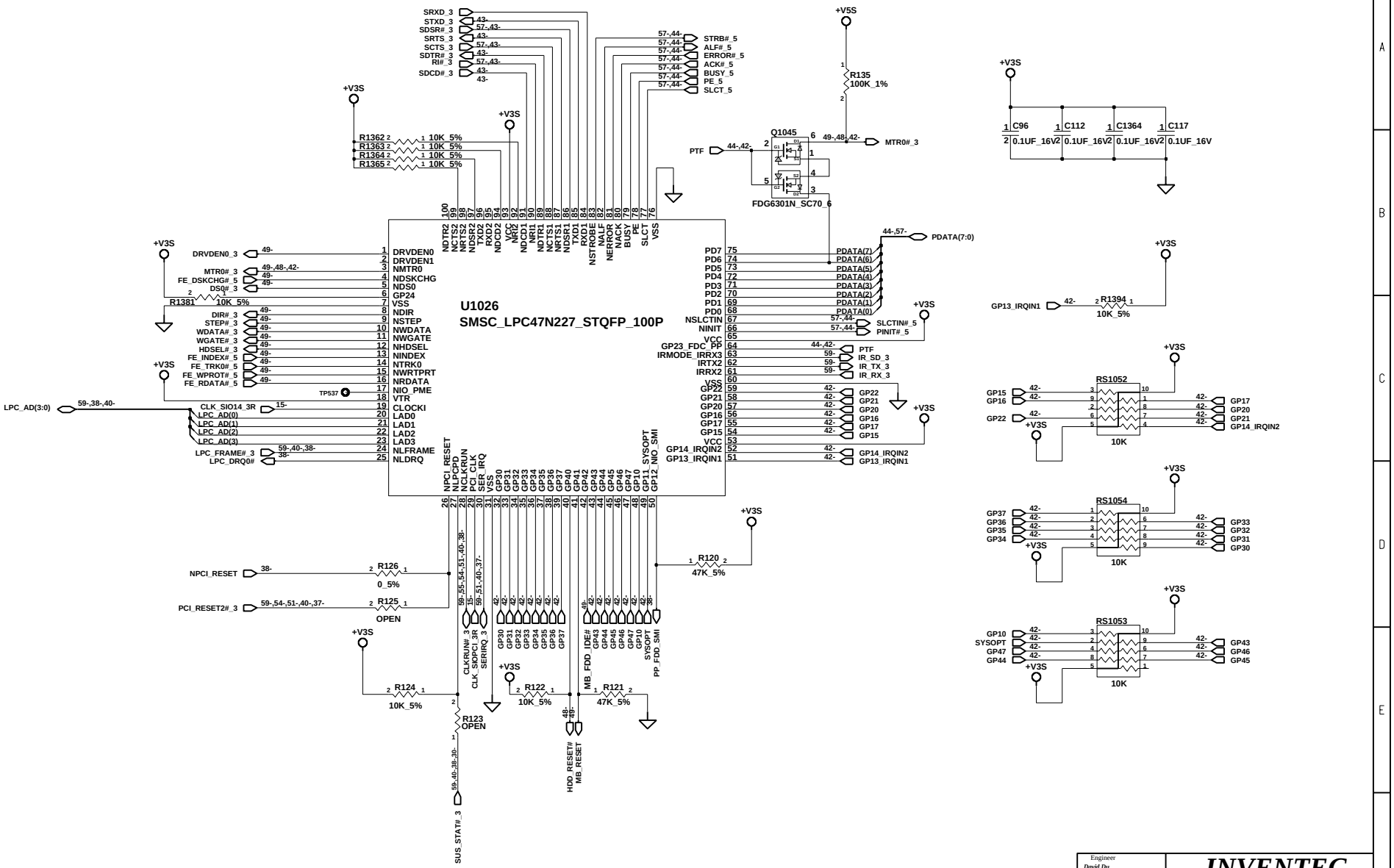


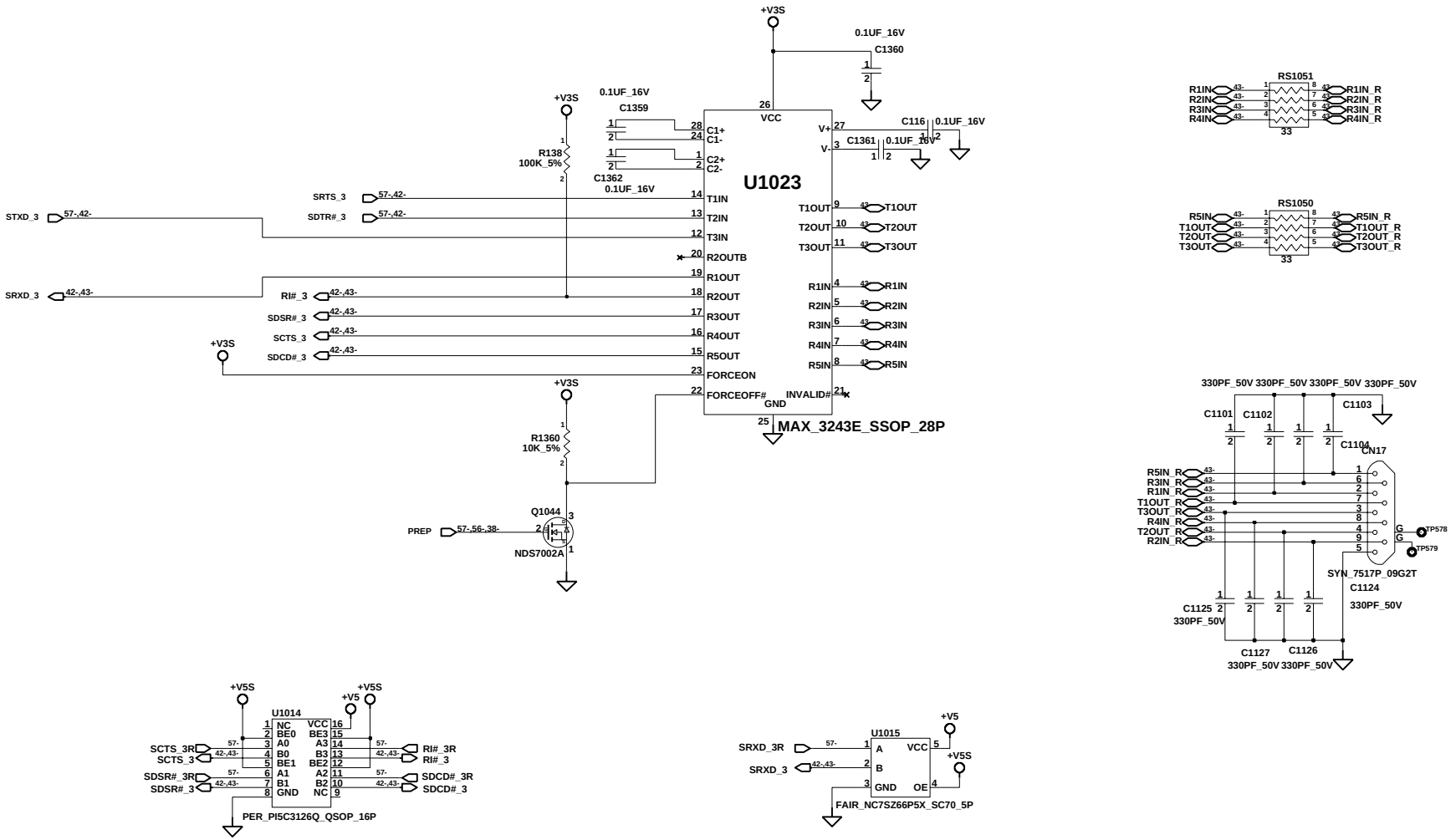






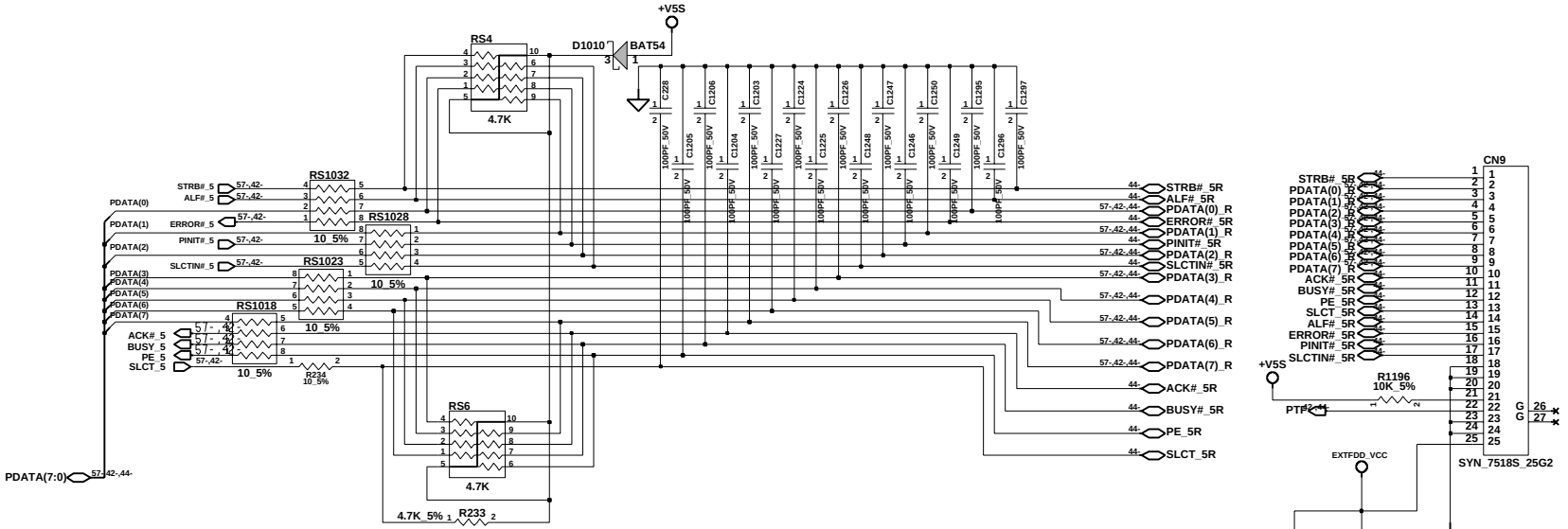


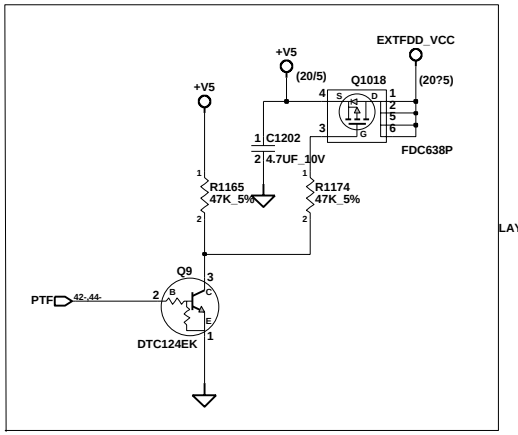

U1026
SMSC_LPC47N227_STQFP_100P
FDG6301N_SC70_6
Q1045
RS1052
RS1054
RS1053
R135 100K_1%
R1362 10K_5%
R1363 10K_5%
R1364 10K_5%
R1365 10K_5%
R1381 10K_5%
R1394 10K_5%
R126 0_5%
R125 OPEN
R124 10K_5%
R122 10K_5%
R121 47K_5%
R120 47K_5%
R123 OPEN
C96 0.1UF_16V
C112 0.1UF_16V
C1364 0.1UF_16V
C117 0.1UF_16V
+V3S
+V5S
PDATA(7:0)
LPC_AD(3:0)
SUS_STAT#_3
HDD_RESET#
MB_RESET
PP_FDD_SMI
INVENTEC
DIAMOND
SUPER I/O
Engineer David Du
Drawn by David Du
R&D CHK
DOC CTRL CHK
MFG ENGR CHK
Changed by EE2
Date Changed Thursday, July 31, 2003
Time Changed 10:56:25 am
QA CHK
Size A3
VER A02
Model Number PC8803
Sheet 42 of 67



Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet 43 of 67
EE2	Thursday, July 31, 2003	10:56:30 am		A02	PC8803	

Engineer: David Du	INVENTEC
Drawn by: David Du	
R&D CHK	TITLE: DIAMOND
DOC CTRL CHK	SERIAL PORT&IR
MFG ENGR CHK	Size: A3

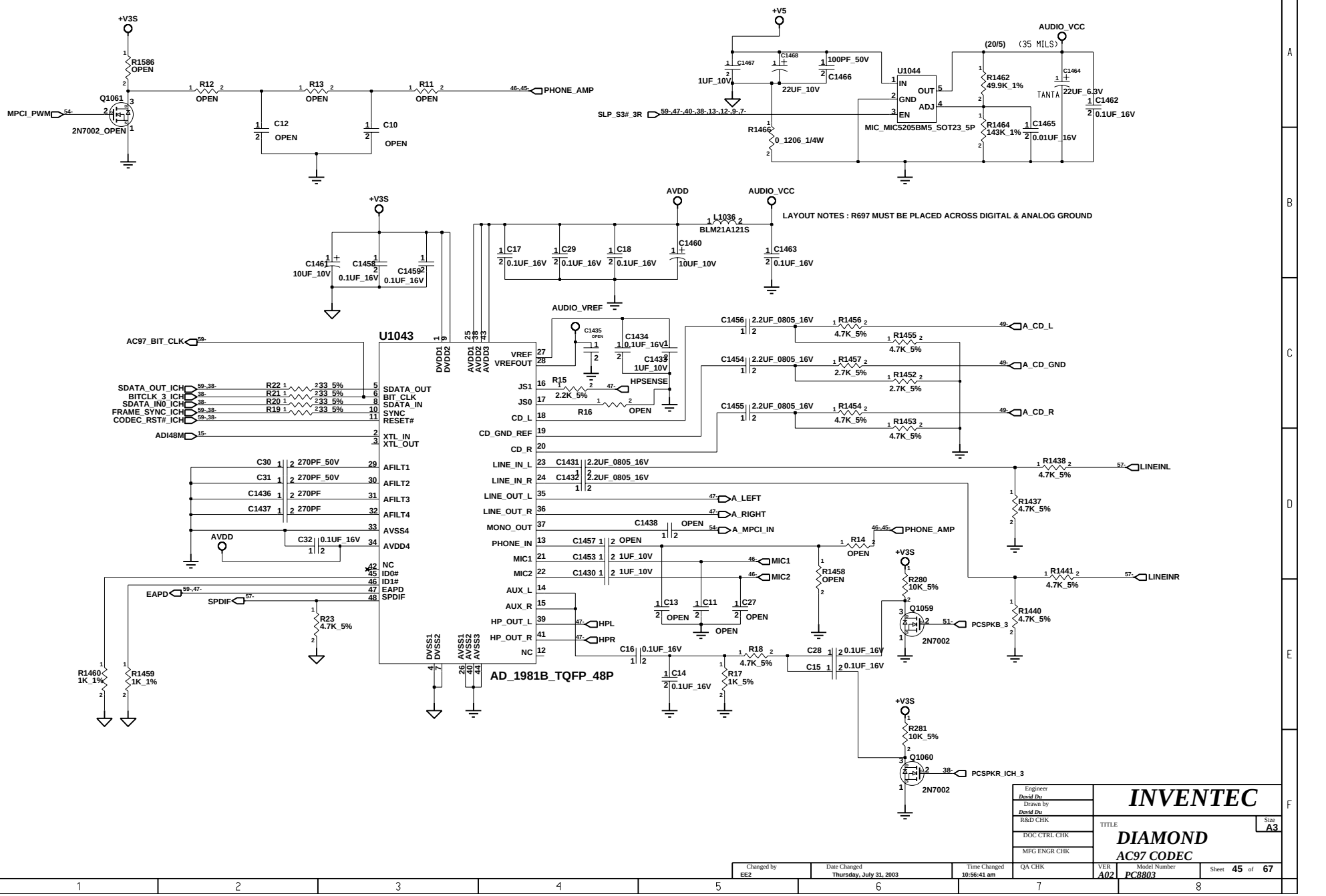




LAYOUT NOTES : PUT THESE CIRCUIT CLOSE TO PARALLER PORT

Changed by EE2	Date Changed Thursday, July 31, 2003	Time Changed 10:56:35 am	QA CHK	VER A02	Model Number PC8803	Sheet 44 of 67

Engineer David Du	INVENTEC
Drawn by David Du	
R&D CHK	TITLE: DIAMOND
DOC CTRL CHK	PARALLER PORT
MFG ENGR CHK	Size A3



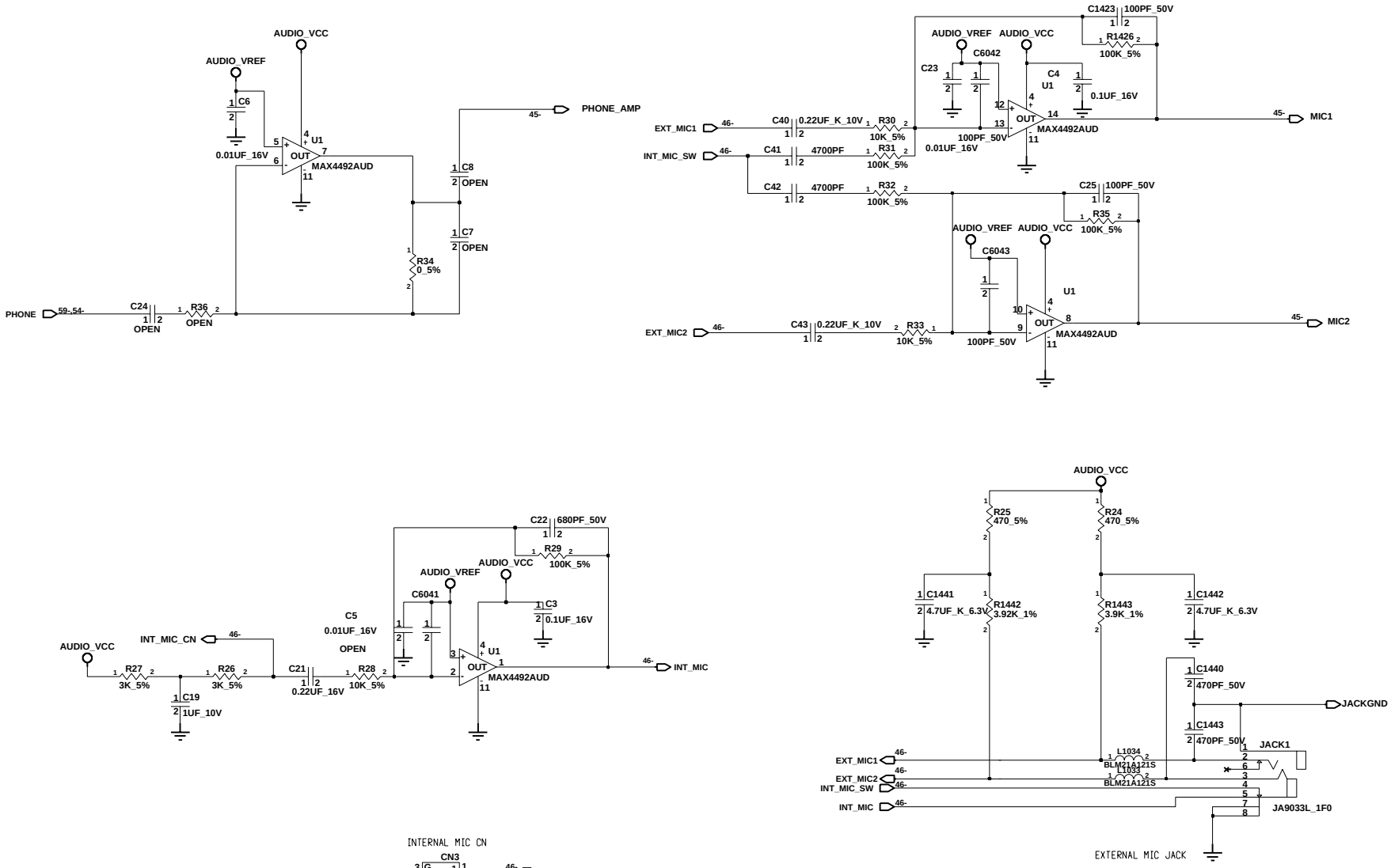
Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet 45 of 67
EE2	Thursday, July 31, 2003	10:56:41 am		A02	PC8803	

Engineer	David Du
Drawn by	David Du
R&D CHK	
DOC CTRL CHK	
MFG ENGR CHK	

INVENTEC

TITLE: **DIAMOND**
AC97 CODEC

Size: A3



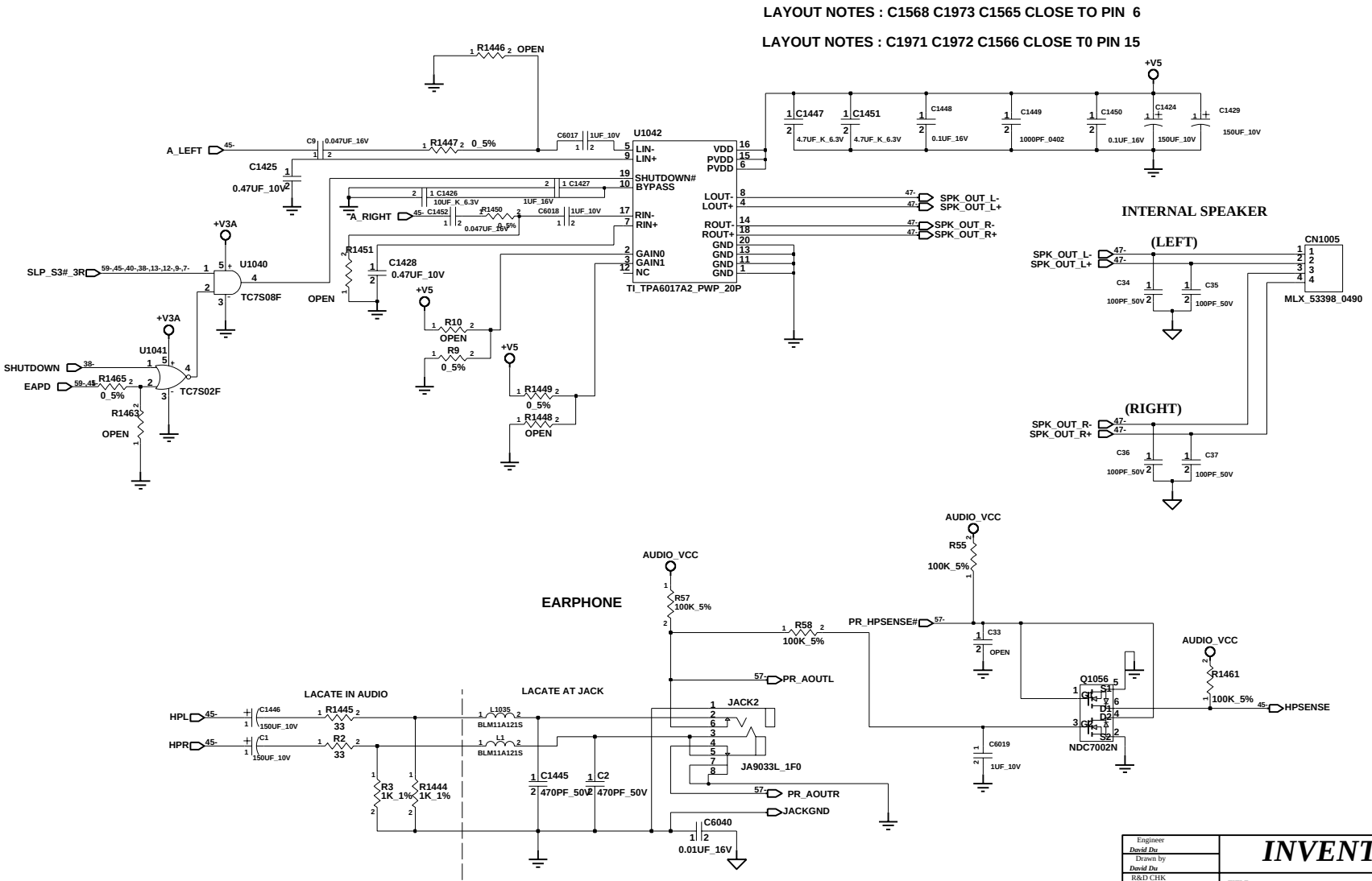
Engineer	David Du
Drawn by	David Du
R&D CHK	
DOC CTRL CHK	
MFG ENGR CHK	

INVENTEC

TITLE: **DIAMOND**
EQ&MIC JACK

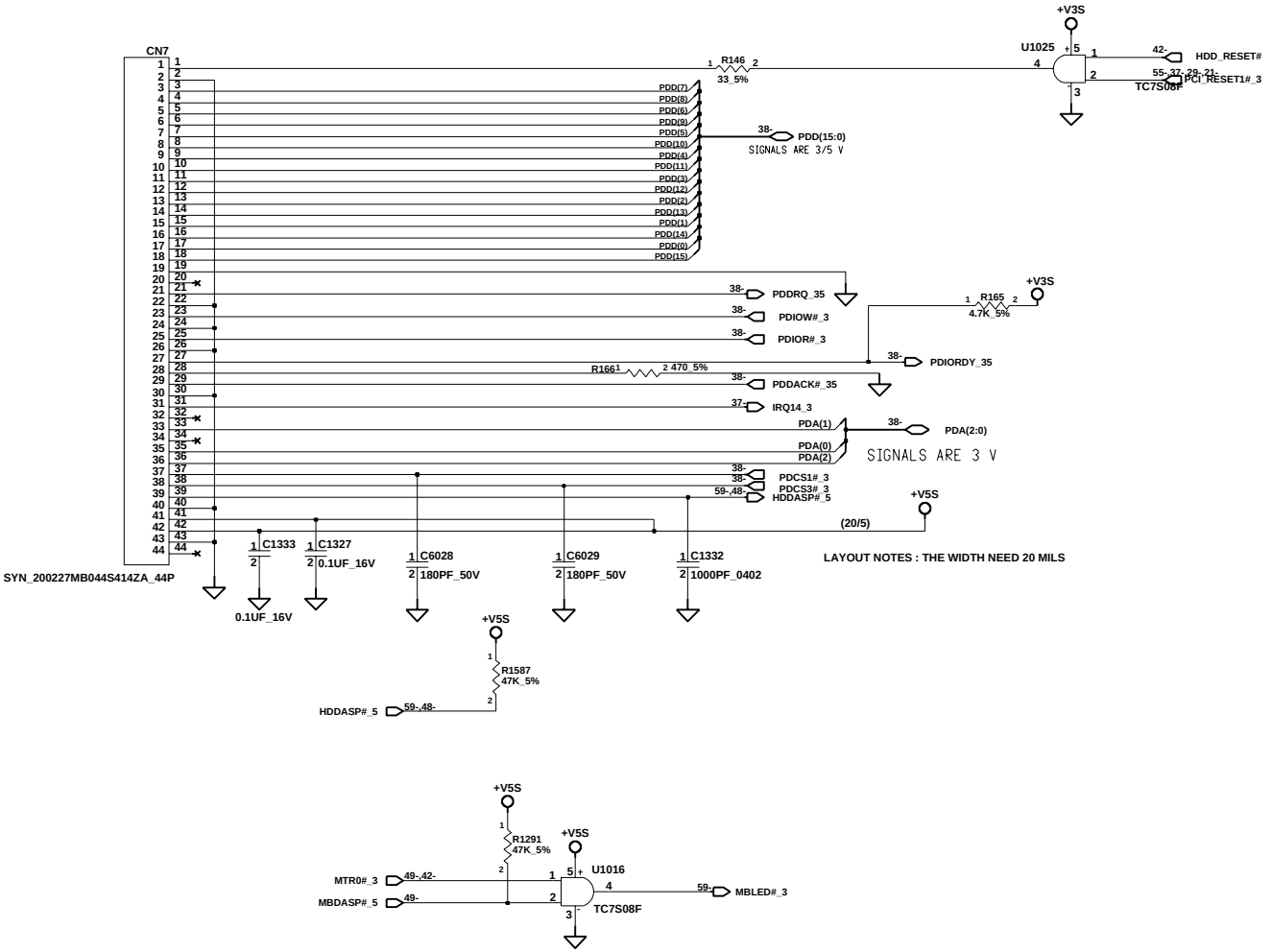
Size: A3

Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet
EE2	Friday, August 15, 2003	1:50:14 pm		A02	PC8803	46 of 67



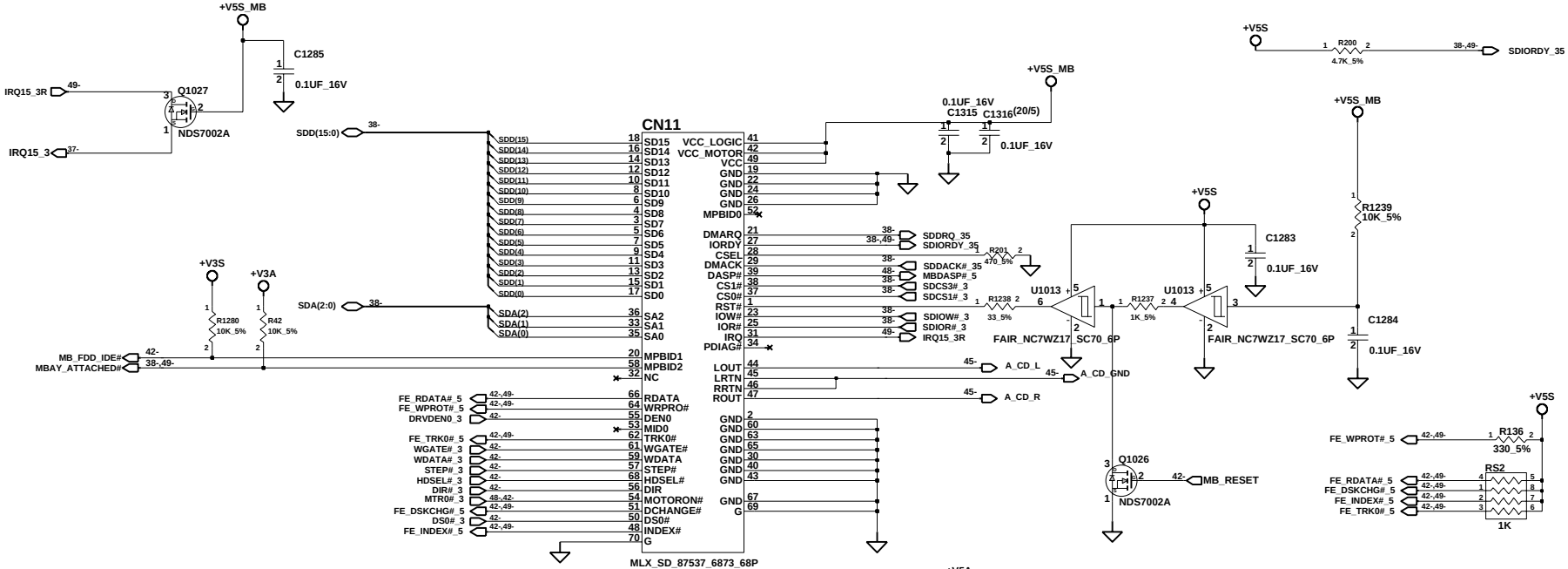
Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet 47 of 67
EE2	Wednesday, August 13, 2003	10:34:34 pm		A02	PC8803	

Engineer: David Du	INVENTEC
Drawn by: David Du	Size: A3
R&D CHK	TITLE: DIAMOND
DOC CTRL CHK	AUDIO AMP & HP JACK
MFG ENGR CHK	



Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet
EE2	Thursday, July 31, 2003	10:57:01 am		A02	PC8803	48 of 67

Engineer: David Du	INVENTEC	
Drawn by: David Du	TITLE: DIAMOND	Size: A3
R&D CHK	HDD CONN	
DOC CTRL CHK		
MFG ENGR CHK		



LAYOUT NOTES :

SIGNAL NAME	TRACE WIDTH (MILS)
+V5S_MB	60
MBPWR	60

Changed by: EE2	Date Changed: Thursday, July 31, 2003	Time Changed: 10:57:06 am	QA CHK	VER: A02	Model Number: PC8803

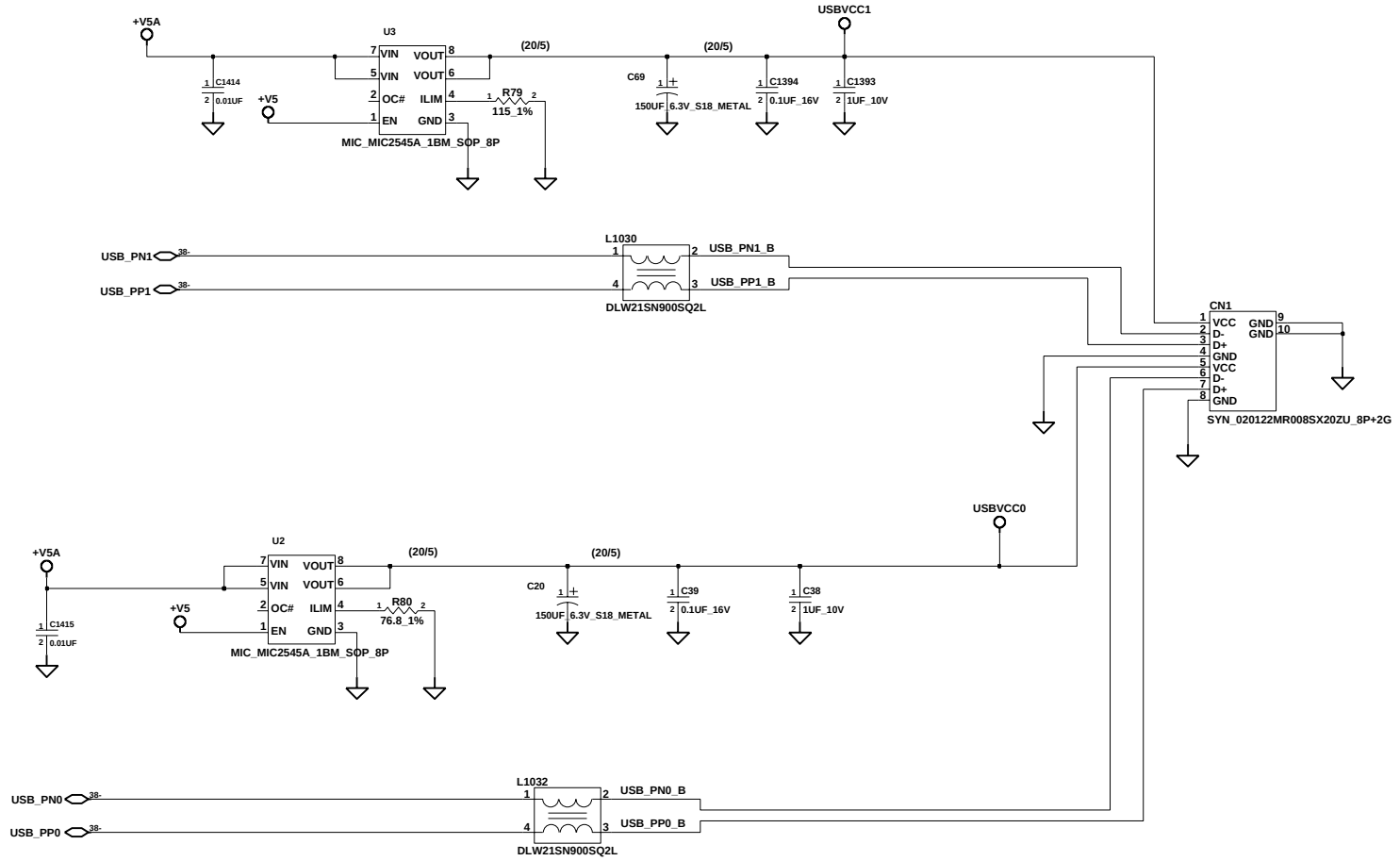
Engineer: David Du
Drawn by: David Du
R&D CHK
DOC CTRL CHK
MFG ENGR CHK

INVENTEC

TITLE: **DIAMOND**
MULTIBAY CONN

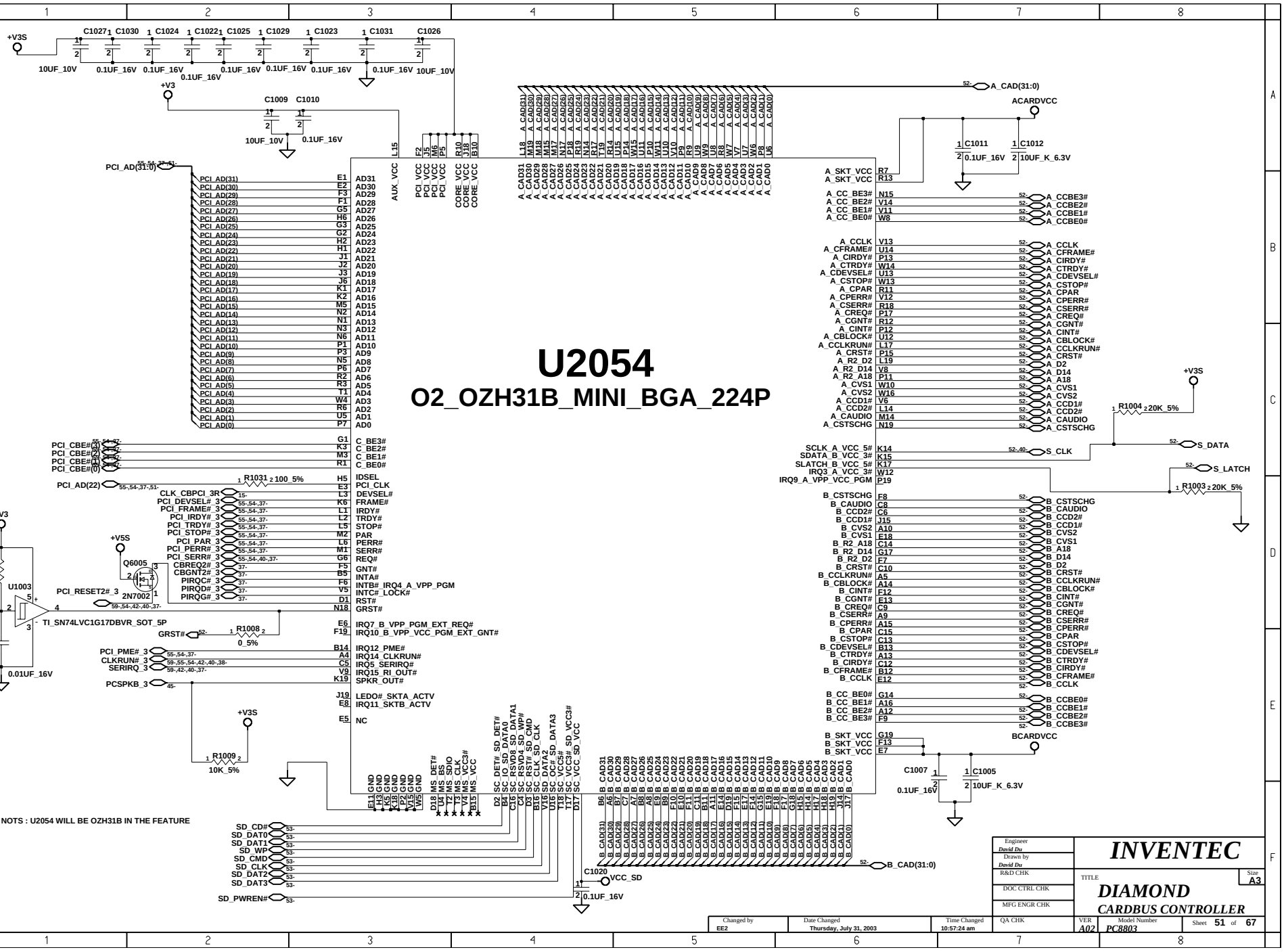
Size: A3

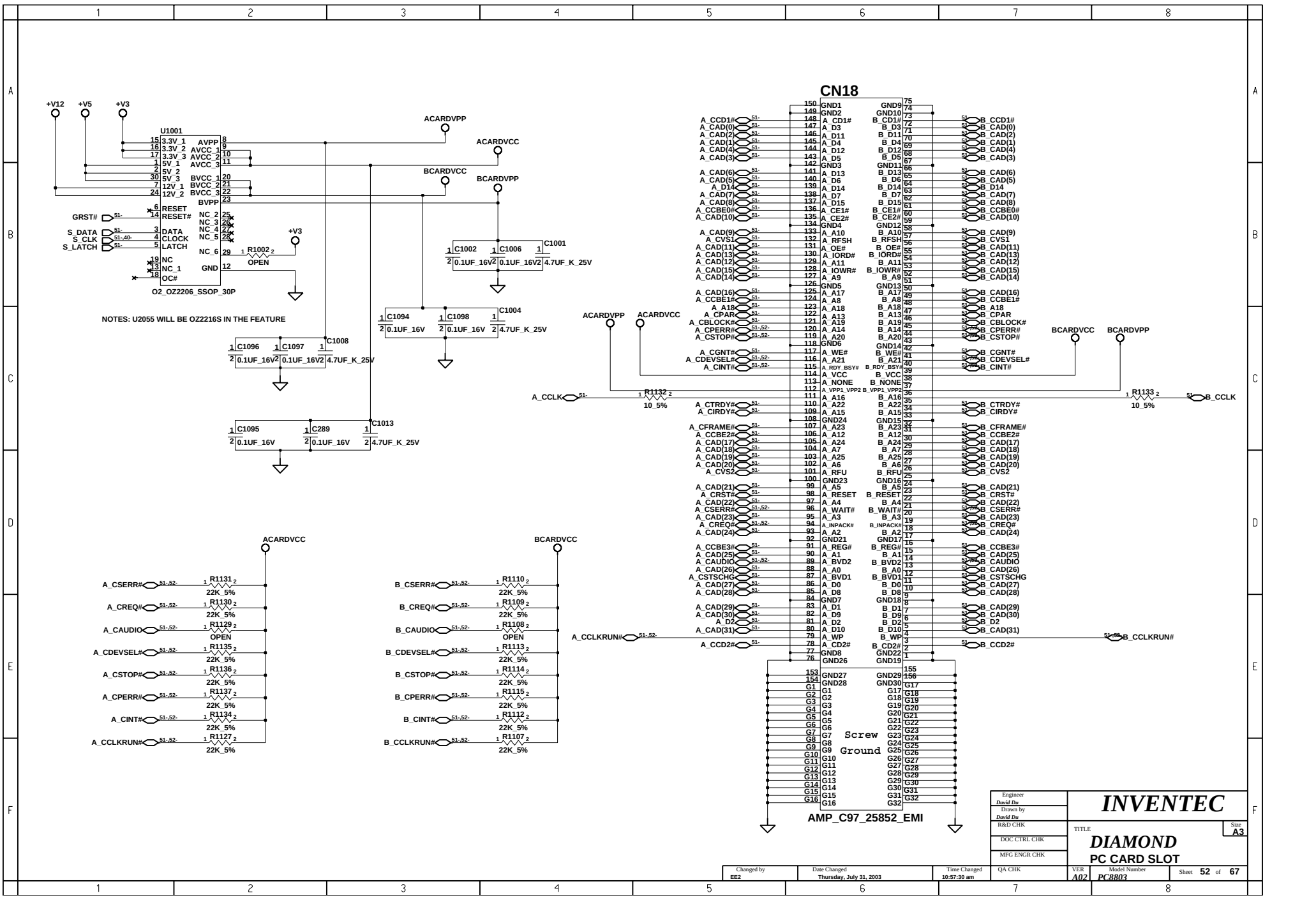
Sheet 49 of 67

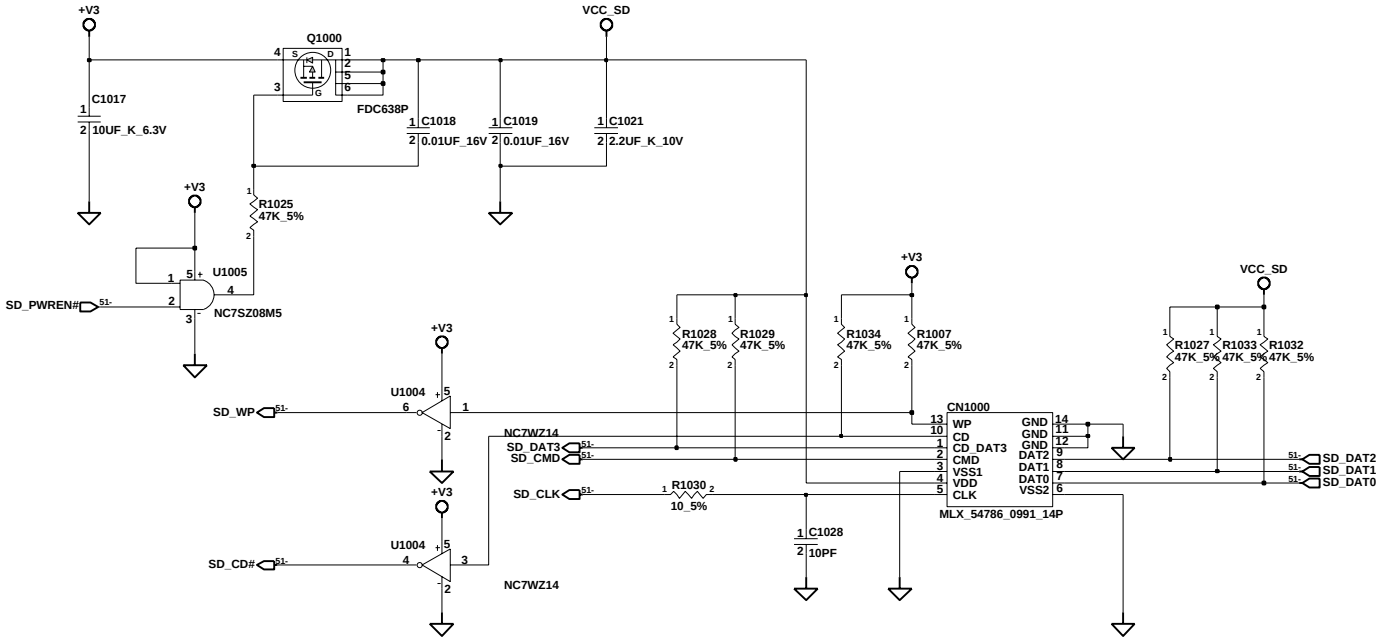


Changed by EE2	Date Changed Thursday, July 31, 2003	Time Changed 10:57:19 am	QA CHK	VER A02	Model Number PC8803	Sheet 50 of 67

Engineer David Du	**INVENTEC**	
Drawn by David Du		
R&D CHK	TITLE **DIAMOND**	Size A3
DOC CTRL CHK	**USB&IR CONN**	
MFG ENGR CHK		

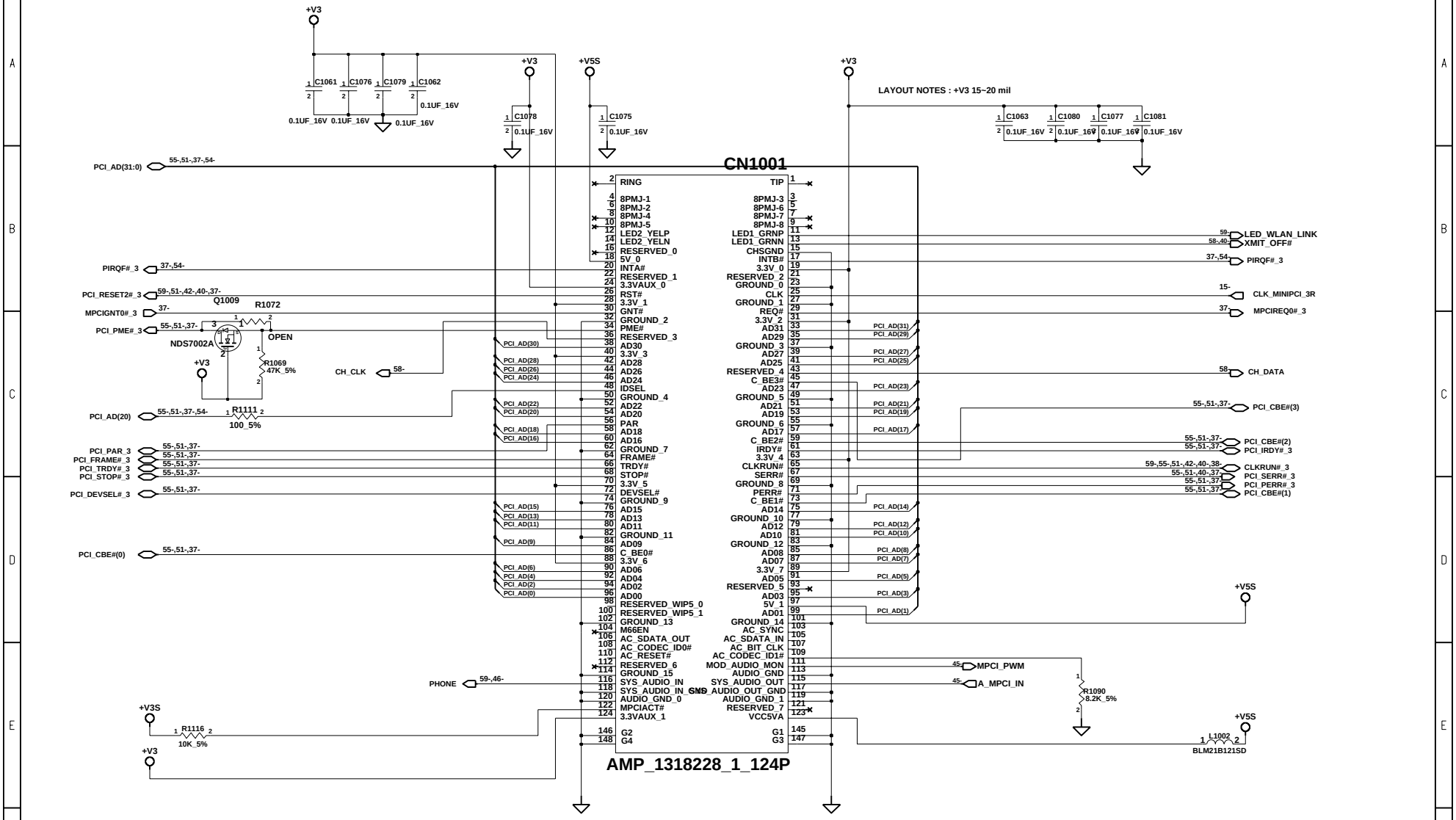

U2054
O2_OZH31B_MINI_BGA_224P
NOTS : U2054 WILL BE OZH31B IN THE FEATURE
INVENTEC
DIAMOND
CARDBUS CONTROLLER
Engineer
David Du
Drawn by
David Du
R&D CHK
DOC CTRL CHK
MFG ENGR CHK
QA CHK
Size
A3
Changed by
EE2
Date Changed
Thursday, July 31, 2003
Time Changed
10:57:24 am
VER
A02
Model Number
PC8803
Sheet
51
of
67


+V12
+V5
+V3
U1001
O2_OZ2206_SSOP_30P
GRST#
S_DATA
S_CLK
S_LATCH
+V3
R1002
OPEN
ACARDVPP
ACARDVCC
BCARDVCC
BCARDVPP
C1001
C1002
C1006
0.1UF_16V
4.7UF_K_25V
C1004
C1094
C1098
C1008
C1096
C1097
C1013
C1095
C289
NOTES: U2055 WILL BE OZ2216S IN THE FEATURE
CN18
AMP_C97_25852_EMI
Screw
Ground
R1131
R1130
R1129
R1135
R1136
R1137
R1134
R1127
R1110
R1109
R1108
R1113
R1114
R1115
R1112
R1107
22K_5%
R1132
R1133
10_5%
A_CCLKRUN#
B_CCLKRUN#
A_CCD2#
B_CCD2#
INVENTEC
DIAMOND
PC CARD SLOT
Engineer
David Du
Drawn by
David Du
R&D CHK
DOC CTRL CHK
MFG ENGR CHK
Changed by
EE2
Date Changed
Thursday, July 31, 2003
Time Changed
10:57:30 am
QA CHK
Size
A3
VER
A02
Model Number
PC8803
Sheet
52
of
67



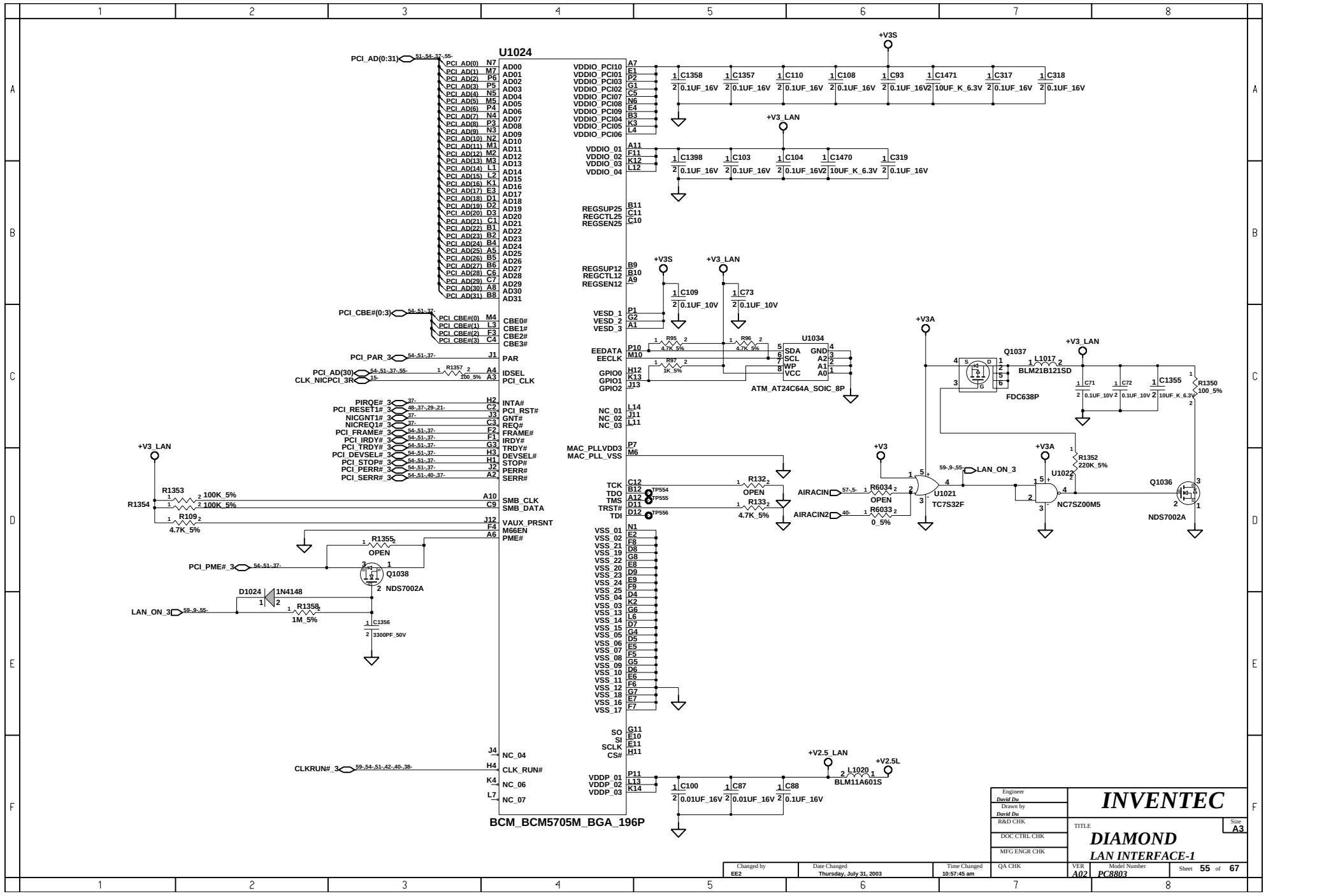
Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet 53 of 67
EE2	Thursday, July 31, 2003	10:57:35 am		A02	PC8803	

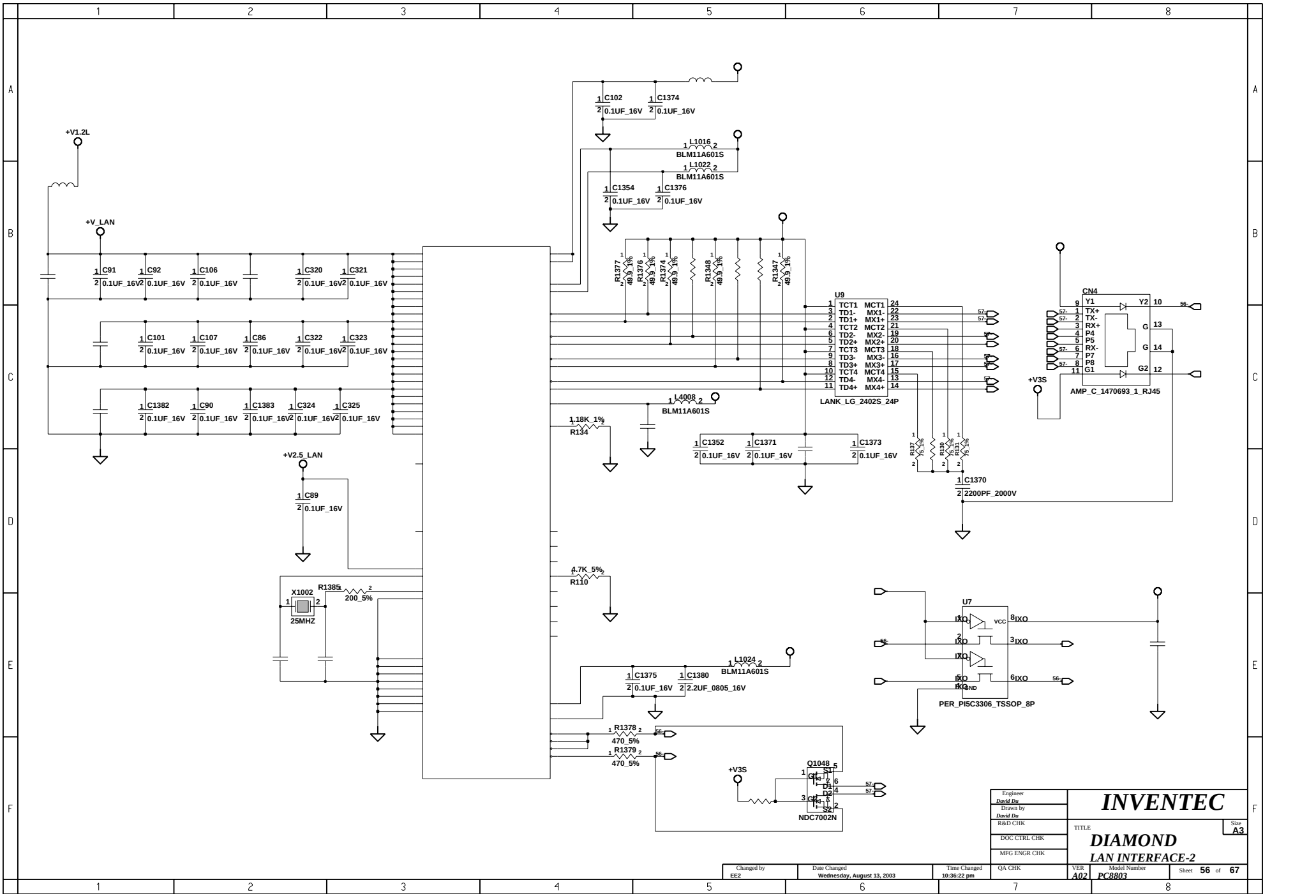
Engineer: David Du	INVENTEC
Drawn by: David Du	
R&D CHK	TITLE: DIAMOND
DOC CTRL CHK	SD CARD CONN
MFG ENGR CHK	Size: A3



Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet 54 of 67
EE2	Thursday, July 31, 2003	10:57:40 am		A02	PC8803	

Engineer: David Du	INVENTEC
Drawn by: David Du	TITLE: DIAMOND MINIPCI CONN
R&D CHK	Size: A3
DOC CTRL CHK	
MFG ENGR CHK	


U1024
BCM_BCM5705M_BGA_196P
U1034
ATM_AT24C64A_SOIC_8P
Q1037
FDC638P
U1021
TC7S32F
U1022
NC7SZ00M5
Q1036
NDS7002A
Q1038
NDS7002A
D1024
1N4148
L1017
BLM21B121SD
L1020
BLM11A601S
+V3S
+V3_LAN
+V3A
+V3
+V2.5_LAN
+V2.5L
LAN_ON_3
INVENTEC
DIAMOND
LAN INTERFACE-1
PC8803
A02
Engineer: David Du
Drawn by: David Du
R&D CHK
DOC CTRL CHK
MFG ENGR CHK
QA CHK
Changed by: EE2
Date Changed: Thursday, July 31, 2003
Time Changed: 10:57:45 am
Size: A3
Sheet 55 of 67



Changed by: EE2	Date Changed: Wednesday, August 13, 2003	Time Changed: 10:36:22 pm	QA CHK	VER: A02	Model Number: PC8803

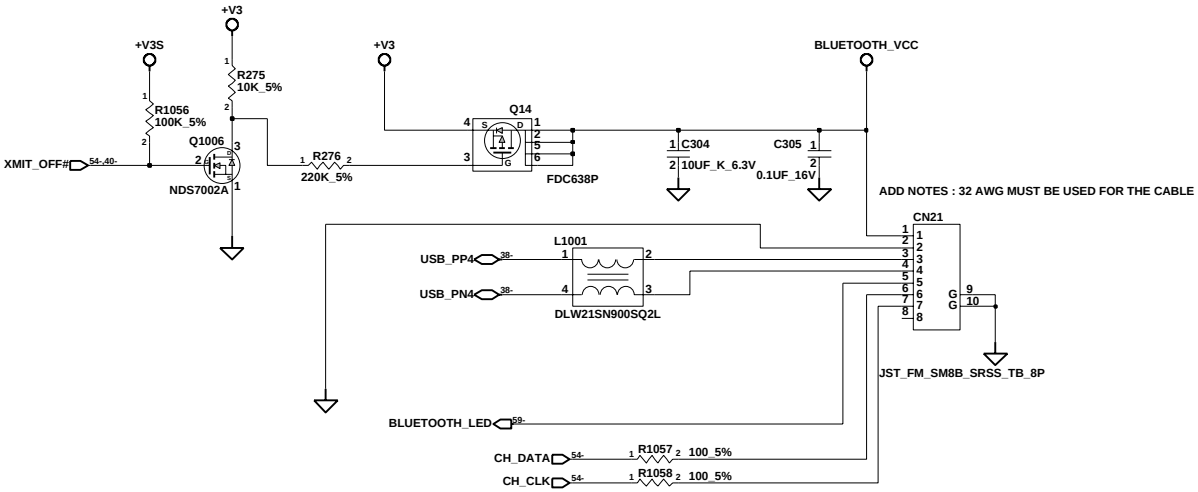
Engineer: *David Du*
Drawn by: *David Du*
R&D CHK
DOC CTRL CHK
MFG ENGR CHK

INVENTEC

TITLE: ***DIAMOND***
LAN INTERFACE-2

Size: A3

Sheet 56 of 67



Changed by: EE2	Date Changed: Thursday, July 31, 2003	Time Changed: 10:58:02 am	QA CHK	VER: A02	Model Number: PC8803

Engineer: *David Du*
Drawn by: *David Du*
R&D CHK
DOC CTRL CHK
MFG ENGR CHK

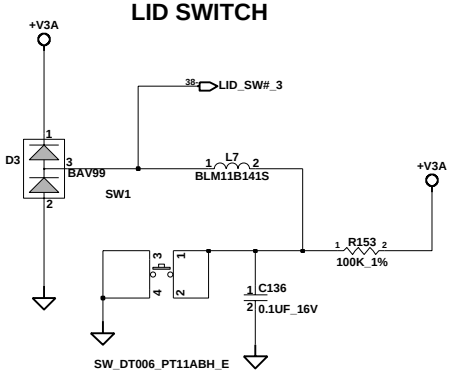
INVENTEC

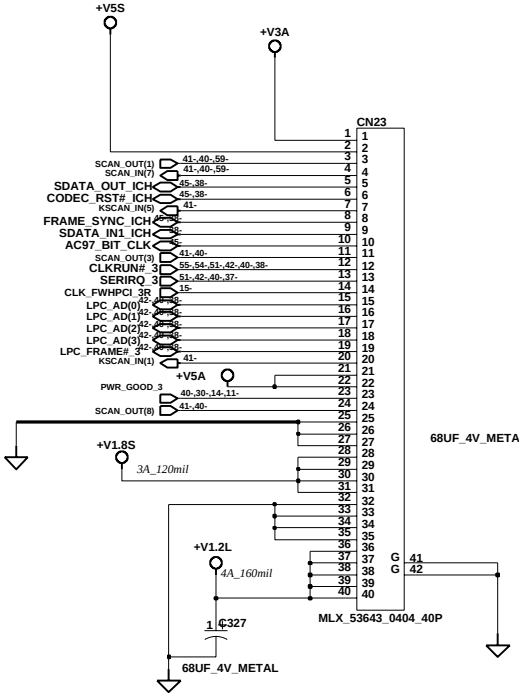
TITLE: ***DIAMOND***
BLUETOOTH

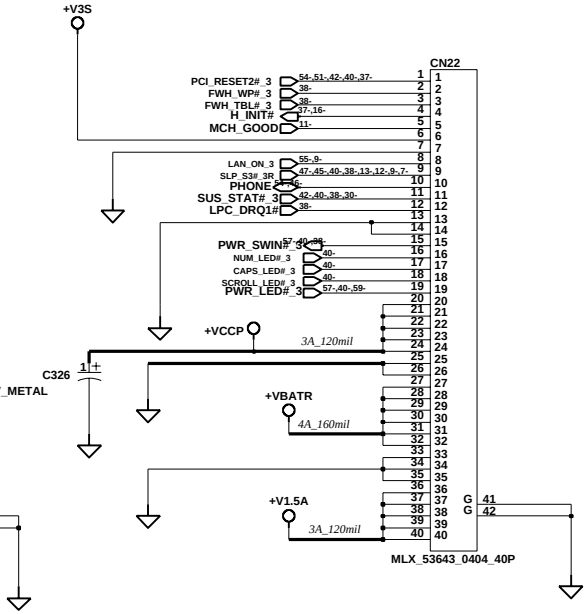
Size: A3

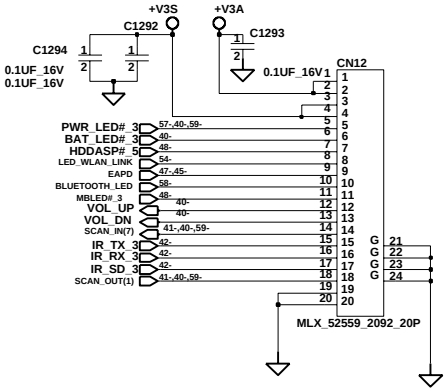
Sheet 58 of 67

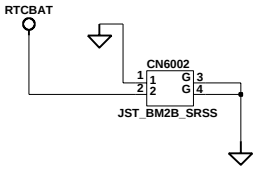
LID SWITCH

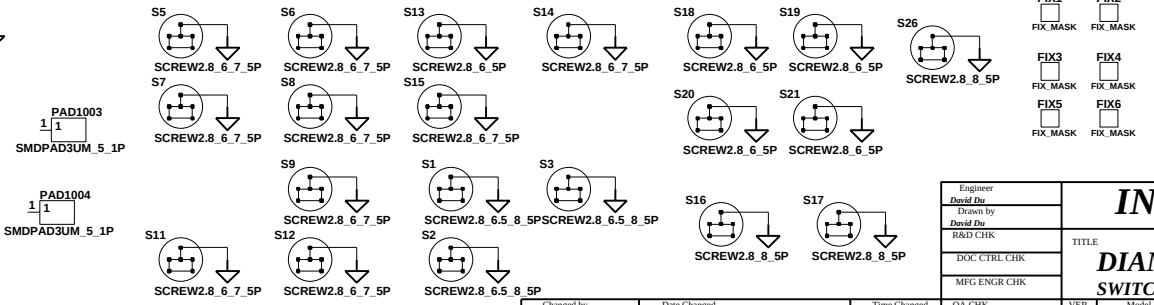












Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number	Sheet
EE2	Thursday, July 31, 2003	10:58:07 am		A02	PC8803	59 of 67

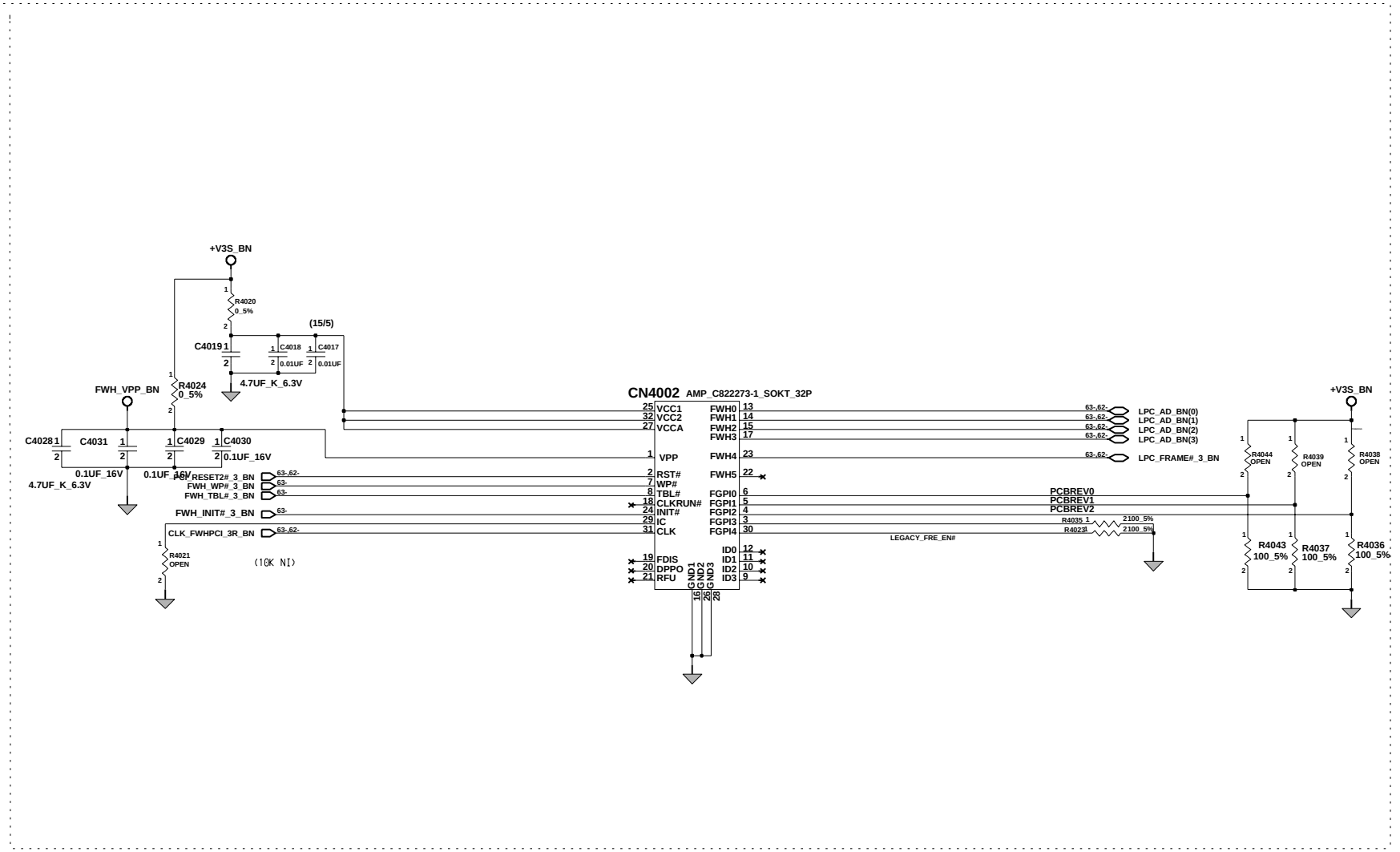
Engineer	David Du
Drawn by	David Du
R&D CHK	
DOC CTRL CHK	
MFG ENGR CHK	

INVENTEC

TITLE: **DIAMOND**

SWITCH BUTTON & LED

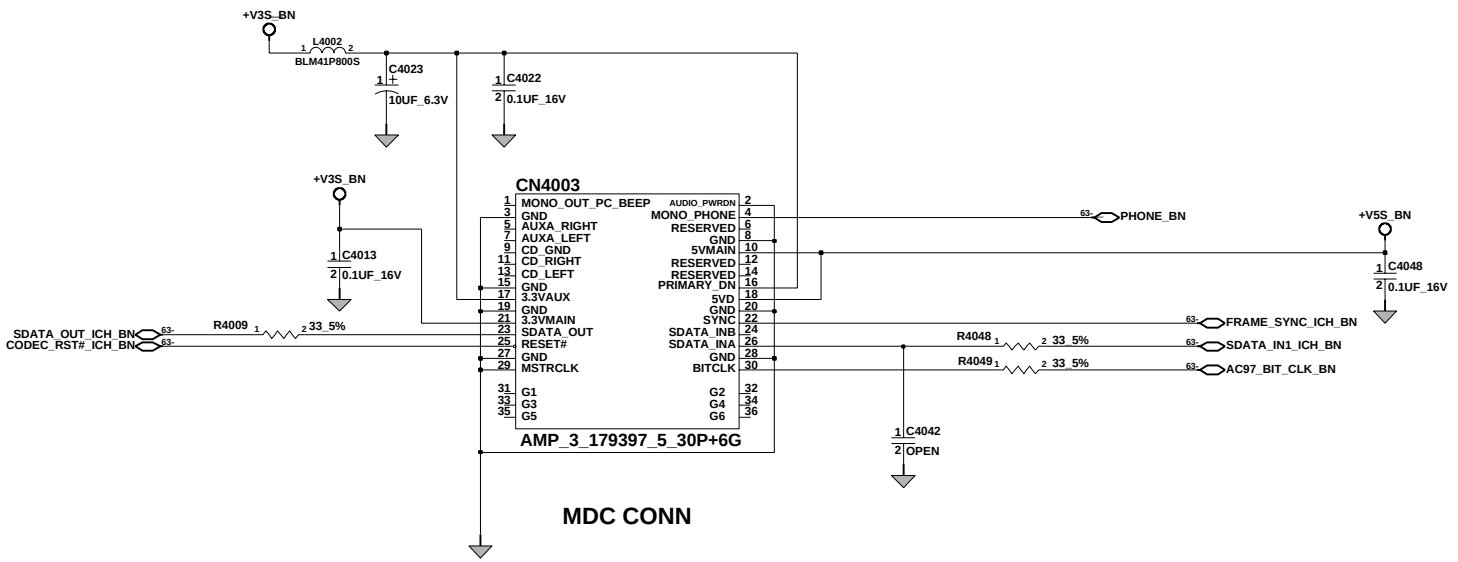
Size: A3



FHW ON BUTTON DAUGHTER BOARD

Changed by EE2	Date Changed Thursday, July 31, 2003	Time Changed 10:58:37 am	QA CHK	VER A02	Model Number PC8803	Sheet 60 of 67

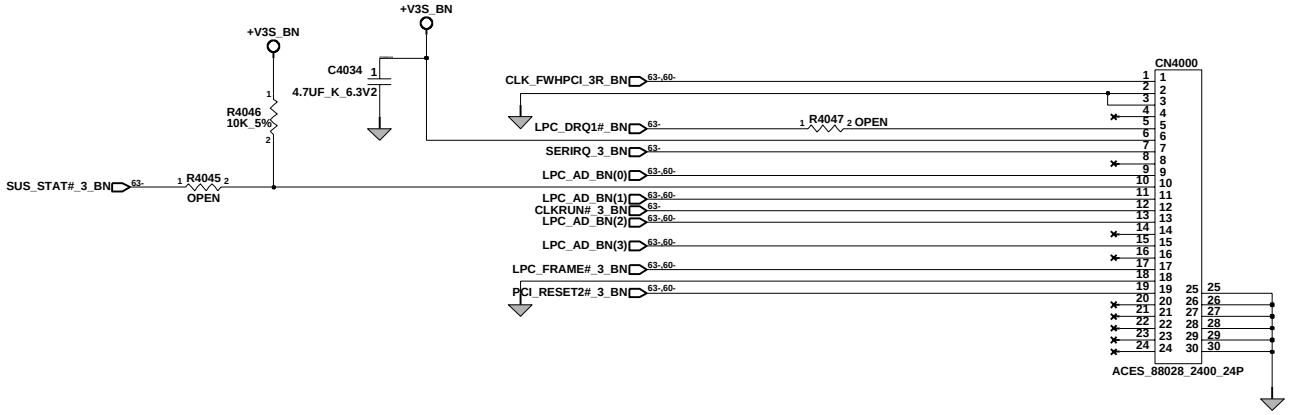
Engineer *David Du*	INVENTEC	
Drawn by *David Du*	TITLE *DIAMOND* *FHW*	Size A3
R&D CHK		
DOC CTRL CHK		
MFG ENGR CHK		



MDC CONN ON BUTTON DAUGHTER BOARD

Changed by: EE2	Date Changed: Thursday, July 31, 2003	Time Changed: 10:58:42 am	QA CHK	VER: A02	Model Number: PC8803	Sheet 61 of 67

Engineer: David Du	INVENTEC	
Drawn by: David Du	TITLE: DIAMOND	Size: A3
R&D CHK	MDC CONN	
DOC CTRL CHK		
MFG ENGR CHK		

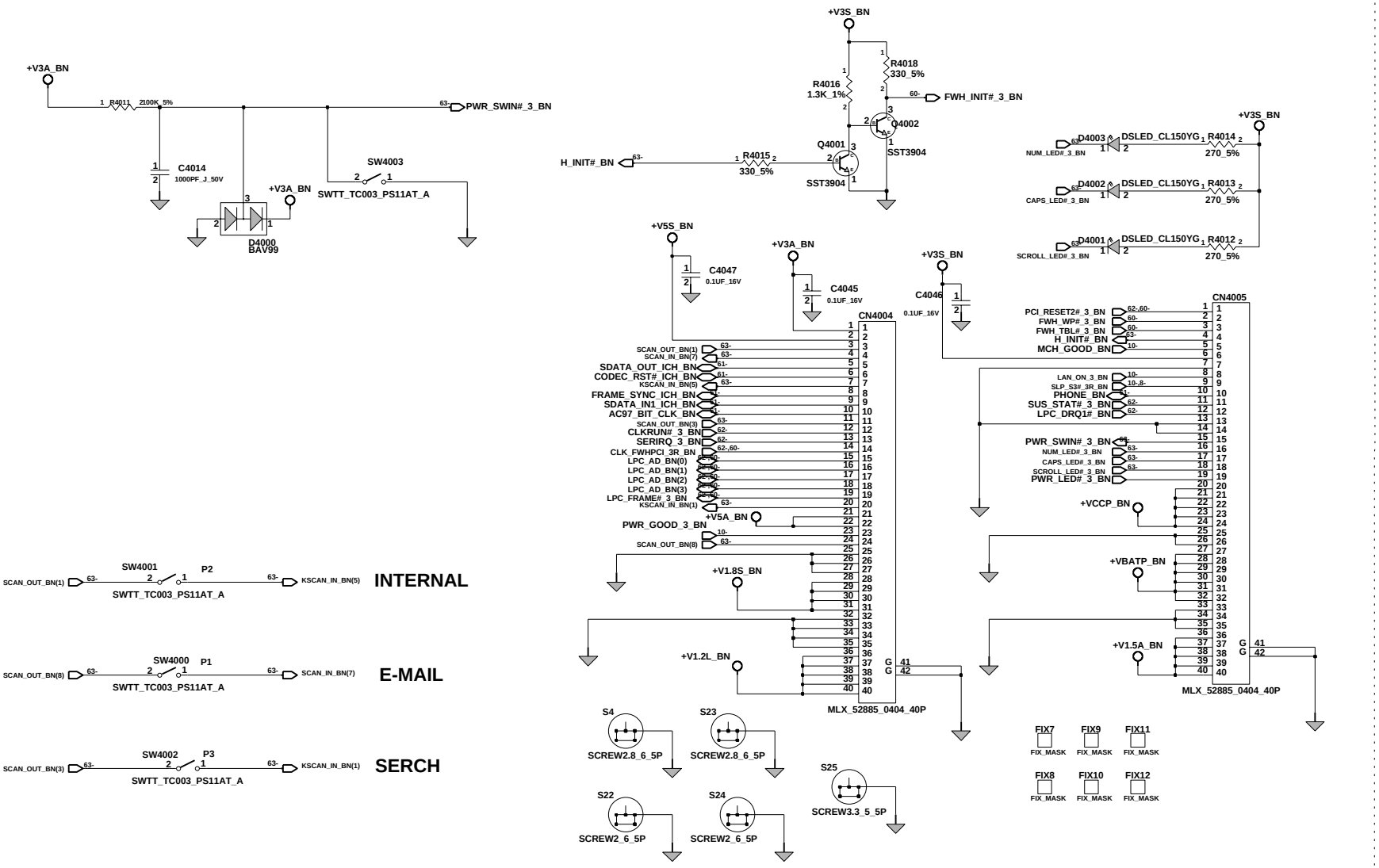


AMP 20PIN CONNECTOR
MOTHERBOARD SIDE
ACES_88028_2400

TCPA CONN ON BUTTON DAUGHTER BOARD

Changed by: EE2	Date Changed: Thursday, July 31, 2003	Time Changed: 10:58:47 am	QA CHK	VER: A02	Model Number: PC8803	Sheet 62 of 67

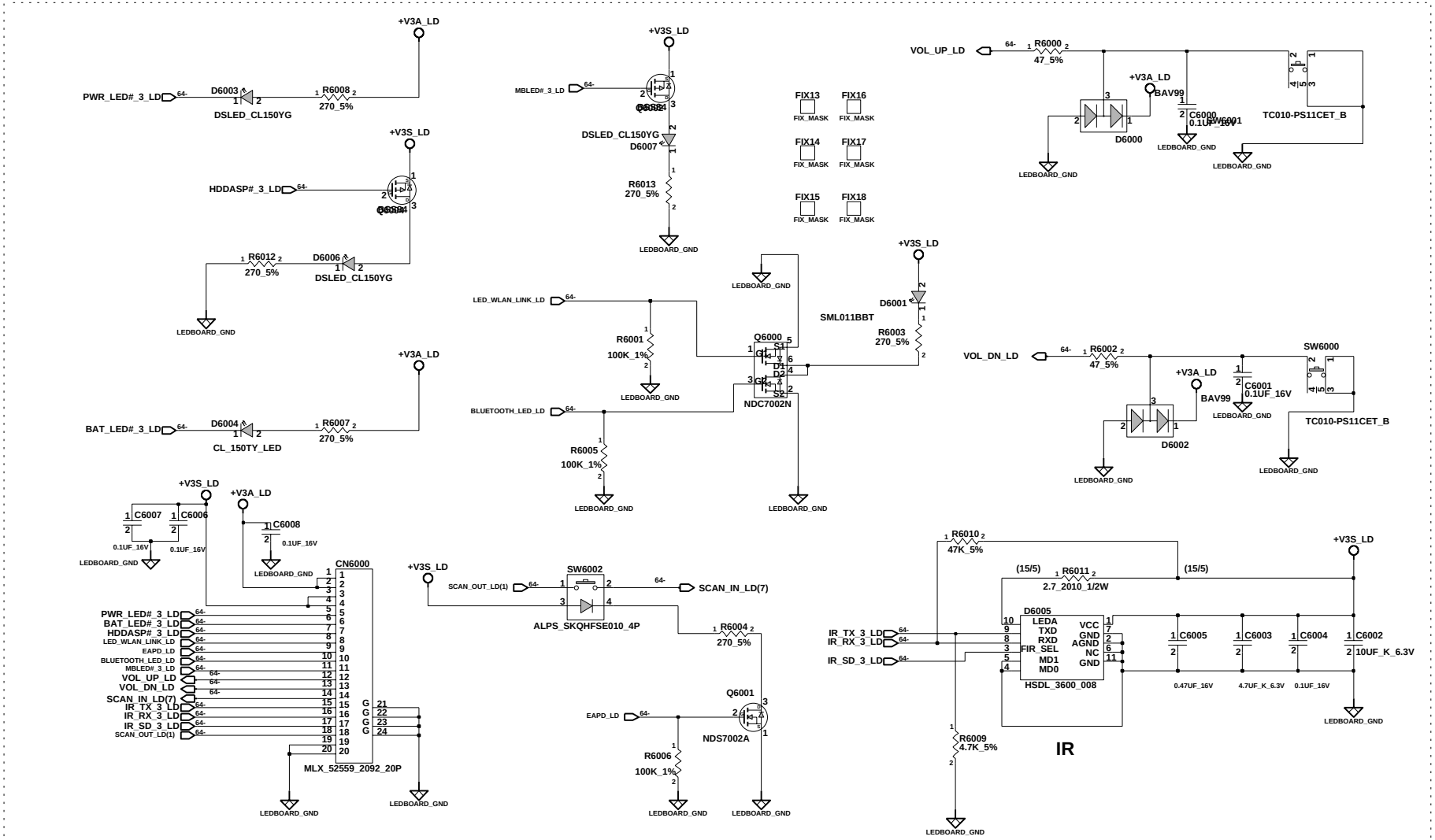
Engineer: David Du	INVENTEC	
Drawn by: David Du	TITLE: DIAMOND TCPA	Size: A3
R&D CHK		
DOC CTRL CHK		
MFG ENGR CHK		



SWITCH & CONN & LED ON BUTTON DAUGHTER BOARD

Changed by: EE2	Date Changed: Thursday, July 31, 2003	Time Changed: 10:58:55 am	QA CHK	VER: A02	Model Number: PC8803

Engineer: David Du	INVENTEC
Drawn by: David Du	
R&D CHK	TITLE: DIAMOND
DOC CTRL CHK	CONN & SWITCH & LED
MFG ENGR CHK	Size: A3
	Sheet 63 of 67



LED & VOL BUTTON ON LED DAUGHTER BOARD

Changed by	Date Changed	Time Changed	QA CHK	VER	Model Number
EE2	Thursday, July 31, 2003	10:59:01 am		A02	PC8803

Engineer	David Du
Drawn by	David Du
R&D CHK	
DOC CTRL CHK	
MFG ENGR CHK	

INVENTEC

TITLE: DIAMOND
LED & VOL BUTTON

Size: A3

Sheet 64 of 67

SHEET 10

1.R1096 change from 10K to 200K and add CAP (0.01UF) to GND

NOTE: For ATI power sequence

SHEET 11

1. Add PM_VGATE DELAY CIRCUIT: U18, D23, R156, C90, C107

2. Add CAP (0.1UF) in R2024 pin1 to GND

NOTE: For PCI_RESET, PCI_CLOCK , PM_VGATE TIMING

SHEET 12

1. Del PAD2 , PAD515 and to short power

SHEET 13

1. R1122 change from 220K to 100K and add CAP(0.1UF) in R1122 pin1 to GND
2. Add CAP (OPEN) in Q25 pin3 to GND

NOTE: For ATI power sequence

SHEET 29

1. Add SERIES RESISTORS on SB_STBS(R208), ADSTBS_0(R211), ADSTBS_1(R212)

2. Add SERIES RESISTORS DBI_LO(R198), DBI_HI(R197)

3. Change R72 from 10K to 20K

4.Add TEST POINT in GPIO(4:7) GPIO(9:14)

5. Add 1K pull down on GPIO8

6. Add 1K pull down RESISTOR to prvent leakage for some ZV_LCDDATA, ZV_LCDCNTL pin

(ZV_LCDDATA(5:15), (18:19), ZV_LCDCNTL(0:3))

7. Add 10K pull down RESISTOR to prevent leakage for ZV_LCDDATA(16:17)

NOTE: Above items for ATI recommed

8. R153,R154, change form 1K to 3K

NOTE: For +V3S power rail (ATI- VREFG)

9. Add SERIES RESISTORS STOP(R192) DEVSEL(R199) TRDY(R200) IRDY(R217) FRAME(R216)
10. Add SERIES RESISTORS WBF(R205) RBF(R215) AD_STBF_0(R202) AD_STBF_1(R203) SB_STBF(R204)
11. R189 , R190 PIN2 TO GND
12. R72 PIN 1 TO LINK +V3S

SHEET 30

1. Add 0 OHM series resistor on the path of osc output to X'TAL and XTALOUT

NOTE: Select frequency enter chalnnel

SHEET 29,30, 31, 32

1. Change for ATI M9+X , M10 dual footprint schematic

SHEET 36

1. Add R219 TO JUMP M10 OR M9+X

SHEET 38

1. U509 -V19 (PM_VGATE/VRMPWRGD) change from PM_VGATE to SB_VGATE
2. Add U509 GPIO42 resistors R236 pull up +V3S and R237 pull down to GND

NOTE: To chose memory type

SHEET 41

1. Rotate CN8 (the touch-pad conect):180 point of view and pin5,6 link to GND, pin7,8 link to CN7 pin3
2. Change CN7 (stick point) geoetric (layout pattern)

SHEET 49

1. Change R155 from 10K to 100K

SHEET 50

1. Change CN2068 for mechanical

SHEET 56

1. Change CN2062 point name

NOTE: Modify LAN signal to right

SHEET 63

1.Change SW4 for mechanical

SHEET 64

1.Change D21 for mechanical

CHANGE LIST FEBRUARY 12

SHEET 11

1.Change C65 TO CONNECTED TO PIN2 OF R2024(REF.RUBY)

2.CHANGE R1061 TO OPEN(REF.RUBY)

SHEET 13

1.CHANGE R1122 FROM 220K TO 100K (REF.RUBY)

SHEET 29

1. ADD NOTE TO M9+X , M10 R153, R154 USED OR NOT

2. Change AGPTEST DBI_LO & DBI_HI TO +V1.5ATIAGP RAIL TO MATCH RUBY

SHEET 30

1.OPEN R2102 TO MATCH RUBY SCH (REF.RUBY)
2.ADD NOTE : R687&R1766 FOR M9+X OR M10 USED
3. ADD 10K PULL-UP +V3S TO AUXWIN PIN AF26 (REF.RUBY)

SHEET 32

1.ADD L20 NFM41P11C204 TO MATCH RUBY SCH (REF.RUBY)
2.CHANGE L19,L1062, L1056, L1063, L1064, L1059,&L1060 TO MATCH RUBY SCH (REF.RUBY)
3.CHANGE L5/L4,&L12 TO MATCH RUBY SCH(REF.RUBY)

SHEET 38

1. CHANGE GPIO42 TO VMEM_CFG3 AND PULL UP TO +V3S & 8.2K TO GROUNDTO MATCH RUBY SCH (REF.RUBY)
2. ADD INVERTER BETWEEN PIN H22 (GPIO39) AND U2038 (PIN26) TO MATCH RUBY SCH (REF.RUBY)

SHEET 50

1.CHANGE USB POWER FROM +V5A TO +V5 TO MATCH RUBY SCH(REF.RUBY)

CHANGE LIST FEBRUARY 14

SHEET 33

1. CHANGE DEPEND RESISTORS PACKAGE TO LAYOUT ROUTES

CHANGE LIST FEBRUARY 15

SHEET 10

1. ADD U22 SCHMITT BUFFER FROM PIN2 R1096
AFTER THE RC CIRCUIT TO PIN10 OF U2008

SHEET 29

1. MAKE A NOTE R1957 R1958 TO INSTALL M9+X OR M10 BE USED
2. MAKE A NOTE R1956 TO INSTALL M9+X OR M10 BE USED

SHEET 31

1. CHANGE C1982 C1983 TO 10 PF THESE TWO CAPACITORS VALUSE CAN BE TUNED
2. ADD SERIES RESISTORS R237 R238 ON EACHOF THE TWO SIGNAL (DIMB_0 DIMB_1)
MAKE A NOTE TO ONLY INSTALL THESE RESISTORS FOR ELPIDA MEMORY

SHEET 32

1. ADD L21 TO POWER PIN(P8, Y8, AC11,AC20,Y23, L23,H20,H11)
MAKE A NOTES: THIS RAIL HAVE JUMPER TO SWITCH FOR +V1.5S FOR M10 , +V1.8S FOR M9+X

SHEET 33

1. CHANGE THE TITLE , THESE IS A TYPO" VEDIO". IT SHOULD BE " VIDEO"

CHANGE LIST FEBRUARY 17

SHEET 32

1. OPEN L12 AND L21 PUT BLM21A121S TO USED M9+X +V1.8S

SHEET 42

1. OPEN L12 AND L21 PUT BLM21A12. CHANGE SIGNAL NAME R252 PIN2 FROM PCI_RESET TO NPCI_RESET

CHANGE LIST FEBRUARY 20

SHEET 11

1. U2010 PIN2(TON) TO ADD RESISTOR (OPEN) THEN PULL UP TO +VCCP

SHEET 13

1. CHANGE C66 VAULE FROM 22UF TO 68UF
2. ADD THE CAP 10UF C91 PIN1 TO LINK Q26 (PIN1,2,5,6) C91 PIN2 TO GND

SHEET 20

1. CHANGE U2019 , U2329 FROM ADM1032 TO ADM1031 TWO PACKAGE THERMAL MONITOR

CHANGE LIST FEBRUARY 21

SHEET 20

1. ADD THE BLUETOOTH SIGNAL AT CN1020 PIN43(CH_DATA) , PIN36(CH_CLK)

SHEET 20

2. REMOVE C1671 C1673 C1674 C1675 THEN C1672 PIN2 CHAISY GND CHANGE TO DIGITAL GND

SHEET 58

1. ADD TWO SIGNAL AT CN1036 PIN6(CH_DATA) , PIN7(CH_CLK) THEN ADD TWO SERIES RESISTORS
100 OHM TO THESE PIN

CHANGE LIST FEBRUARY 23

1. ADD RESISTOR 100K PULL UP TO +V3S ANOTHER PIN TO LINK Q53 PIN2(XMIT_OFF#)

CHANGE LIST FEBRUARY 27

SHEET 20

1. ADD BYPASS CAPACITORS C308, C309, C310 (220P)

SHEET 25 26

1. ADD BYPASS CAPACITORS , C311, C313 (150UF)

Changed by: EE2	Date Changed: Thursday, July 31, 2003	Time Changed: 10:59:06 am	QA CHK	VER: A02	Model Number: PC8803	Sheet 65 of 67

Engineer: David Du	INVENTEC	
Drawn by: David Du		
R&D CHK	TITLE: DIAMOND	Size: A3
DOC CTRL CHK	CHANGE LIST	
MFG ENGR CHK		

FOR SI SCHEMATIC CHANGE

SHEET 17

1. FOR PSI# PULL HIGH +VCCP TO OPEN R1089

SHEET 38

1. REMOVE PWM_GOOD_3 SIGNAL DELETE R1419

SHEET 40

1. R1049 FOR PCI_SERR#_3 PULL DOWN TO OPEN
2. KBC INITIAL NEED TO PULL DOWN 100K RESISTOR R1
3. ADD U4004 , D6008 , R6014 , C6011 , C6012 THIS CIRCUIT TO FINE_TUNE POWER ON SEQUENCE TIME

SHEET 41

1. RNET FOR KAHUNA PROGRAMMING. NEED ONLY FOR FLASH RECOVERY SATURATION

SHEET 50

1. REMOVE USB_OC#0 U2 , U3 PIN_2 TO OPEN
2. U2 , U3 PIN_1(EN) CONNECT +5V
3. U2 , U3 PIN_5 CONNECT +V5A
4. REMOVE L3 AND TO SHORT

SHEET 52

1. REMOVE R1126 , R1128 TO A_CCLKRUN# , B_CCLKRUN# SIGNAL PULL UP REPEAT

SHEET 64

1. ADD Q6004 TO PREVENT THE CURRENT FROM D6006 INTO HDDLED LOGIC OUTPUT

2003/03/18 update

SHEET 15

1. ADD R6013 RESISTOR OPEN TO JUMP CLK_SIO14_3R & ADI48M

SHEET 40

1. ADD C6013 18P CAPICATOR BYPASS TO LOW_BAT#_3 THIS SIGNAL

2003/03/19 update

SHEET 20

1. ADD U4005 , R6016 , Q6005 , D6009 , C6015 , C6014 THIS CIRCUIT AND CN6001 TO SECOND FAN

SHEET 59

1. REMOVE CN22_PIN-5-RTCBAT
2. ADD CN6002 TO RCT CONNECTOR FOR MECHICNAL CHANGE

SHEET 63

1.REMOVE CN4001 RTC CONNECTOR , CN1010_PIN-5-RTCBAT FOR MECHICNAL CHANGE

SHEET 64

1. CHANGE SW6001 AND SW6000 FOR MECHICNAL MODIFY
2. CHANGE D6005 I.R. SIZE FOR MECHICNAL MODIFY

2003/03/24 update

SHEET 31 & 34

1. ADD DDR_CSB1# SIGNAL FOR 8M*32 VRAN USED
AND ADD R6016 RESISTOR DEPEND TO LINK U1018 & U1019 PIN M4

SHEET5

1.CHANGE R1068 VALUE FROM 150K TO 10K AND R1070 VALUE
FROM 20K TO 1.3K FOR AC DETECT LEVEL

SHEET15

1.CHANGE R6015 VALUE FROM OPEN TO 33 AND R1361 VALUE
CHANGE TO OPEN FOR AC'97 FRQUENCE CHANGE 14.318MHZ

SHEET42

1.CHANGE R126 VALUE TO 0 AND R125 VALUE CHANGE
FROM 0 TO OPEN FOR SUPER I/O USED_NPCI_RESET

SHEET44

1.CHANGE R1165_PIN-1 VOLTAGE FROM +5VA TO +V5

SHEET45

1.CHANGE R1460 FROM OPEN TO 1K FOR AC'97 CHOOSE FRQUENCE

2003/03/26 update

SHEET13

1. CHANGE P MOS Q1041 , Q1046 , Q10 , Q11 HIGH CURRENT AND LOW RDS ON

SHEET36

1. CHANGE P MOS , Q1050 HIGH CURRENT AND LOW RDS ON

SHEET 40

1. REMOVE R1409 , D1028 AND TO SHORT PCI_SERR#_3

SHEET 50

1. REMOVE L2 AND TO SHORT

SHEET 52

1. CHANGE R1132 , R1133 FROM 47 OHM TO 22 OHM FOR A_CCLK & B_CCLK TIMING

SHEET 55

1. CHANGE VALUE C1358 , C1357 , C1398 FROM 0.01UF TO 0.1UF FOR BCM MODIFY

2. ADD CAPACITORS C6015 , C6014 10UF FOR BCM MODIFY

2003/03/30 update

SHEET 5

1. ADD R6021 RESISTOR OPEN TO MODIFY OCP

SHEET 33

1. CHANGE THE VALUE OF SERIAL TERMINATION RESISTORS ON THE DATA ,
DQM AND QS LINES TO 22-OHMS

SHEET 34

1. CHANGE THE VALUE OF SERIAL TERMINATION RESISTOR OF QS LINE TO 22-HHMS

SHEET 40

1. R105 SHOULD BE OPEN TO FINE TUNE KBC POWER UP SEQUENTIAL

SHEET 55

1. ADD TWO 0.1uF DECAPS C6018 AND C6019 TO +V3S RAIL
2. ADD 0.1uF DECAPS C6019 TO +V3_LAN

SHEET 56

1. ADD SIX 0.1uF DECAPS C6021 , C6022 , C6023 , C6024 , C6025 , C6026 TO +V_LAN RAIL

2003/04/03 update

SHEET 45

1. ISOLATE PCSPKB_3 USING A 2N7002 WITH DRAIN CONNECTED TO +V3S THROUGH A 10K PULL-UP ,
SOURCE TO GROUND , & GATE TO PCSPKB_3 . PIN 2 OF C28 SHOULD BE CONNECTED TO DRAIN OF 2N7002 ,
WHICH FEEDS THE AUX INPUT OF CODEC

2. ISOLATE PCSPKB_ICH_3 USING A 2N7002 WITH DRAIN CONNECTED TO +V3S THROUGH A 10K PULL-UP ,
SOURCE TO GROUND , & GATE TO PCSPKB_ICH_3 . PIN 2 OF C15 SHOULD BE CONNECTED TO DRAIN OF 2N7002 ,
WHICH FEEDS THE AUX INPUT OF CODEC

SHEET 46

1. ISOLATE PHONE USING A 2N7002 WITH DRAIN CONNECTED TO +V3S THROUGH A 10K PULL-UP ,
SOURCE TO GROUND , & GATE TO PHONE . PIN 1 OF C24 SHOULD BE CONNECTED TO DRAIN OF 2N7002 ,
WHICH FEEDS THE AUX INPUT OF CODEC

SHEET 47 , 64

1. CHANGING FDD ACTIVITY LED TO MULTI-BAY ACTIVITY LED
AND HDD ACTIVITY LED TO JUST THE FIXED HDD

2003/04/05 update

SHEET 31

1. CAPS C120 AND C125 ON CLOCK LINES CHANGE TO 10NF

SHEET 57

1. THE HPD CIRCUIT REQUIRES A 2.5V ZENER (IN PARALLEL WITH R1264) , AND ADDITIONAL 20K IN SERIES.
SEE THE ATI REFERENCE DESIGN BOARD

Changed by: EE2	Date Changed: Thursday, July 31, 2003	Time Changed: 10:59:10 am	QA CHK	VER: A02	Model Number: PC8803	Sheet 66 of 67

Engineer: David Du	INVENTEC	
Drawn by: David Du	TITLE: DIAMOND CHANGE LIST	Size: A3
R&D CHK		
DOC CTRL CHK		
MFG ENGR CHK		

FOR PV SCHEMATIC CHANGE

SHEET 30

1. ADD R6026 , R6027 , R6028 75 OHM RESISTORS TO PULL DOWN TO CRT SIGNAL R G B

SHEET 35

1. ADD C6020 , CC6021 , C6022 CAPACITORS PULL DOWN TO OPEN
2. CHANGE L4 , L5 , L6 FROM BEAD TO 0 OHM RESISTORS

SHEET 45

1. CHANGE C1435 FROM 4.7UF TO OPEN
2. SWAP MPCI_PWM THIS SIGNAL FROM C24 PIN1 TO Q1061 PIN 2 THAN Q1061 PIN 3 SWAP TO LINK R12 PIN 1

SHEET 46

1. SWAP PHONE THIS SIGNAL FROM R12 PIN1 TO C24 PIN 2
2. CHANGE C5 , C6 , C23 FROM 0.1UF TO 0.01UF

SHEET 51

1. ADD Q6005 NMOS TO CLEAR BACKWARD CURRENT

SHEET 57

1. KB_DAT_5 , KB_CLK_5 , EM_DAT_5 , EM_CLK_5, THIS SIGNAL
   ADD R6021 , R6022 , R6023 , R6024 , RESISTORS TO PULL UP +V5S
2. CHANGE L8 , L9 , L13 FROM 110 OHM FITTER BEAD TO 39 NH FITTER BEAD
3. CHANGE L4007 , L4006 , L4005 FROM 75 OHM FITTER BEAD TO 110 NH FITTER BEAD
4. CHANGE C137 , C138 , C165 FROM OPEN TO 18PF CAPICATORS

FOR PV-R SCHEMATIC CHANGE 2003/07/15

SHEET 9

1. CHANGE CAPACITOR C230 IMPROVE VGA/VCC BYPASS CAPACITOR ESR

SHEET 54

1. CHANGE CAPACITOR C1075 , C1076 , C1077 , C1078 , C1079 , C1080 , SIZE FROM 0603 TO 0402
   TO FIX INTEL WIRELESS CARD TO TOUCH CAPACITOR MECHANICAL ISSUE

SHEET 56

1. CHANGE RESISTOR R134 VALUE FROM 1.24K TO 1.21K TO INCREASE THE SIGNAL VOLTAGE LEVEL TO MEETING THE IEEE SPEC.
2. CHANGE R1351 FROM 4.7OHM TO BLM11A601S FITTER BEAD (L4008) TO INCREASE THE SIGNAL VOLTAGE LEVEL TO MEETING THE IEEE SPEC.

SHEET 57

1. ADD CAPACITOR C328 , C329 , C330 TO 150PF FOR EMI SOLUTION

SHEET 58

1. MOVE R1057 , R1058 FOR LAYOUT PLACEMENT BECAUSE MECHANICAL COVER HAVE TO SHORT

SHEET 59

1. CLEAR PAD1003 LINK TO GND FOR EMI SOLUTION
2. ADD PAD1004 FOR THERMAL

SHEET 63

1. DELETE R4010 AND LED_POWER_SWITCH CIRCUIT CHANGE TO SWITCH CIRCUIT FOR MECHANICAL REQUIREMENT

SHEET 64

1. IR_PIN11 LINK TO GND

LAYOUT NOTE:

1. CLK_CBIPCI_3R==>L5 CHANGE TO L3
2. CLK_MINIPCI_3R==>L5 CHANGE TO L3

FOR PVR SCHEMATIC CHANGE 2003/07/17

SHEET 57

1. CHANGE CAPACITOR VALUE C6027, C1335 FROM OPEN TO 100PF FOR EMI SOLUTION
2. ADD CAPACITOR C6030_100PF FOR EMI SOLUTION

SHEET 28

1. ADD CAPACITORS C6031 , C6032 , C6033 , C6034 , C6035 , C6036 , C6037 100PF FOR EMI SOLUTION
2. CHANGE CAPACITOR VALUE C256 FRPM 0.1uF TO 100PF

FOR PVR SCHEMATIC CHANGE 2003/07/23

SHEET 9

1. R1582 , R1583 , R6019 , C1469 , Q1057 , Q1058 CHANGE TO OPEN FOR HP REQUIREMENT

SHEET 20

1. CN24 NOT TO USE

SHEET 47

1. R1465 , R1449 CHANGE VALUE FROM OPEN TO 0 OHM
2. R1463 , R1448 CHANGE VALUE FROM 0 OHM TO OPEN

SHEET 5 54

1. MOVE Q1009 , R1071 FOR LAYOUT PLACEMENT BECAUSE MECHANICAL COVER HAVE TO SHORT

FOR PVR SCHEMATIC CHANGE 2003/07/28

SHEET 5 57

1. Change resistors value R6021,R6022 ,R6023 , R6024 from 4.7k ohm to 1k ohm for KBC PS2 microsoft keyboard to match .

FOR PVR SCHEMATIC CHANGE 2003/07/31

SHEET 32

1. change capacitor value C1268 from 22uF to 68uF

FOR PVR SCHEMATIC CHANGE 2003/08/01

SHEET 35

1. change ferrite-bead from L4, L5 , L6 (BLM11B750S) to resistors R6037 , R6038 , R6039 0 ohm

SHEET 57

1. change ferrite-bead from L4007, L4006 , L4005 (111XJBC_110NH) to resistors R6040 , R6041 , R6042 0 ohm
2. change ferrite-bead from L8, L9 , L13 (111XJBC_39NH) to ferrite-bead L8 , L9 , L10 BLM11B750S

FOR MV B SCHEMATIC CHANGE 2003/08/13

SHEET 32

1. ADD P-MOS Q6006 PIN4 LINK TO SLP_S3_5R , PIN5,8 LINK TO +AGP_V3S & PIN1, 2, 3, 6, 7, LINK TO +V3S
2. REMOVE L1014

SHEET 20

1. CHANGE CAPACITOR VALUE C286 2200pF TO NON-INSTALL

SHEET 38

1. ADD C6039 0.1UF BYPASS CAPACITOR .

SHEET 47

1. CHANGE RESISTOR R6035 0 OHM TO C6040 0.01UF CAPACITOR .

SHEET 56

1. CHANGE RESISTOR VALUE R134 FROM 1.21K TO 1.18K OHM.

FOR MV B SCHEMATIC CHANGE 2003/08/15

SHEET 5

FOR OCP circuit add R6043(0ohm) between U1002-8 and U1002-5
add R6044(open) and D6009(BAT54C) between U1002-13 and U1002-5

SHEET 46

1. ADD CAPACITOR C6042_100PF between U1 pin 12 , pin 13 .
2. ADD CAPACITOR C6041 open between U1 pin 2 , pin 3 .
3. ADD CAPACITOR C6043_100PF between U1 pin 9 , pin 10 .

FOR MV B SCHEMATIC CHANGE 2003/08/21

SHEET 6

1. CHANGE RESISTOR VALUE R1581 FROM 2.7K OHM TO 3.3K OHM FOR BATTERY SOLUTION

SHEET 9

1. CHANGE CAPACITOR VALUE C6016 FROM OPEN TO 12000PF FOR ATI RECOMMEND

SHEET 13

1. CHANGE CAPACITOR VALUE C187 FROM 0.1UF TO 39000PF FOR ATI RECOMMEND

Changed by: EE2	Date Changed: Thursday, August 21, 2003	Time Changed: 11:48:22 am	QA CHK	VER: A02	Model Number: PC8803	Sheet 67 of 67

Engineer: David Du Drawn by: David Du R&D CHK DOC CTRL CHK MFG ENGR CHK	INVENTEC	Size: A3
	TITLE: DIAMOND CHANGE LIST	